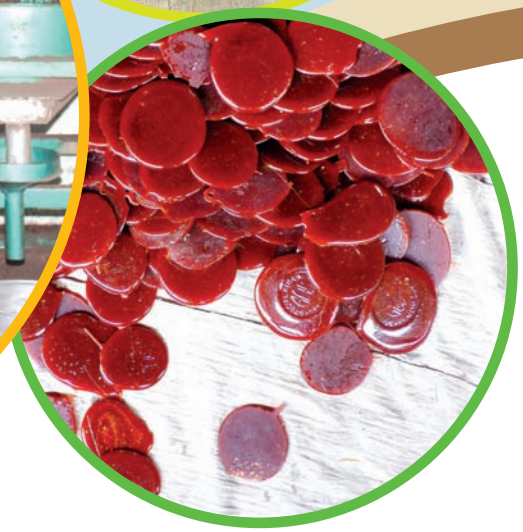
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# Lac, Plant Resins and Gums Statistics 2016 : At a Glance







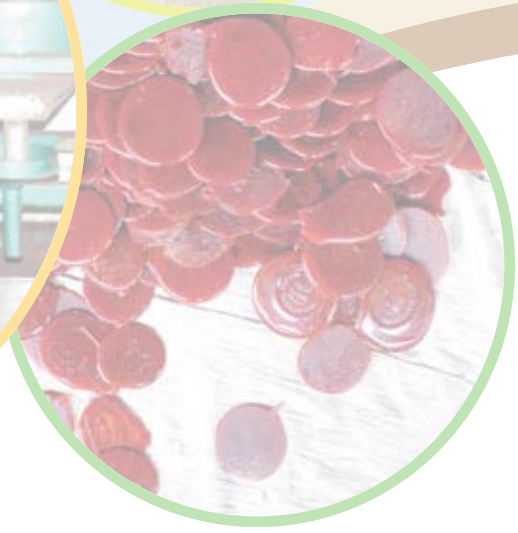
**ICAR-Indian Institute of Natural Resins and Gums**
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***Cover page photographs***
Left to right :
**Left Circle** - Lac processing plant
**Right Circle (Top)** - *Sterculia urens* (Plant gum exudate)
**Right Circle (Bottom)** - *Button* lac (Hand made shellac)

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# FOREWORD

India is one of the largest producers of natural resins, gums and gum-resins (NRGs) along with China, Indonesia, Russia and Brazil. Our country is the world leader in production of *guar*, *karaya* and psyllium gums as well as lac. The present publication, 'Lac, Plant Resins and Gums Statistics 2016: At a Glance' contains statistics on geographical distribution, production, price, processing, export and import of natural resins and gums including lac, pine resins, *guar* gum, gum *karaya*, *dhawda* gum, Olibanum, *etc*. The production of gums in important states, *viz*, Chhattisgarh, Madhya Pradesh, Gujarat, Rajasthan and Andhra Pradesh; production of pine resins in Uttarakhand and Himachal Pradesh; export and import of natural resins and gums is included to give a clear and comprehensive picture of the importance of these at national and international level.

Documentation of reliable and updated data on a major production and market related information would be helpful for the stakeholders to plan their operation in time for collection and disposal of the produces. This publication covers comprehensively the Natural Resins and Gums (NRGs) in India, focusing on their existing markets and potential. It would serve a wider section of the society mainly business communities, organizations/federations, public policy leaders, academia and many more.

In this issue, content is presented in five major headings covering introduction; methodology; production, processing and trade; policy implications and conclusions. Information on market trend is essential for better organization of Indian producing states to meet the demand through better organization of their local/regional/national commercial channels from production to export. Further, stabilization of the market with appropriate stocks, quality control of exported products at global level and adequate remunerative price support at production level in India is also covered.

The statistics presented in this publication offer valuable insights of NRG sector. I am sure that the information and data contained in this bulletin would be useful to all the stakeholders of NRG sector. Authors have made every effort to distill and condense a very large and diverse topic into and approachable volume. My appreciation goes to authors for their efforts made in bringing out this bulletin. Suggestions and inputs are sought from stakeholders for improvement of this publication in future.



**(KK Sharma)**

Director

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# CONTENTS


| Sl. No. | Particulars | Page No. |
|---|---|---|
| | ***LIST OF UNITS AND CHAPTERS*** | |
| **FOREWORD** | | 3 |
| **ACKNOWLEDGEMENT** | | 4 |
| **CONTENTS** | | 5-8 |
| **I.** | **INTRODUCTION** | 9-13 |
| **II.** | **METHODOLOGY** | 14-17 |
| 1. | Sampling design and survey area | 15 |
| **III.** | **PRODUCTION, PROCESSING AND TRADE** | 18-67 |
| (A) | NATURAL RESINS | 18-38 |
| 2. | Lac - a fauna based natural resin | 19 |
| 3. | Pine resin - a flora based resin | 30 |
| 4. | Copal - a flora based resin | 36 |
| 5. | Dammar batu - a flora based resin | 37 |
| 6. | Mastic gum - a flora based resin | 38 |
| (B) | NATURAL GUMS | 38-60 |
| 7. | Gum *karaya* - Plant exudate based natural gum | 40 |
| 8. | *Dhawda* gum/Gum ghatti - Plant exudate based natural gum | 44 |
| 9. | Gum arabic - Plant exudate based natural gum | 46 |
| 10. | Asian gum - Plant exudate based natural gum | 48 |
| 11. | African gum - Plant exudate based natural gum | 50 |
| 12. | Jhingan gum - Plant exudate based natural gum | 50 |
| 13. | Saja - Plant exudate based natural gum | 50 |
| 14. | Tragacanth (Adracanth) - Plant exudate based natural gum | 50 |
| 15. | Xanthum gum - A fauna based natural gum | 51 |
| 16. | Other gums - Plant exudate based natural gums | 51 |
| 17. | Casia tora seeds - Seed based natural gum | 53 |
| 18. | *Guar* gum - Seed based natural gum | 54 |
| (C) | NATURAL GUM RESINS | 60-64 |
| 19. | Asafoetida | 60 |
| 20. | Salai | 61 |
| 21. | Myrrh | 63 |
| 22. | Dikamali | 64 |
| (D) | NATURAL RESINS AND GUMS | 64-67 |
| 23. | Export and import of Natural Resins and Gums during 2015-16 | 65 |
| **IV.** | **POLICY INTERVENTIONS** | 68-69 |
| 24. | Mechanism for marketing of Non - Wood Forest Products (NWFPs) through Minimum Support Price (MSP) and development of value chain for NWFPs | 68 |
| **V.** | **CONCLUSIONS** | 70-71 |

| Sl. No. | Particulars | Page No. |
|---|---|---|
| | ***LIST OF TABLES*** | |
| 1. | World EXIM trade aggregation of lac, natural gums, resins, gum-resins and balsam | 13 |
| 2. | Details of surveys conducted | 15 |
| 3. | Distribution of the sample size during the survey | 16 |
| 4. | Total NRG production and trade during 2015-16 | 18 |
| 5. | Lac production in India during 2015-16 | 20 |
| 6. | Top ten lac producing districts in the country | 21 |
| 7. | Major lac processing centers in India during 2015-16 | 23 |
| 8. | Quantity of sticklac processed in India during 2015-16 | 24 |
| 9. | Markets with annual arrival of over 500 tons | 24 |
| 10. | Lac processing centers in the major lac producing districts of India | 25 |
| 11. | Export of lac and its value added products from India during 2014-16 | 27 |
| 12. | Top destination of Indian lac during 2015-16 | 27 |
| 13. | Export of lac during previous eight years | 28 |
| 14. | Commercially tapped sources of pine resin: species and country of production | 30 |
| 15. | State wise annual pine resin production and their contribution in the total production | 32 |
| 16. | Quality criteria for different grades of the rosin | 33 |
| 17. | Grade-wise price of rosin during 2015-16 | 34 |
| 18. | Destination and source wise EXIM scenario of rosin during 2015-16 | 35 |
| 19. | Source wise EXIM scenario of copal during 2015-16 | 37 |
| 20. | Destination and source wise EXIM scenario of dammar batu during 2015-16 | 37 |
| 21. | Destination and source wise EXIM scenario of mastic gum during 2015-16 | 38 |
| 22. | Gum production in major gum producing states of India (in tons) | 39 |
| 23. | Gum *karaya* production in major gum producing states of India | 41 |
| 24. | Destination and source wise EXIM scenario of gum *karaya* during 2015-16 | 42-43 |
| 25. | *Dhawda* gum production in major gum producing states of India | 45 |
| 26. | Destination and source wise EXIM scenario of gum arabic during 2015-16 | 47 |
| 27. | Destination and source wise EXIM scenario of asian gum during 2015-16 | 49 |
| 28. | Export of african gum during 2015-16 | 50 |
| 29. | Destination and source wise EXIM scenario of Tragacanth (Adracanth) during 2015-16 | 50 |
| 30. | Destination and source wise EXIM scenario of xanthum gum during 2015-16 | 51 |
| 31. | Production of other gums in India | 52 |
| 32. | Collection price of other gums in India | 53 |

| Sl. No. | Particulars | Page No. |
|---|---|---|
| 33. | All India area, production and yield of *guar* seed and potential of *guar* gum production | 54 |
| 34. | State wise area, production and yield of *guar* seed during previous seven years | 55 |
| 35. | Various varieties of processed products of *guar* gum for industrial applications | 56 |
| 36. | Top agri-export commodities during previous three years | 57 |
| 37. | Export of *guar* gum from India during previous years | 58 |
| 38. | Destination and source wise EXIM scenario of asafoetida during 2015-16 | 61 |
| 39. | Destination and source wise EXIM scenario of olibanum during 2015-16 | 62 |
| 40. | Destination and source wise EXIM scenario of myrrh during 2015-16 | 64 |
| 41. | Total NRGs production | 65 |
| 42. | Total NRGs export from and import in India during 2015-16 | 66 |
| ***LIST OF ANNEXURES*** | | |
| ***Annexure I***. Sticklac production scenario in India during 2015-16 | | 72-73 |
| ***Annexure II.*** Lac producing areas in the country | | 74-76 |
| ***Annexure III.*** Gums producing districts and areas | | 76-78 |
| ***Annexure IV.*** Legal categorization of major gums produced in various states of India | | 79-80 |
| ***LIST OF FIGURES*** | | |
| 1. | World export flow of lac, natural gums, resins, gum-resins and balsams during the decade | 10 |
| 2. | World import flow of lac, natural gums, resins, gum-resins and balsams during the decade | 10 |
| 3. | Breakup of the value of World export aggregation of NRGs during 2015 | 11 |
| 4. | Breakup of the value of World import aggregation of NRGs during 2015 | 12 |
| 5. | Composition of stakeholders under survey programme | 16 |
| 6. | Composition of stakeholders under survey programme | 17 |
| 7. | Product/major group-wise production share of NRGs | 18 |
| 8. | State-wise share in lac production during 2015-16 | 20 |
| 9. | Crop-wise share in total lac production during 2015-16 | 20 |
| 10. | Lac production in India during last five years | 21 |
| 11. | State-wise share in lac processing during 2015-16 | 22 |
| 12. | Lac processing in India during last five years | 23 |
| 13. | Trends in price of sticklac and seedlac over the time | 26 |
| 14. | Direction of the trade of lac and its value added products | 28 |
| 15. | Trends in export of lac based products from India | 29 |
| 16. | Product-wise export share of various lac based products | 29 |

| | | |
|---|---|---|
| 17. | Pine resin production and its value in Himachal Pradesh and Uttarakhand | 31 |
| 18. | Pine resin production in Himachal Pradesh and Uttarakhand | 32 |
| 19. | Price pattern of various grades of pine resin based products | 34 |
| 20. | Destination wise scenario of rosin from India during 2015-16 | 36 |
| 21. | Source wise scenario of rosin in India during 2015-16 | 36 |
| 22. | Year wise production scenario of natural gums in India | 40 |
| 23. | State-wise average production of gum *karaya* | 41 |
| 24. | Gum *karaya* production in India | 42 |
| 25. | Export of gum *karaya* from India | 43 |
| 26. | Destination wise scenario of gum *karaya* from India during 2015-16 | 44 |
| 27. | Source wise scenario of gum *karaya* in India during 2015-16 | 44 |
| 28. | Year wise production scenario of gum *ghatti* in India | 45 |
| 29. | Destination wise scenario of gum *arabic* from India during 2015-16 | 46 |
| 30. | Source wise scenario of gum *arabic* in India during 2015-16 | 48 |
| 31. | Destination wise scenario of *asian* gum from India during 2015-16 | 49 |
| 32. | Total production of other gums across various states in India | 52 |
| 33. | International price level of different grades of *guar* gum | 56 |
| 34. | Trends in production potential and export of *guar* gum from India | 58 |
| 35. | Composition of *guar* gum export from India | 58 |
| 36. | Destination wise scenario of *guar* meal during 2015-16 | 59 |
| 37. | Destination wise scenario of *guar* gum Refined split during 2015-16 | 59 |
| 38. | Destination wise scenario of *guar* gum treated and pulverized during 2015-16 | 59 |
| 39. | Destination-wise quantitative share of the *guar* gum exported from India | 60 |
| 40. | Destination-wise quantitative share of the olibanum exported from India | 63 |
| 41. | Trend in export of natural resins and gums from India during last five years | 65 |
| 42. | Trend in import of natural resins and gums in India during last five years | 67 |

***LIST OF PHOTOS***

| | |
|---|---|
| ***Photo 1.*** Lac processing plant at Chattisgarh, India | 19 |
| ***Photo 2.*** Natural resin (Plant Kingdom) | 30 |
| ***Photo 3.*** Natural gum : Gum *Ghatti* (Grade-1) | 39 |
| ***Photo 4.*** Natural seed gum : *Guar* gum | 54 |

# INTRODUCTION

While the benefits from forests are numerous, the tangible benefits like timber, bamboo, fuel wood, fodder, Non-Wood Forest Products (NWFPs), *etc*., are quantifiable. Intangible benefits like maintenance of ecological balance, conservation of soil and moisture, regulating the water flow, carbon sequestration, *etc*., are not quantified but are of great significance. Forest is an important sector having a significant contribution to the Indian economy. According the Forest Survey of India, 2015 forest cover is about 21.34% of the total geographical area of the country. Forest produce based industries contribute to 1.2% of India's Gross Domestic Product (Economic Survey, Ministry of Finance, 2011). In India, there are around 1.73 lakh villages located in and around forests (MoEF, 2006). Though there is no official census figures for the forest dependent population in the country, different estimates put the figures from 275 million (World Bank, 2006) to 350- 400 million (MoEF, 2009).

Indian subcontinent is a major hub of biodiversity. Several forest products have significant importance in social and economic life in tropical areas. These forest products are classified into wood and NWFPs. NWFPs include natural resins, gums and exudates, leaves (*tendu*), turpentine, perfumery oils from roots, stumps and fruits of various tree species. These are also natural source of spices, medicines, dyes and tannins. Most of NWFPs are exported, but some are consumed locally also. In many areas, NWFPs are more profiTable than timber; despite of having more value, the sustainable management of forests has been traditionally focused on timber production. However, in the recent decades the interest for NWFPs have been increased, as a result of the international shift to multifunctional sustainable forest management, which aims at optimizing the provision of multiple goods and services, while maintaining the equilibrium of forest ecosystems. Within this framework, the promotion and utilization of NWFPs is identified as a priority area by the FAO. However, the particularities in harvesting these products distinguish their management from that of timber. For example, some NWFPs have short harvesting period and products perish soon after this period. Additionally, their frequent, uncontrollable and illegal harvest may have negative effects on the forest ecosystem.

The global macroeconomic landscape is currently chartering a rough and uncertain terrain characterized by weak growth of world output. The situation has been exacerbated by; (i) declining prices of a number of commodities, with reduction in crude oil prices being the most visible of them, (ii) turbulent financial markets (more so equity markets), and (iii) volatile exchange rates. These conditions reflect extreme risk-aversion behaviour of global investors, thus putting many, and in particular, commodities exporting economies under considerable stress. Even in these trying and uncertain circumstances, India’s growth story has largely remained positive on the strength of domestic absorption, and the country has registered a robust and steady pace of economic growth in 2015-16 as it did in 2014-15.

India’s share in world GDP has increased from an average of 4.8% during 2001-07 to 6.1% during 2008- 13 and further to an average of 7.0% during 2014 to 2015 in current Purchasing Power Parity

(PPP) terms (International Monetary Fund Report, 2016). The growth rates in agriculture have been fluctuating at 1.5% in 2012-13, 5.6% in 2013-14, (-) 0.2% in 2014-15 and 0.7% in 2015-16. (Economic Survey, 2016-17). Among the agriculture and allied sectors, crops including fruits and vegeTables account for about 61.0% of the GVA; the rest by the allied sectors consisting of livestock products, forestry and fisheries. The share of forestry and loggings to the GVA (at 2011-12 prices) of the country has also been declining and it is fluctuating between 1.58 % in 2013-14 to1.2 % in 2014-15 and further likely to be decline in 2015-16 also.

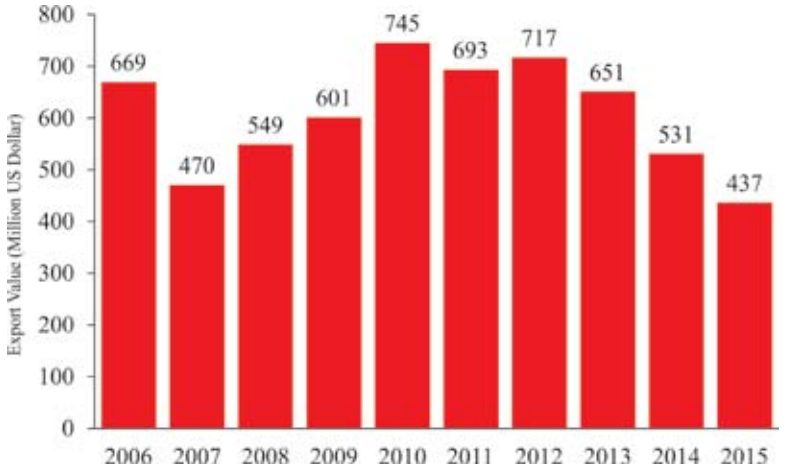



Figure 1. World export flow of lac, natural gums, resins, gum-resins and balsams during the decade

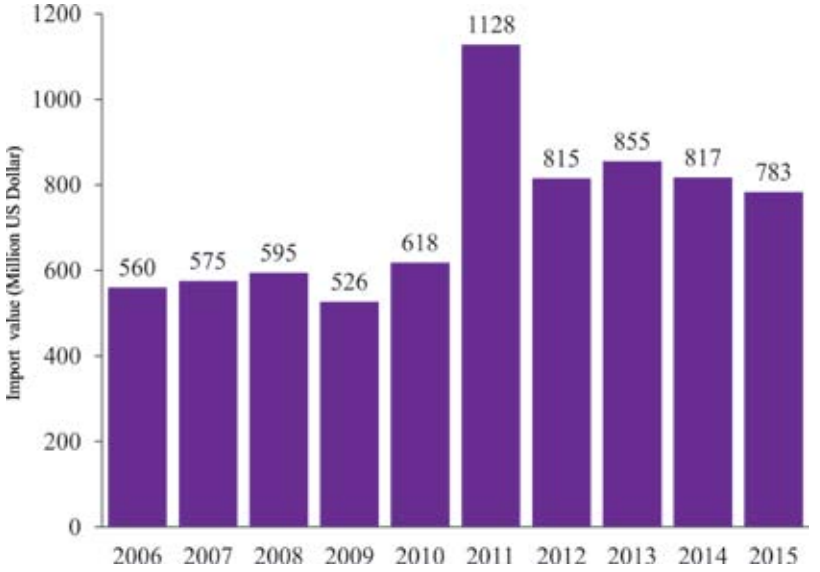



Figure 2. World import flow of lac, natural gums, resins, gum-resins and balsams during the decade

India ratified the Paris Agreement on 2nd October 2016. India's comprehensive NDC target is to lower the emissions intensity of GDP by 33 to 35% by 2030 from 2005 levels, to increase the share of non-fossil fuels based power generation capacity to 40% of installed electric power capacity by 2030, and to create an additional (cumulative) carbon sink of 2.5–3 GtCO2 e through additional forest and tree cover by 2030 (Economic Survey, 2016-17).

According International Trade Centre (ITC) calculations based on UN COMTRADE statistics, the world trade aggregation of lac, natural gums, resins, gum-resins and balsams during 2015 was about 1220.16 million US dollars. Out of this, the world export aggregation of lac, natural gums, resins, gum-resins and balsams during 2015 was about 436.70 million US dollars. Share of NRGs in India's total import (₹27370.87 billion in 2014-15 and ₹24902.98 billion in 2015-16) has been increased from 0.04% (₹11.18 billion) in 2014-15 to 0.05% (₹13.34billion) in 2015-16 with an annual acceleration of 19.35%. Share of NRGs in India's total export (₹18963.48 billion in 2014-15 & ₹17163.78 billion in 2015-16 has been decreased from 0.62% (₹118.95 billion) in 2014-15 to 0.33 %( ₹56.72 billion) in 2015-16 with an annual deceleration of 52.32 % (Department of Commerce, GOI, 2016). A decadal data (2006-2015) on world EXIM aggregation of lac, natural gums, resins, gum-resins and balsams were analyzed and presented in Figure 1 and Figure 2. Since 2012, deceleration in the value of world export aggregation was observed and stagnation was found in the value of world import aggregation during the similar period.

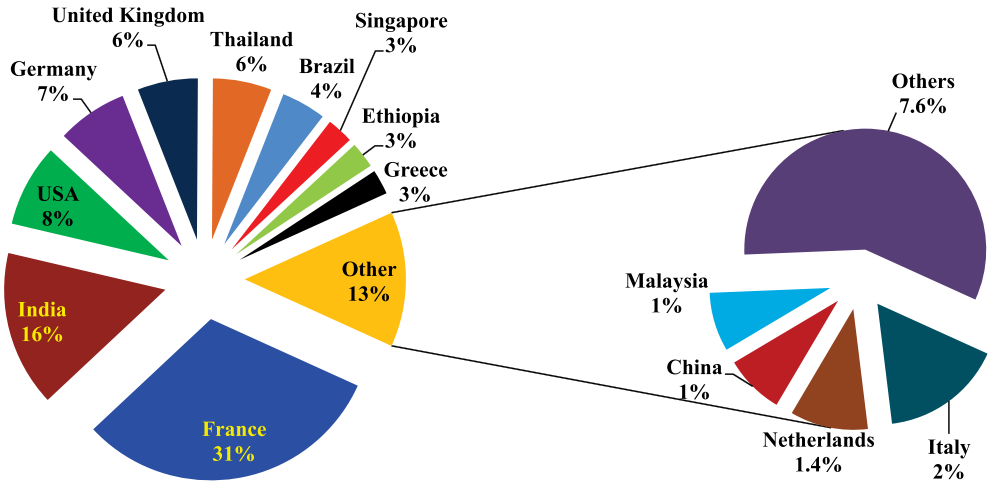



Figure 3. Break up of the value of World export aggregation of NRGs during 2015

Major suppliers of NRGs contributing about 92 % share in international market are France (31.2%), India (15.6%), USA (8.3%), Germany (7.1%), UK (6.1%), Thailand (5.9%), Brazil (4.5%), Singapore (2.7%), Ethiopia (2.6%), Greece (2.5%), Italy (2.2%), Netherlands (1.4%), China (1.1%) and Malaysia (1.0%). Rests of the 7.6% NRGs are supplied from 83 countries across the world (Figure 3). Similarly, the world import aggregation of lac, natural gums, resins, gum-resins and balsams

during 2015 was about 783.45 million US dollars. Major importers of NRGs contributing about 80% share in the international market are India (17.0 %), France (13.6 %), United States of America (11.2 %), Germany (5.2 %), China (4.7 %), Italy (3.9 %), Netherlands (3.4 %), United Kingdom (3.3 %), Portugal (2.8%), Russian Federation (2.6%), Singapore (2.5%), Saudi Arabia (2.4%), Spain (2.3%), Brazil (2.1%), Ireland (2.0%), Japan (1.6%), Switzerland (1.4%), and Thailand (1.3%). Rests of the 16.6% demand of NRGs aroused from the 115 countries across the world (Figure 4).

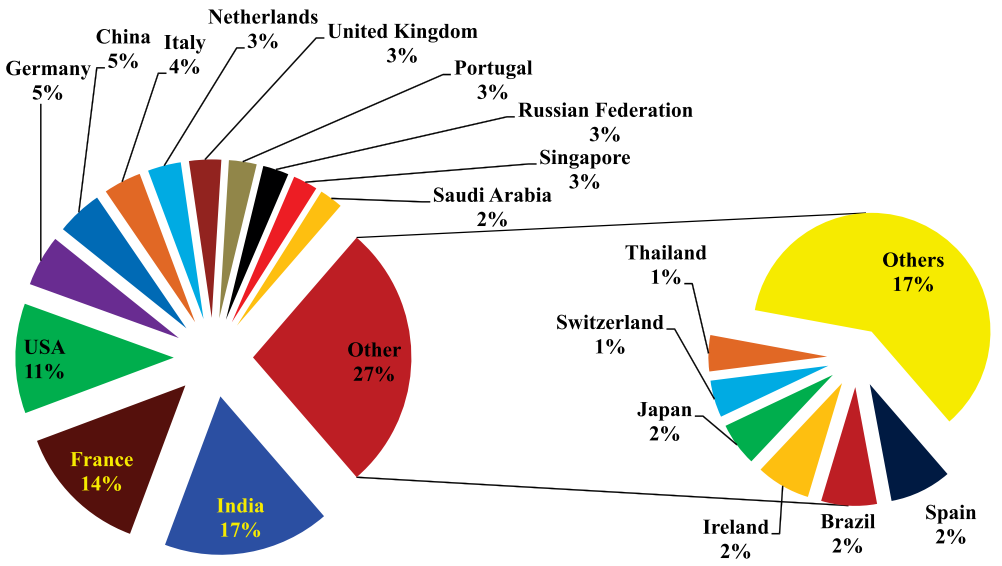



Figure 4. Breakup of the value of World import aggregation of NRGs during 2015

About 70% of the NWFP collection in India takes place in the tribal belt of the country (Mitchell *et al.*, 2003). Around 55% of employment in forestry sector is attributed to this sector alone (Joshi, 2003). NRGs of commercial importance like lac (*Kerria lacca* Kerr), pine resin (*P. roxburghii* Sarg.), *guar* gum (*Cyamopsis tetragonoloba* L.), gum *karaya* (*Sterculia urens* Roxb.), *dhawada* gum (*Anogeissus latifolia* Roxb.), tamarind gum (*Tamarindus indica* L.), *char /piyar* gum (*Buchanania lanzan* Spreng.) and *babool* gum (*Acacia nilotica* L.) are produced in India. India holds monopoly in international trade over some of the NRGs such as lac, gum *karaya* and *guar* gum.

NWFPs based on their chemical composition may be classified in three categories namely natural resins, natural gums and gum resins. Natural resins are solid or semi-solid materials, usually a complex mixture of organic compounds called terpenoides, which are insoluble in water but soluble in certain organic solvents. Resins are secretion of several plants, particularly coniferous trees.

Resins are used in the production of varnishes, adhesives and food glazing agents. These are also used as raw material for synthesis of incense and perfume. This group of natural resins includes lac secreted by an insect *K. lacca* (Kerr) and plant originated products like rosin, copal and dammer. Solidified resin from which the volatile terpene components have been removed by distillation

is known as rosin. Natural gums are polysaccharides of natural origin, capable of causing a high viscosity in the solution. Most often these gums are found as exudates from woody elements of plants or in seed coatings. In the food industry these are used as thickening, gelling and emulsifying agents and stabilizers. These are also used as adhesives, binding agents, crystal inhibitors, clarifying agents, encapsulating agents, flocculating, foam stabilizers, swelling agents, *etc.* Natural gums can be classified according to their origin. Firstly, tree exudates *e.g.* gum *arabic*, gum *ghatti*, gum *tragacanth*, gum *karaya*, *guar* gum, locust bean gum, *chicle* gum, *dammar*, *mastic* gum, psyllium gum and spruce gum. Secondly, originated from seaweeds *e.g.* agar and *carrageenan* and thirdly, produced by bacterial fermentation *e.g. gellan* gum and *xanthum* gum. They can also be classified as uncharged or ionic polymers (polyelectrolyte).

**Table 1. World EXIM trade aggregation of lac, natural gums, resins, gum-resins and balsams**
*(Value in Million US$)*

| Year | Export | Import | Re-Export | Re-Import |
|---|---|---|---|---|
| 2006 | 669.12 | 559.72 | 11.77 | 0.75 |
| 2007 | 470.47 | 575.00 | 11.82 | 1.50 |
| 2008 | 548.82 | 594.65 | 13.26 | 0.12 |
| 2009 | 601.20 | 526.39 | 19.94 | 0.24 |
| 2010 | 745.48 | 617.75 | 15.46 | 0.20 |
| 2011 | 693.28 | 1127.70 | 15.12 | 0.55 |
| 2012 | 716.89 | 814.98 | 17.88 | 0.24 |
| 2013 | 650.94 | 854.70 | 17.06 | 0.31 |
| 2014 | 530.78 | 816.86 | 10.43 | 0.37 |
| 2015 | 436.70 | 783.45 | 5.87 | 0.58 |

*Source: ITC calculations based on UN COMTRADE statistics.*

Gum-resins are the natural mixtures of gums and resins in variable proportions therefore possess properties of both the groups. They contain traces of essential oils and are partly soluble in water. They have a penetrating and characteristic odour and taste and obtained from the plants. Olibanum/ *salai* gum (*Boswellia serrata*), guggal *(Commiphora wightii),* myrrh, asafoetida, *etc.* are the major gum resins of national importance.

# METHODOLOGY

Reliable and periodical assessments of quantities and values of production and market outlook studies at the national level of NWFPs are essential to decision-makers for policy formulation and governing the sustainable development of the sector. Appropriate and biometrically valid inventories of non-wood forest resources are an essential prerequisite for their sustainable management and harvesting. Timely and accurate estimation of production may be helpful for the stakeholders to plan their operations in time. Accuracy in production estimate would be helpful in precision planning by all concerned. Besides knowing the present status, it would be helpful in regulating imports, planning for enhanced exports, reasonable prices and reliability in supply of lac based products. However, such statistics do not yet exist for most countries, neither is the already available data comparable among countries. In most countries, the current coverage and quality of existing information is inadequate for policy analysis and decision making at national level.

From the lessons learned by implementing this national survey, specific case studies to improve NRGs data gathering methodologies are being carried out in selected states (Jharkhand, West Bengal, Chhattisgarh and Madhya Pradesh). Hence, under the research project "Market research for production and marketing of natural resins and gums" a standard reporting format has been developed to assist in recording statistics on the production and trade of major NRGs. This approach was used in order to compile NRGs state profiles for all states of India. A brief summary of the assessment is also being published annually as a separate chapter of the Agricultural Research Data Book, ICAR-IASRI, New Delhi.

The objectives of present methodology are to estimate and update the production, processing and value addition of NRGs at national level and to compile the latest EXIM data. The approach used in the present methodology was through survey of local traders and processors, as all NRGs produced in India are collected/procured through the local traders/societies/forest departments. This report presents an overview of quantity and values for selected forest products and services from Indian forests. Results highlight a strong concentration of value in *guar* gum production. The economic value of NWFPs however is only partly reflected on the market and recorded by official statistics. Underestimation by existing statistics might be due to several reasons, including: (i) the public-good nature of many products/services and consequent difficulties in estimating them; (ii) the fact that data recorded for official statistics are not always complete (for example they do not always cover all the region as in the case of NWFPs); and (iii) the fact that a certain proportion of the NRGs is not mirrored by official data because some products are traded through informal channels and markets.

Big traders/federations are limited in number but they have close contact with the primary purchaser who have knowledge of present crop condition and expected output. Survey of all big traders was helpful in estimation. Further, all the produced products pass through the processing units. Processors use the current harvested or stocked or imported crop/produce. Survey of processing units was helpful in estimating quantity of processed products at national level and validation of production data. Information on market arrival during seasons at important markets in India was

collected through survey of identified major markets. Regular contacts were also made with the persons/ organizations related to NRGs in India through correspondence, phone and personal visit for collection and updating of data. Production estimation was made by the survey of selected processors, exporters, importers and markets (traders). EXIM data on NRGs & related products were obtained from Directorate General of Commercial Intelligence and Statistics (DGCI&S), Kolkata. In the Indian context, the data relating to India's exports and imports by commodities is officially compiled and published by the DGCI&S, Government of India. The data is compiled in accordance with the Harmonized Commodity Description and Coding System, also known as the Harmonized System (HS) of tariff nomenclature, developed and maintained by the World Customs Organization (WCO) and adopted by the Government of India as Indian Trade Classification, commonly called ITC (HS) codes. Five schedules / questionnaire were framed for collection of data and information during the field surveys *i.e.* survey of markets, survey of processing centers, crop-wise arrival in the market, survey of importers and processing at processing centers. Production year was considered from April to March (Financial year) and final production assessment completed by the month of April. Data and other related information for estimation of production and processing was collected throughout the year. Tabular analysis and pictorial presentation has been used to analyze the survey data. Validation of production data at national level was made by secondary data on quantity processed at national level, export and import figures.

## Sampling design and survey area

National level information and data on NRGs were collected from primary and secondary sources. Surveys were made in various NRG producing areas for collection of data throughout the year.

**Table 2. Details of the surveys conducted**

| States/Country | Districts |
|---|---|
| Andhra Pradesh | Vishakhapatnam and East Godavari |
| Assam | Kamrup and Dhemaji |
| Bihar | Muzaffarpur and East Champaran |
| Chhattisgarh | Balarampur, Bilaspur, Dhamtari, Korba, Kanker, Korea and Raipur |
| Delhi | Chandni Chauk |
| Gujarat | Bharuch, Surat and Vadodara |
| Jharkhand | Bokaro, Dhanbad, Giridih, Gumla, Khunti, Ranchi, Simdega and West Singhbhum |
| Karnataka | Bangaluru and Uttara Kannada |
| Madhya Pradesh | Balaghat and Seoni |
| Maharashtra | Bhandara and Gondia |
| NE region | Garo hills, Imphal, East Kameng, Barapani and Kohima |
| Odisha | Balasore and Sundergarh |
| Rajasthan | Jaipur, Ajmer and Jodhpur |
| Uttar Pradesh | Allahabad |
| Uttarakhand | Bilaspur |
| Telangana | Hyderabad |
| West Bengal | Kolkata and Purulia |
| Nepal | Kathmandu |

The requisite data were collected from respondents at various NRGs markets and processing centers. For updating the information and data, regular telephonic contacts were also made with the respondents.

**Table 3. Distribution of the sample size during the survey**

| States/ Country | No. of districts | Farmers/ Forest dwellers | Market functionaries | Processors/ Manufacturers/ Exporters | Govt. officials/ NGOs/ Other key informants | Total |
|---|---|---|---|---|---|---|
| Andhra Pradesh | 2 | 0 | 3 | 0 | 2 | 5 |
| Assam | 2 | 80 | 1 | 0 | 6 | 87 |
| Bihar | 1 | 10 | 10 | 10 | 12 | 42 |
| Chhattisgarh | 6 | 90 | 4 | 6 | 15 | 115 |
| Delhi | 1 | 0 | 10 | 0 | 27 | 37 |
| Gujarat | 1 | 0 | 2 | 1 | 2 | 5 |
| Jharkhand | 7 | 100 | 11 | 6 | 22 | 139 |
| Karnataka | 2 | 10 | 0 | 2 | 6 | 18 |
| Madhya Pradesh | 2 | 50 | 1 | 4 | 2 | 57 |
| Maharashtra | 2 | 20 | 1 | 4 | 1 | 26 |
| NEH | 5 | 20 | 0 | 2 | 77 | 99 |
| Odisha | 2 | 20 | 1 | 2 | 2 | 25 |
| Rajasthan | 3 | 20 | 5 | 29 | 13 | 67 |
| Telangana | 1 | 0 | 0 | 0 | 8 | 8 |
| Uttar Pradesh | 1 | 20 | 2 | 13 | 7 | 42 |
| Uttarakhand | 1 | 0 | 0 | 2 | 4 | 6 |
| West Bengal | 2 | 26 | 3 | 7 | 9 | 45 |
| Nepal | 1 | 0 | 6 | 0 | 0 | 6 |
| Others | 5 | 40 | 1 | 1 | 52 | 94 |
| Total | 47 | 506 | 61 | 89 | 267 | 923 |

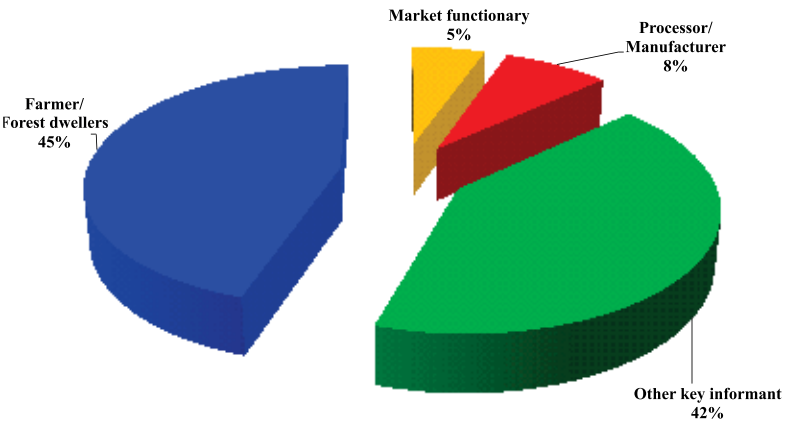



Figure 5. Composition of stakeholders under survey programme (Category wise)

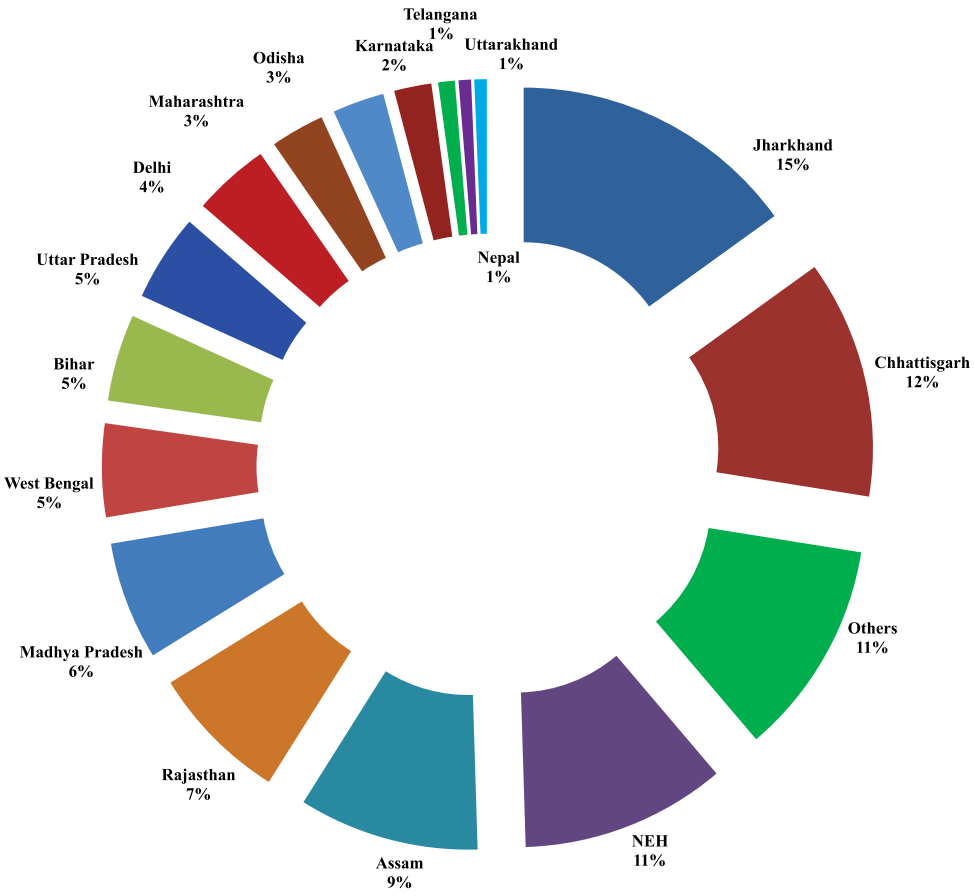



Figure 6. Composition of stakeholders under survey programme (State-wise)

Surveys were conducted in 47 districts of 18 states of India and Nepal covering 923 stakeholders including 267 institutions (Govt. Officials/ NGOs/ Other key informant), 506 farmers/forest dwellers, 61 market functionaries (traders/wholesalers), 89 processors/manufacturer/exporters/importers and resource persons were interacted through visits and telephonic conversations during 2015-16. These respondents were directly/indirectly concerned with NRG production, processing and value addition over the country. Name of the states and districts which were covered under survey and sample size surveyed during the year is presented in Table 2 and Table 3, respectively. Category wise and state-wise composition of selected stakeholders is presented in Figure 5 and Figure 6.

# PRODUCTION, PROCESSING AND TRADE IN NRGS

NRG production level in India during 2015-16 was estimated comparatively lower (8.43 lakh tons) than previous year (11.60 lakh tons). Production and trade of NRGs including *guar* gum, lac, pine resin, *karaya* gum, *dhawda* gum and other natural resins and gums is depicted in Table 4 and Figure 7.

**Table 4. Total NRG production and trade during 2015-16 (quantity in tons)**

| Name of product | Production | Share (%) | Export | Share (%) | Import | Share (%) |
|---|---|---|---|---|---|---|
| *Guar* gum* | 817042 | 97.05 | 256677.18 | 94.21 | 633.13 | 0.66 |
| Lac | 18746 | 1.99 | 7668.42 | 2.81 | 0.00 | 0.00 |
| Pine resin | 5726 | 0.76 | 204.33 | 0.08 | 41962.57 | 43.53 |
| Karaya gum | 100 | 0.02 | 230.79 | 0.08 | 771.56 | 0.8 |
| Other NRGs | 1563 | 0.18 | 7681.5 | 2.82 | 53020.39 | 55.01 |
| **Total** | **843178** | **100.00** | **272462.22** | **100.00** | **96387.65** | **100.00** |

**Estimation of guar gum is based on the conversion of total guar seed production*

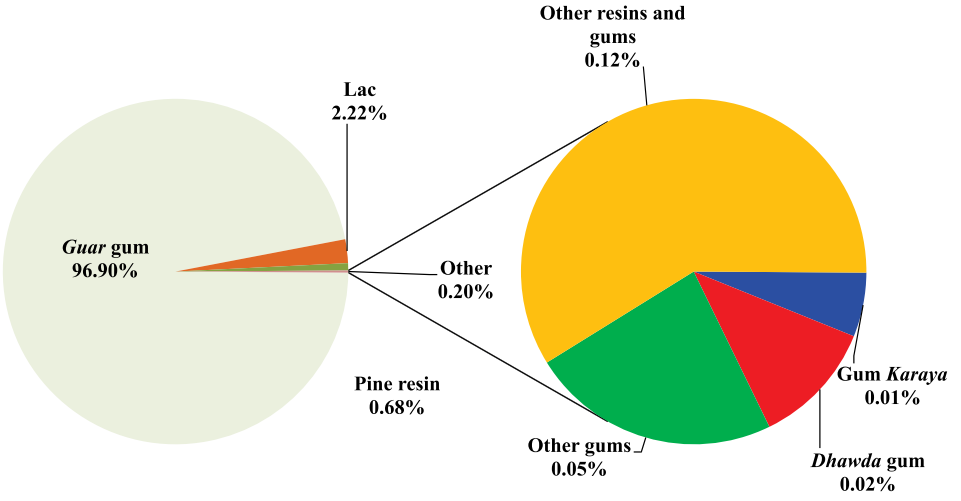



Figure 7. Product/major group-wise production share of NRGs

It is evident from the Table 4 that *guar* gum has a lion's share in total NRG production and consequently in export also. For *guar* gum, India is the leading producer, processor and exporter country in the world. However, a negligible quantity has been imported for research and other purposes. Similarly, India is the largest producer, processor and exporter of the lac. Export quantity of lac has been increased by 16.73% in comparison to previous year export. However, the value of exported lac was declined by 23.23%. Only less than one percent production of the total NRGs is contributed by pine resins, gum *karaya* and other NRGs. Export under gum *karaya* and other NRGs has exceeded the production data as the exported quantity supplied from the carry over stock and secondly, raw material imported and exported after processing in India. Overall, there was

about 50% decline in total quantity of NRGs exported and quantity in import basket was increased by 11.81% over the previous year. But, price fall of *guar* gum in international market resulted with a decline of 180% in total foreign exchange earnings as comparison to 2014-15. The detailed information about state-wise production processing and trade of natural resins (Lac, pine resin), natural gums (Gum *karaya*, *guar* gum, gum *ghatti*, *gum arabic* ) and natural gum resins (Olibanum, myrrh, asafoetida) is presented below.

## Lac – a fauna based natural resin

Lac is a natural resin secreted mainly by Indian lac insect, *K. lacca* (Kerr) which thrives on the tender twigs of specific host trees *viz.*, *palas* (*Butea monosperma*), *ber* (*Ziziphus mauritiana*), *kusum* (*Schleichera oleosa*), *Flemingia semialata*, *Ficus spp. etc. Rangeeni* and *kusmi* are the two strains of lac insect which are classified based on preference of the insect for specific host plants. Raw lac is the source of three valuable products *i.e.* resin, dye and wax.



Photo 1. Lac processing plant at Chattisgarh, India

Lac cultivation is an important source of income for livelihood of the forest and sub-forest dwellers of Jharkhand, Chhattisgarh, Madhya Pradesh, West Bengal, Maharashtra, Odisha and parts of Uttar Pradesh, Andhra Pradesh, Gujarat and NEH region. Lac production is highly labour intensive process and provides employment to both men and women dwelling in forest and sub-forest areas of these states. It is a highly remunerative crop, paying high economic returns to the farmers and also to foreign exchange of the country through its export. Lac is mainly produced in India, Thailand, Indonesia, parts of China, Myanmar, Philippines, Vietnam, Cambodia *etc.* and India is the largest producer of lac in the world.

## Lac production in India

During 2010s, the efforts in terms of policy, research and development regarding upliftment of lac sector was percolated at grass root level. Consequently, the interest of stakeholders had got the vital support. At present only less than 5% lac host trees are under the lac cultivation. It is assumed that there are some bottlenecks hampering the growth of the sector. Estimation of lac production is required by the Government, lac-based industries, lac traders, entrepreneurs and exporters. The lac growers, processors, traders, exporters and policy makers can plan their operations in time by using timely and accurate estimations. The cultivation of lac on a large number of hosts of different kinds, its collection by numerous small growers, variations in the yield depending on the type and size of the host, cultivation practices and climatic conditions are the major factors influencing the estimation of lac production. Accuracy in production estimate would be helpful in precision planning by all concerned. Besides knowing the present status, it would be helpful in regulating imports, planning for enhanced exports, reasonable prices and reliability in supply of lac based products.

On the basis of survey in the markets of different lac producing districts, the estimated national production of sticklac during 2015-16 was approximately 18,746 tons comprising *rangeeni* (7,597 tons) and *kusmi* (11,149 tons) sticklac. Among the lac growing states, Jharkhand state ranks 1st followed by Chhattisgarh, Madhya Pradesh, Maharashtra and Odisha. These five states contribute around 93% of the national lac production (Figure 8). Contribution of Jharkhand in national lac production is about 53% followed by Chhattisgarh (17%), Madhya Pradesh (12%), Maharashtra (8%) and Odisha (3%). Among the different cropping season crops, *jethwi* crop was ranked 1st with the contribution of 32% followed by *aghani* (27%), *baisakhi* (24%) and *katki* (17%) in total lac production.

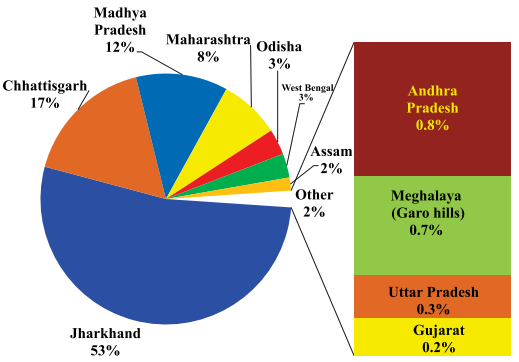



Figure 8. State-wise share in lac production during 2015-16

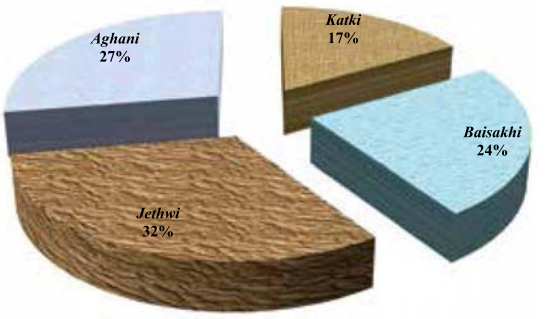



Figure 9. Crop-wise share in total lac production during 2015-16

**Table 5. Lac production in India during 2015-16 (in tons)***

| Name of states / Districts | Name of lac crop@ | | | | Total |
|---|---|---|---|---|---|
| | *Baisakhi* | *Jethwi* | *Katki* | *Aghani* | |
| Jharkhand | 825 | 4935 | 785 | 3405 | 9950 |
| Chhattisgarh | 622 | 765 | 600 | 1200 | 3187 |
| Madhya Pradesh | 1670 | 66 | 425 | 61 | 2222 |
| Maharashtra | 860 | 0 | 605 | 0 | 1465 |
| Odisha | 65 | 118 | 75 | 365 | 623 |
| West Bengal | 150 | 62 | 235 | 135 | 582 |
| Assam | 65 | 0 | 250 | 0 | 315 |
| Andhra Pradesh | 60 | 5 | 100 | 7 | 172 |
| Meghalaya | 120 | 0 | 7 | 0 | 127 |
| Uttar Pradesh | 15 | 0 | 40 | 0 | 55 |
| Gujarat | 20 | 20 | 3 | 5 | 48 |
| **Total** | **4472** | **5971** | **3125** | **5178** | **18746** |

*@**Baisakhi:** - Summer season crop of rangeeni; **Jethwi:** - Summer season crop of kusmi; **Katki:** - Rainy season crop of rangeeni; **Aghani:** - Winter season crop of kusmi; $ Andhra Pradesh including Telengana;* See Annexure I for details.*

In comparison to previous year, the production of *ranggeni* as well as *kusmi* crop was increased by 8% and 12%, respectively. The increase in production was comprised by 33 %, 12 % and 4 % for *jethwi*, *baisakhi* and *katki* crops, respectively. However, a decline of 6 % was observed in case of *aghani* crop .Overall, the total lac production for the year 2015-16 has increased by 10 % in comparison to 2014-15. Lac production scenario in India is presented in Table 5 and the top ten lac producing districts in the country are enlisted in Table 6. Share of different crops at national level is presented in Figure 9. Lac production in India during previous five years is depicted in Figure 10.

**Table 6. Top ten lac producing districts in the country**

| District (States) | 2014-15 | Rank | 2015-16 | Rank |
|---|---|---|---|---|
| Ranchi (Jharkhand) | 2530 | 1 | 3190 | 1 |
| Simdega (Jharkhand) | 1910 | 2 | 2020 | 2 |
| Khunti (Jharkhand) | 1380 | 3 | 1610 | 3 |
| Gumla (Jharkhand) | 1330 | 4 | 1605 | 4 |
| Seoni (Madhya Pradesh) | 1165 | 5 | 900 | 7 |
| Gondia (Maharashtra) | 1100 | 6 | 1100 | 5 |
| Balaghat(Madhya Pradesh) | 882 | 7 | 741 | 9 |
| West Singhbhum (Jharkhand) | 860 | 8 | 825 | 8 |
| Korba (Chhattisgarh) | 750 | 9 | 1010 | 6 |
| Kanker(Chhattisgarh) | 510 | 10 | 705 | 10 |

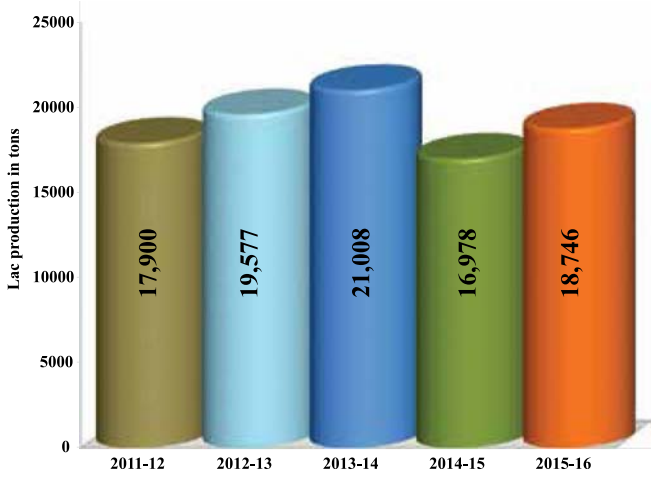



Figure 10. Lac production in India during previous five years

The overall, production of the lac in the country has been estimated to be 18,746 tons which is higher than the previous year production (16,978 tons). During, current year 2015-16, the lac production tends to be the average production (18841 tons) during last five years. It is interesting to mention that the production level of lac had increased from the lowest level of 9,035 tons during 2010-11 to 18,746 tons (about 100 % increases) during 2015-16. It reached up to the highest level of 21,008 tons during the second year of XII plan period. Although, during 2014-15, it had declined due to

various biotic and abiotic stresses. District wise lac production statistics and major lac producing areas in the country have been presented in *Annexure I* and *Annexure II*, respectively.

**Lac processing and value addition in India**

Sticklac in small quantity or lots is generally sold in the rural markets (*haats)* by the lac growers. Rural markets (*haat*) in remote lac growing areas operate once or twice in a week. Lac growers, after harvesting sticklac sell to *paikars* (primary purchasers). Lac growers, nearer to lac processing units sell their produce directly to processers. After collection of scraped lac in the course of the market day by the *paikars* is sold to the wholesaler in the same market or nearby processing centers in bigger lots. Finally the wholesalers sell their purchase to processor. After processing at the primary processing centre, lac is sold for domestic consumption or exported. Different lac processing centers in the country during 2015-16 have been presented in Table 7 and Table 8, respectively.

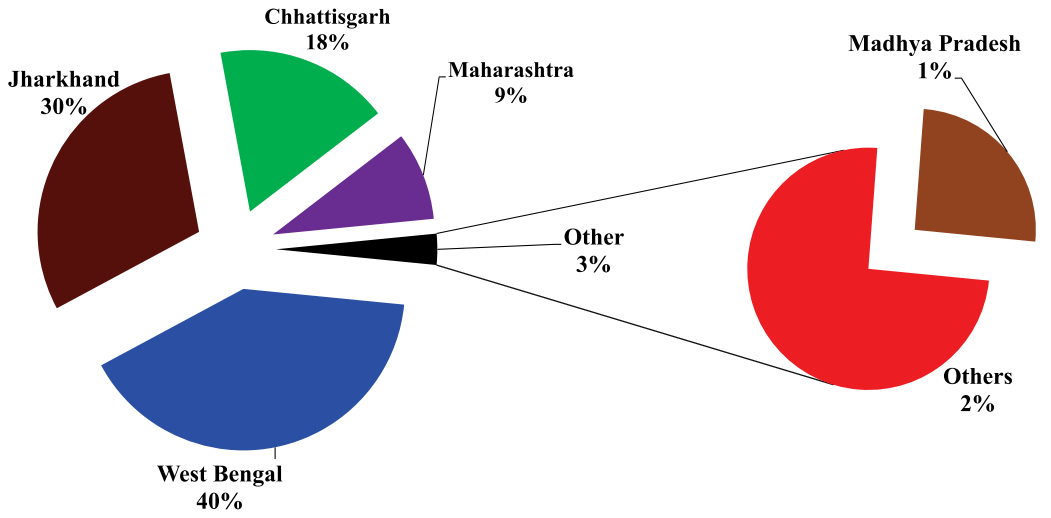



Figure 11. States-wise share in lac processing during 2015-16

On the basis of surveys conducted at different lac processing centers in the country, the total quantity of sticklac processed during 2015-16 was 21571 tons which also included the imported as well as previous years carry over stock in India. Information about share of different states in lac processing is presented in Figure 11 and amount of lac processed in India during previous five years is depicted in Figure 12. Lac processing centers in India and amount of sticklac processed at A total 183 lac processing units were functional in West Bengal (134), Chhattisgarh (31), Jharkhand (16), Maharashtra (6) and Madhya Pradesh (6) during the year 2015-16.

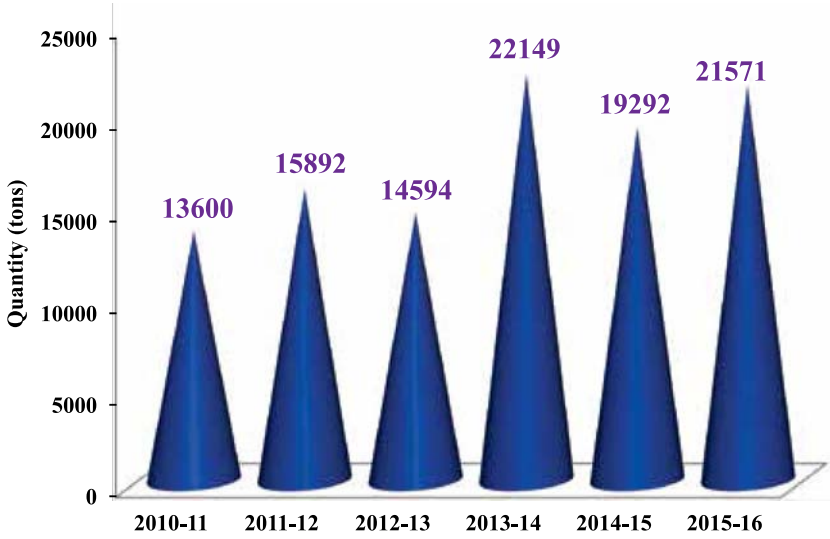



Figure 12. Lac processing in India during last five years

**Table 7. Major lac processing centers in India during 2015-16**

| States | Districts /Centres | No. of processing units | Products made |
|---|---|---|---|
| Chhattisgarh | Dhamtari | 15 | Seedlac, Button lac, Bleached lac, Aleuritic acid |
| | Janjgir-Champa | 5 | Seedlac, Shellac, Bleached lac, Dewaxed Shellac, Lac dye |
| | Kanker | 2 | Seedlac |
| | Korba | 7 | Seedlac, Shellac, Bleached lac, Button lac |
| | Rajnandgaon | 1 | Seedlac, Shellac |
| | Raipur | 1 | Bleached lac, Aleuritic acid |
| Jharkhand | Daltonganj | 2 | Seedlac |
| | Ranchi and Khunti | 10 | Seedlac, Button lac, Shellac, Lac dye, Bleached lac |
| | Simdega | 2 | Seedlac |
| | Saraikela-Kharsawan | 1 | Bleached lac |
| | West Singhbhum | 1 | Black Shellac |
| Madhya Pradesh | Indore | 1 | Seedlac, Bleached lac, Varnish |
| | Balaghat | 2 | Seedlac |
| | Seoni | 2 | Seedlac |
| | Hosangabad | 1 | Seedlac |
| Maharashtra | Gondia | 6 | Seedlac, Shellac, Gasket Shellac Compound, Bleached lac |
| West Bengal | Purulia | 93 | Seedlac, Shellac, Button lac, Bleached lac, Aleuritic acid, lac wax, Dewaxed Decolourised lac |
| | North 24 Paragana | 1 | Aleuritic acid |
| | Others | 20 | Lac based value added products |
| Others | | 10 | Lac based value added products |
| **Total** | | **183** | |

**Table 8. Quantity of sticklac processed in India during 2015-16**

| States | Districts/ Centres | Quantity processed (tons) | % change over last year |
|---|---|---|---|
| Chhattisgarh | Dhamtari | 2091 | -0.4 |
| | Janjgir-Champa | 416 | -63.8 |
| | Kanker | 250 | 127.3 |
| | Korba | 901 | -32.3 |
| | Rajnandgaon | 120 | 50.0 |
| | **Sub total** | **3778** | **-20.8** |
| Jharkhand | Daltonganj | 60 | 200.0 |
| | Ranchi | 1551 | 84.6 |
| | Khunti | 4443 | 26.9 |
| | Saraikela-Kharsawan | 50 | 66.7 |
| | Simdega | 301 | -39.8 |
| | West Singhbhum | 50 | 66.7 |
| | **Sub total** | **6455** | **31.2** |
| Maharashtra | Gondia | 1910 | -2.1 |
| Madhya Pradesh | Indore | 20 | -50.0 |
| | Balaghat | 40 | 33.3 |
| | Seoni | 100 | 100.0 |
| | Hoshangabad | 10 | -66.7 |
| | **Sub total** | **170** | **13.3** |
| West Bengal | Purulia | 7003 | 6.2 |
| | North 24 Paragana | 100 | 150.0 |
| | Others | 1655 | 133.1 |
| | **Sub total** | **8758** | **19.3** |
| Others | Odisha, Karnataka, Uttar Pradesh, Delhi, Tamil Nadu, *etc.* | 500 | 214.5 |
| | **Total** | **21571** | **11.8** |

**Table 9. Markets with annual arrival of over 500 tons**

| States | Primary markets | Secondary markets |
|---|---|---|
| Chhattisgarh | Bhaisama Bazar | Dhamtari, Kathgora and Sakti |
| Jharkhand | Bandgaon, Jaldega and Kolebira | Khunti |
| Maharashtra | Barghat region | Gondia |
| West Bengal | Balarampur, Jhalda and Tulin | Balarampur |

In the processing of lac across the country, West Bengal (40 %), Jharkhand (30 %) and Chhattisgarh (18 %) shared about 90 % followed by Maharashtra (9 %), Madhya Pradesh and others (1%). There were 8 primary and 6 secondary markets existing at national level, in which annual arrival of sticklac was more than 500 tons. Name of the primary and secondary markets with annual arrival of over 500 tons and district wise distribution of lac processing centres across major lac producing districts of India have been presented in Table 9 and Table 10, respectively.

**Trends in price of lac over the time**

The data on prices of *rangeeni* and *kusmi* sticklac and seedlac was collected from the lac markets of Jharkhand (Khunti, Ranchi and Simdega), West Bengal (Balarampur, Purulia), Chhattisgarh (Dhamtari, Kanker and Korba), Maharashtra (Gondia) and Madhya Pradesh (Seoni and Balaghat) on quarterly basis. Trends in the prices of lac have been presented in Figure 13. The prices of *rangeeni* and *kusmi* sticklac as well as seedlac has shown increasing trend from December, 2010 to December, 2012 and decresed afterwards as the policy intervention for price stability by Goverment of India. Market price also started to rise up from December 2015 from its lowest level.

**Table 10. Lac processing centers in the major lac producing districts of India**

| Classifications *( Qty. in tons)* | No. of districts | No. of processing centers | Name of the districts |
|---|---|---|---|
| **> 1,000** | 06 | 20 | Gumla, Khunti, Ranchi & Simdega (Jharkhand); Seoni (Madhya Pradesh) and Gondia (Maharashtra) |
| **500-1000** | 04 | 12 | West Singhbhum (Jharkhand); Balaghat (Madhya Pradesh); Korba and Kanker (Chhattisgarh) |
| **100-500** | 17 | 100 | Palamau, (Jharkhand); Bastar, Janjgir-Champa Bilaspur, Raipur and Rajnandgaon (Chhattisgarh); Purulia, Midnapur (West Bengal); Nabarangpur, Sundergarh (Odisha) and Mandla (Madhya Pradesh); Garhchiroli and Chandrapur (Maharashtra) and Meghalaya (Garo hills) |
| **1- 100** | 22 | 16 | Balasore, Keonjhar, Koraput and Mayurbhanj (Odisha); Bhandara (Maharashtra); Mahasamund, Ambikapur, Surguja, Raigarh, Dhamtari, Durg & Raigarh (Chhattisgarh); Garhwa and Latehar (Jharkhand); Annuppur, Chhindwada, Dindori, Hosangabad, Narshinghpur & Shahdol (Madhya Pradesh) and Bankura (West Bengal). |
| **< 1** | 24 | 35 | Rest of the districts in Assam, Andhra Pradesh, Bihar, Chhattisgarh, Gujarat, Jharkhand, Madhya Pradesh, Meghalaya, Odisha, Uttar Pradesh, Telangana and West Bengal. |
| **Total** | **73** | **183** | |

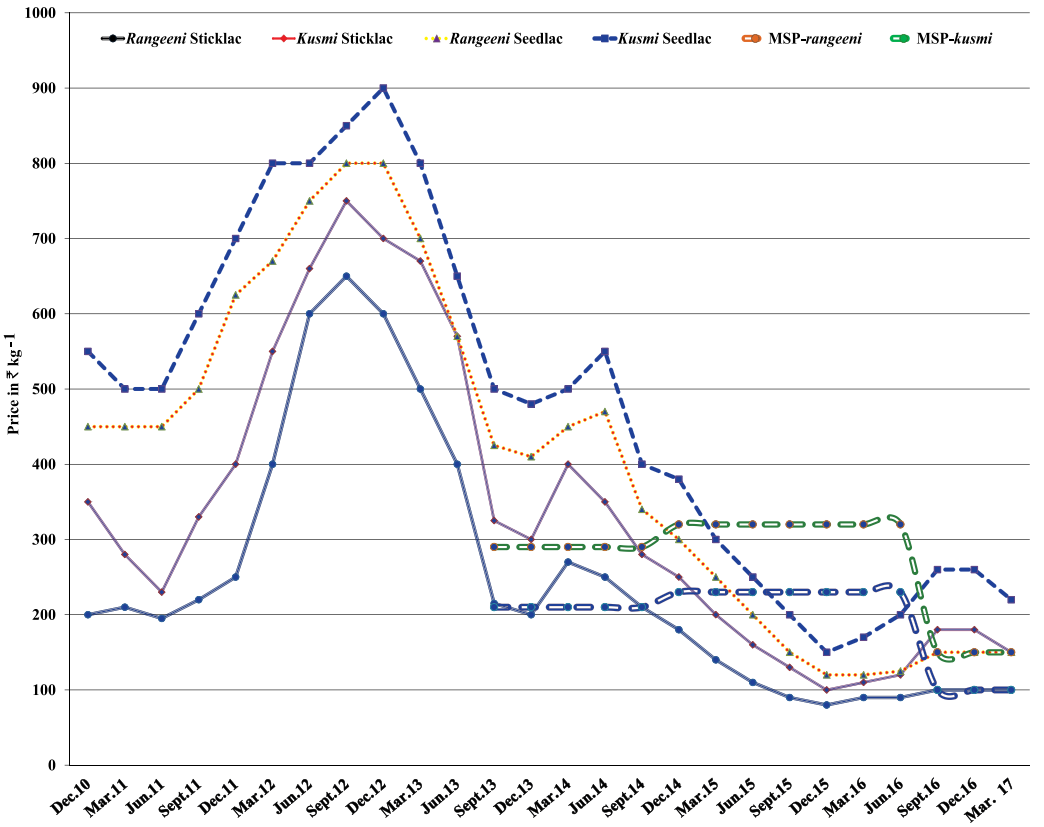



Figure 13. Trends in price of sticklac and seedlac over the time

**International trade of lac and its value added products**

Data on export of lac and its value added products from India were collected from Shellac and Forest Products Export Promotion Council (SHEFEXIL), Kolkata. The total export of lac and its value added products during the year 2015-16 was 7668.42 tons which was valued ₹ 247.55 crores. Details of export in quantity and value, list of top 16 countries importing Indian lac and export of lac has been presented in the Table 11 and Table 12 while direction of the trade, the trend in export of lac in quantity and value during last seven years and share of different items of lac export value from India are shown in Figure 14, Figure 15 and Figure 16, respectively.

Among the export destinations of lac and its value added products Bangladesh (21.3%), USA (15.3%), Pakistan (12.4%), China (9.5%) Germany (8.5%) and Switzerland (7.2%) remained top destinations with more than 70% import from India. Iraq (3.7%), Egypt (3.1%), UK (2.7%) Indonesia (2.7%) and Italy (2.5%) and Spain (2.4%) were the other destinations for Indian lac.

**Table 11. Export of lac and its value added products from India during 2014-16**

| Name of product | Export in 2014-15 | | Export in 2015-16 | |
|---|---|---|---|---|
| | Quantity (tons) | Value (₹ lakh) | Quantity (tons) | Value (₹ lakh) |
| Shellac | 3261.44 | 14650.36 | 2350.99 | 6737.50 |
| Aleuritic acid | 181.12 | 5909.11 | 168.50 | 3669.51 |
| Seedlac | 2690.29 | 8770.25 | 3091.06 | 6225.18 |
| Dewaxed shellac | 183.02 | 1250.94 | 150.57 | 775.23 |
| Bleached lac | 142.89 | 1048.13 | 225.34 | 1275.90 |
| Shellac wax | 15.20 | 116.28 | 14.20 | 64.21 |
| Buttonlac | 0.00 | 0.00 | 1447.81 | 4882.32 |
| Dewaxed decolourised | 0.00 | 0.00 | 2.55 | 21.24 |
| Gasket lac | 0.00 | 0.00 | 6.48 | 16.50 |
| Lac dye | 0.00 | 0.00 | 0.01 | 0.51 |
| Dewaxed bleached shellac | 31.07 | 273.48 | 125.63 | 809.20 |
| Garnet shellac | 52.05 | 214.87 | 17.17 | 54.23 |
| Dewaxed garnet shellac | 0.00 | 0.00 | 48.13 | 112.99 |
| Shellac(kiri) | 12.00 | 15.57 | 0.00 | 0.00 |
| Stick lac | 0.10 | 0.60 | 0.00 | 0.00 |
| Waxy bleacedlac | 0.00 | 0.00 | 20.00 | 110.66 |
| **Total** | **6569.17** | **32249.58** | **7668.42** | **24755.18** |

**Table 12. Top destinations of Indian lac in international market during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | % Share |
|---|---|---|---|
| Bangladesh | 1564.55 | 5271.74 | 21.3 |
| USA | 1182.21 | 3775.90 | 15.3 |
| Pakistan | 1510.01 | 3065.69 | 12.4 |
| China | 431.81 | 2340.86 | 9.5 |
| Germany | 750.73 | 2098.71 | 8.5 |
| Switzerland | 67.50 | 1772.36 | 7.2 |
| Iraq | 397.00 | 921.15 | 3.7 |
| Egypt | 355.50 | 773.53 | 3.1 |
| U.K. | 136.61 | 670.04 | 2.7 |
| Indonesia | 266.25 | 657.35 | 2.7 |
| Italy | 225.40 | 629.41 | 2.5 |
| Spain | 168.90 | 603.86 | 2.4 |
| Japan | 123.08 | 307.78 | 1.2 |
| Haiti | 77.00 | 210.40 | 0.8 |
| France | 36.00 | 163.43 | 0.7 |
| Netherland | 16.00 | 157.94 | 0.6 |
| Others | 359.89 | 1335.02 | 5.4 |
| **Total** | **7668.42** | **24755.18** | **100.0** |

Lac is an important natural resins exported by and also imported from Indonesia, Thailand, *etc*. in India. During 2015-16, exported value showed that 99.96 % contribution in natural resins export were from lac (97.35 %), other resins (1.42 %) and gum rosin (1.19 %).

Average annual export quantity of lac during last 15 years (1998-99 to 2012-13) was 8,249.85 tons valued ₹17419.91 lakh. Average annual demand of lac was 7465.23 tons valued ₹37952.8 lakh during the period 2013-14 to 2015-16. However, during last 15 years growth in exported quantity was negative for lac and total natural resins group as the sharp deceleration in quantity demanded in 2012-13. But, in value terms the same figure has got a momentum of the significant growth rate with a high stability coefficient.

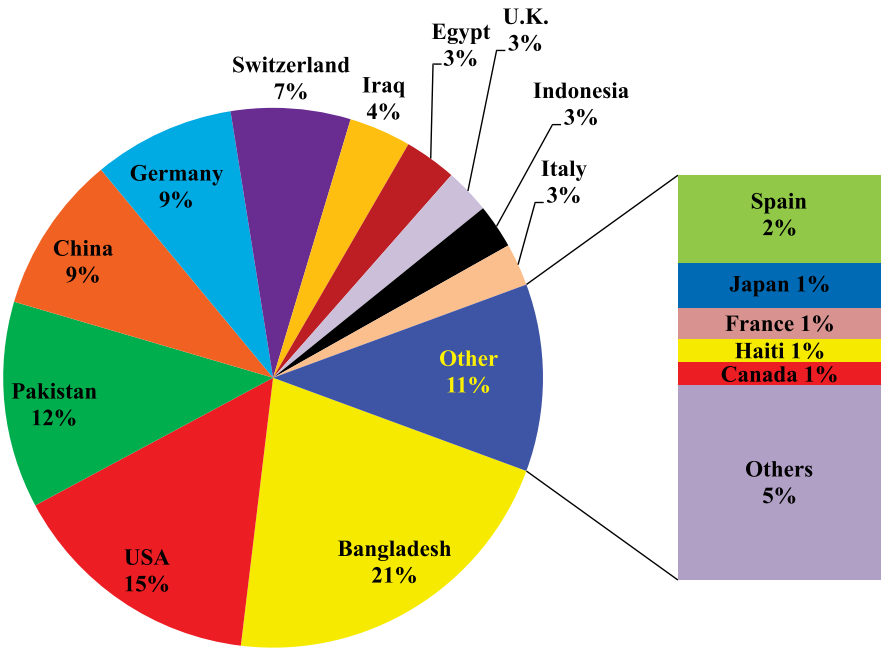



Figure 14. Direction of the trade of lac and its value added products

**Table 13. Export of lac during previous eight years**

| Year | Quantity (tons) | Value (₹ lakh) |
|---|---|---|
| 2008-09 | 6968.42 | 12414.50 |
| 2009-10 | 6422.61 | 11002.33 |
| 2010-11 | 6339.05 | 21112.92 |
| 2011-12 | 6858.21 | 36461.30 |
| 2012-13 | 4361.30 | 48027.58 |
| 2013-14 | 8158.10 | 56853.63 |
| 2014-15 | 6569.17 | 32249.58 |
| 2015-16 | 7668.42 | 24755.18 |

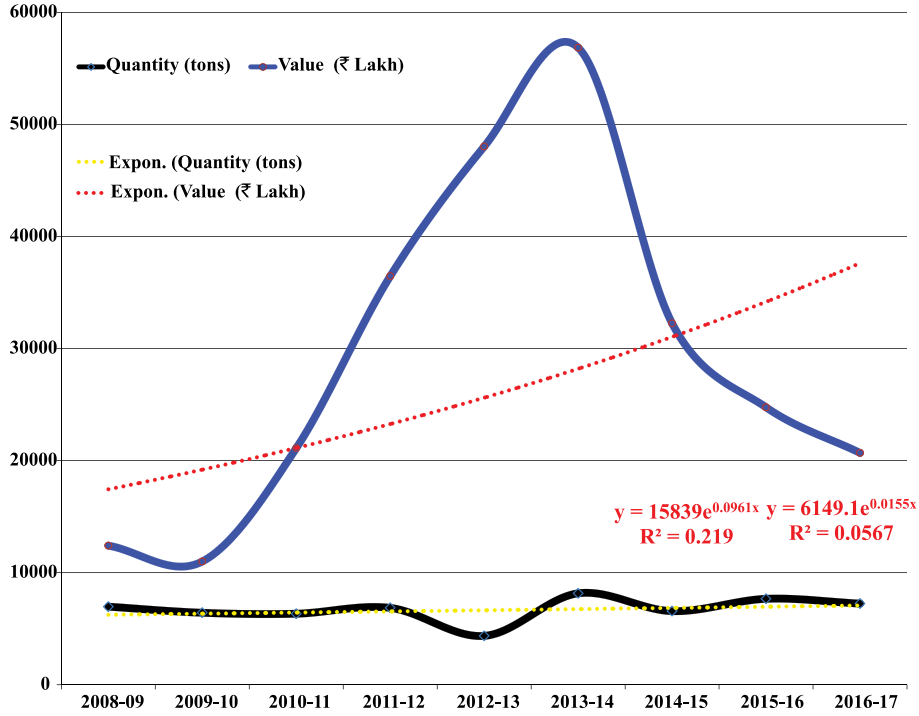



Figure 15. Trends in export of lac based products from India

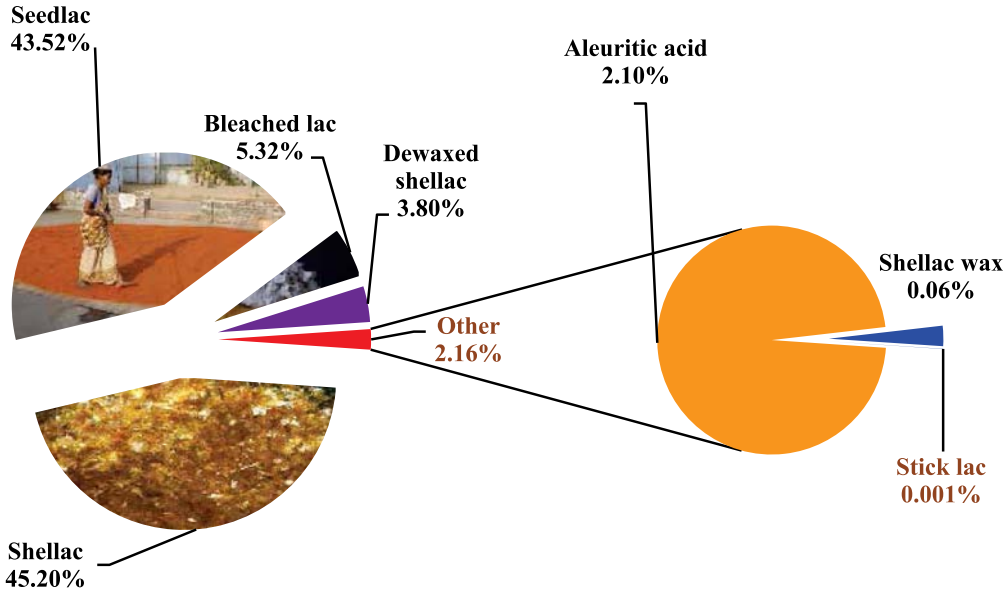



Figure 16. Product-wise export share of various lac based products

## Pine resin –a flora based resin

Rosin is a natural product of pine resins (*Pinus roxburghii* Sarg.). Rosin and its derivatives have been used as paper-sizing agents, emulsifiers, surface coatings, chewing gums, tackifiers in adhesives, insulating materials, and additives for printing inks. Recently, it was also evaluated for application in the pharmaceutical as microencapsulating materials (Lee & Hong, 2002; Mandaogade, Satturwar, Fulzele, Gogte, & Dorle, 2002; Wilbon, Chu, & Tang, 2013).



Photo 2. Natural resin: Plant Kingdom

Pine resins are secretion of plants, particularly coniferous trees. These are used in the production of varnishes, adhesives, and food glazing agents. Extensive *chir* pine forests are found in the Himalayas between an elevation of 1000 to 1900 m. *Chir* pine yields commercially important oleo-resin which forms the raw material for rosin and turpentine oil industry in India. *Chir* pine is widely tapped for resin on commercial basis, particularly in the hills of Himachal Pradesh, Uttarakhand, Jammu & Kashmir and north-eastern states. The northern hill states annually produce around 8,000 to 9,000 tons of raw rosin extracted from pine trees.

**Table 14. Commercially tapped sources of pine resin: species and country of production**

| Sl. No. | Species | Producing country |
|---|---|---|
| | *Pinus caribaea* Morelet | Venezuela, South Africa, Kenya |
| | *P. halepensis* Miller | Greece |
| | *P. kesiya* Royale ex Gordon | People's Republic of China |
| | *P. massoniana* D. Don | People's Republic of China |
| | *P. merkusii* Jungh. & Vriese | Indonesia, Vietnam |
| | *P. oocarpa* Schiede | Mexico, Honduras |
| | *P. pinaster* Aiton | Portugal |
| | *P. radiata* D. Don | Kenya |
| | *P. roxburghii* Sarg. | India, Pakistan |
| | *P. sylvestris* L. | Russia |
| | *P. elliottii* Engelm. | Brazil, Argentina, South Africa, USA, Kenya |

*Source: FAO, Rome, Italy*

Major share of resin production comes from Himachal Pradesh and Uttarakhand. Commercially tapped sources of pine resin are depicted in Table 14. The production of resin in the states during the year 2013-14 was about 8000 tons and about 85 % of this raw material is processed in the Rosin

and Turpentine oil Factories (RTFs). Indonesia also supplies rosin to Indian industries but China is the major supplier of imported product. Indonesian rosin costs around the same as the Chinese product. Both China and Indonesia have captured more than 50% of the Indian market. After the global recession, China had dropped its prices in the Indian market. The annual requirement of rosin in the country was 40,000-50,000 tons.

### Pine resin production

World production of turpentine is approximately 3.3 lakh tons from all sources; almost 1.0 lakh tons (30%) is estimated to be gum turpentine, and the bulk of the remainder is sulphate turpentine. In India resin production and its value during 2007-08 to 2015-16 is depicted in Figure 17.

### Resin production policy

Himachal Pradesh Forest Development Corp (HPFDC), one of the major players in rosin business in the country, is facing tough competition as the buyers are opting for the imported product. China is marketing rosin at much lower rates in the Indian market due to global recession. It has dumped its surplus stocks in India.

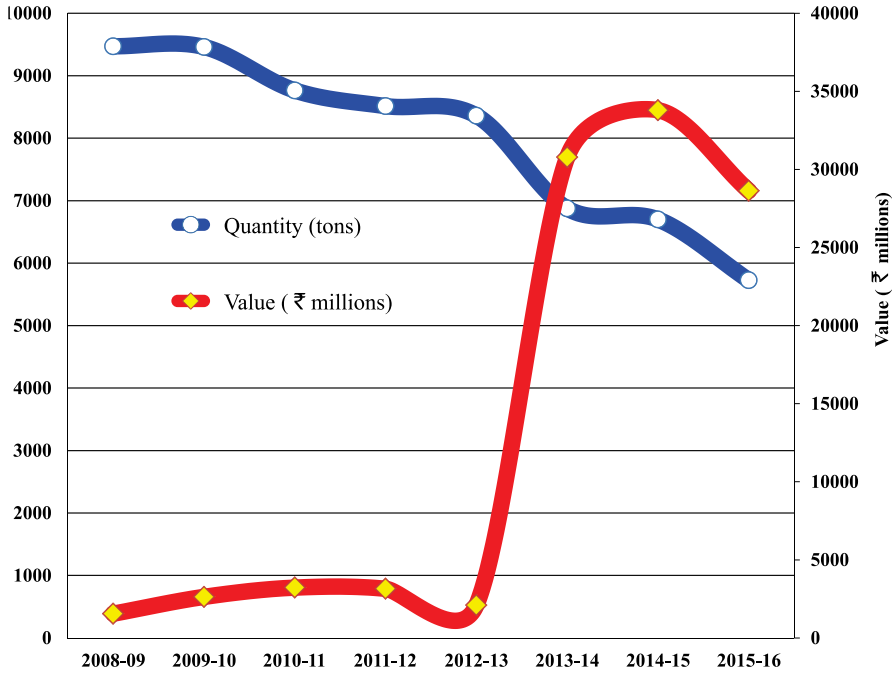



Figure 17. Pine resin production and its value in Himachal Pradesh and Uttarakhand

Total annual production of rosin is about 1.0 million tons world-wide. Of this, it is estimated that almost 60 %, is gum rosin; most of the remainder, about 35 % is tall oil rosin and the rest is wood rosin. State wise annual pine resin production and their contribution in the total production is given in Table 15 and graphical illustration presented in Figure 18.

A total of 50% of resin is sold by open auction to units registered in Uttarakhand according to their processing capacity and 25% of the resin is sold by open auction on all India basis. Balance 25% is sold by open auction amongst the units of Khadi-Gramodyog, Co-operatives, Kumaon Mandal Vikas Nigam and Garhwal Mandal Vikas Nigam according to their processing capacity. If some quantity of resin is left unsold as per the above arrangements it may also be sold by all India open auction.

**Table 15. State wise annual pine resin production and their contribution in the total production**

| Year | Uttarakhand | Himachal Pradesh | Total |
|---|---|---|---|
| 2007-08 | 1930(18.5) | 8514(81.5) | 10444(100.0) |
| 2008-09 | 1875 (19.8) | 7596 (80.2) | 9471(100.0) |
| 2009-10 | 1961 (20.7) | 7500 (79.3) | 9460 (100.0) |
| 2010-11 | 1764 (20.1) | 7000 (79.9) | 8764 (100.0) |
| 2011-12 | 1614 (19.0) | 6900 (81.0) | 8514(100.0) |
| 2012-13 | 1467 (17.5) | 6894 (82.5) | 8361 (100.0) |
| 2013-14 | 1486 (21.6) | 5389 (78.4) | 6875 (100.0) |
| 2014-15 | 1442 (21.5) | 5258 (78.5) | 6699 (100.0) |
| 2015-16 | 1562(27.3) | 4164(72.7) | 5726(100.0) |
| **Average** | **1678 (20.3)** | **6579 (79.7)** | **8257(100.0)** |

*Figures in parentheses are the percentage of the total*

**Uttarakhand:** To ensure transparency in allotment of resin to different agencies, resin policy has been formulated and issued in 2003.

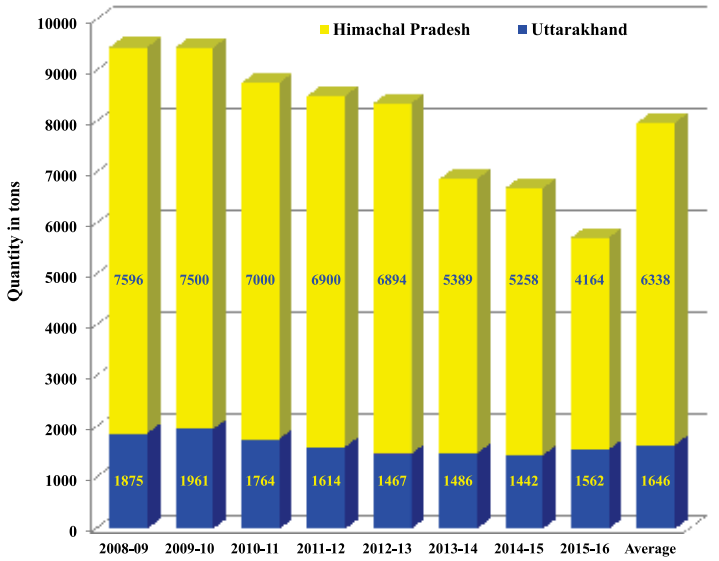



Figure 18. Pine resin production in Himachal Pradesh and Uttarakhand

**Himachal Pradesh:** The hill state annually produces around 6500 tons of raw rosin extracted from pine trees, grown mainly in lower hills of Hamirpur, Una, Mandi, Solan and Sirmaur districts. Exploitation of resin in the Pradesh is done by the Himachal Pradesh State Forest Corporation, Shimla. About 15.76 lakh blazes were handed over to H.P. State Forest Corporation for tapping during the year from state forests. The quantity have to be extracted by the Corporation was 58251 quintals for which the total royalty payable by the H.P. State Forest Corporation amounted to ₹10.25 Crore. As per decision of the Pricing Committee of Government of Himachal Pradesh, a rate of Rs. 65/- per blaze was fixed for the year.

**Forest based industry:** There are resin processing units functioning in private sector in Hamirpur Circle particularly in Una and Dehra Divisions and two in public sector at Bilaspur and Nahan. The HP State Forest Corporation: The Himachal Pradesh State Forest Corporation functioned principally as a harvesting agency during the year. The entire exploitation works of timber, resin, fuel-wood, charcoal, *etc*. from the state forests were handled by the Forest Corporation. The standing volume marked to the HP State Forest Corporation, for commercial felling during the year, was 145984 $m^3$. Besides 1576148 resin blazes were also handed over to the Forest Corporation for tapping of resin. The Forest Corporation also regulated the resin exploitation from the private areas in the State (Administrative Report 2015-16, HP Forest Department).

## Pine resin processing in India

Himachal Pradesh State Forest Development Corporation Limited deals mainly with marketing of timber, fuel wood, pulpwood, bamboo, *khair* and resin.

**Table 16. Quality criteria for different grades of the rosin**

| Sl. No. | Rosin | Grade | Full Name |
|---|---|---|---|
| 1 | Pale | X | Extra White |
| | | WW | Water White |
| | | WG | Window Glass |
| | | N | Nancy |
| 2 | Medium | M | Mary |
| | | K | Kitty |
| | | H | Harry |
| 3 | Dark | D | Dark |
| | | B | Black |

The entire government resin tapping work is being done by the Corporation, through modern techniques (Rill method as of now). It is graded and sold on the basis of colour, the palest shades of yellow-brown being the better quality. Quality criteria and specifications are described in Table 16.

The two Resin & Turpentine Oil Factories, located at Bilaspur and Nahan operational since four decades, have both skilled and unskilled workers to undertake resin processing and other diversification activities. Rosin and turpentine oil and other subsidiary products like phenyl, varnish, black japan, *etc*. are produced from the collected resin. Rosin is the major product obtained from pine resin. It remains behind as the residue after distillation of the turpentine oil. It is a brittle, transparent, glassy solid. It is insoluble in water but soluble in many organic solvents.

## Pine resin market and price

In any analysis of world production and trade in gum naval stores, the volume of trade taking place in crude resin needs to be estimated. Until recently, this did not need to be considered as all resin was processed at origin and rosin and turpentine were the primary products of trade. However, trade in crude resin has developed over years as the capacity for tapping has fallen in some of the traditional producing countries, notably Portugal and India. These countries have excess processing capacity which can be utilized fully if a sustainable method of crude resin extraction is applied extensively. The absence of capital costs therefore enables the processors of imported resin to sell the outputs (gum rosin and turpentine) at a price which only need covers raw material and processing costs, freight, drums and profit.

**Table 17. Grade-wise price of rosin during 2015-16**

| Rosin Category | Grade | Full name | Rate (₹/Kg) | | | |
|---|---|---|---|---|---|---|
| | | | Nahan-TP barrels | Nahan-PGI barrels | Bilaspur-TP barrels | Bilaspur-PGI barrels |
| Pale | X | Extra White | 90 | 91 | 89 | 90 |
| | WW | Water White | 85 | 86 | 84 | 85 |
| | WG | Window Glass | 84 | 85 | 83 | 84 |
| | N | Nancy | 83 | 84 | 82 | 83 |
| Medium | M | Mary | 82 | 83 | 81 | 82 |
| | K | Kitty | 81 | 82 | 80 | 81 |
| | H | Harry | 80 | 81 | 79 | 80 |
| Dark | D | Dark | 77 | 78 | 76 | 77 |
| | B | Black | 67 | 68 | 66 | 67 |

The sale of crude resin also allows producing countries with a surplus to earn extra revenue without investing in additional capacity for processing. Price of turpentine oil (vegeTable) per litre (naked) ex-factory on cash basis ranged from ₹ 66 to ₹ 91. The price of per kilogram rosin is given in Table 17 and graphically shown in Figure 19. This price is an ex-factory price including packing charges excluding taxes and duties *etc*.

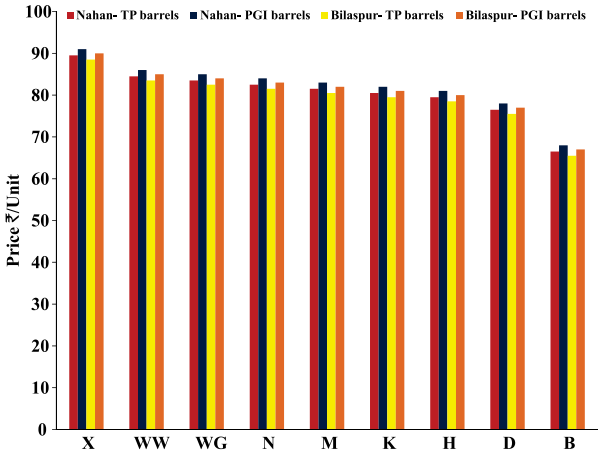



Figure 19. Price pattern of various grades of Pine resin based products

## International trade of pine resin

Details of export in quantity and value, list of countries importing resin from India and details of resin supply from various countries during 2015-16 is presented in the Table 18. The total export of rosin during the year 2015-16 was 204.32 tons which was valued ₹ 303.30 lakh. Similarly, total import of rosin during the year 2015-16 was 41962.58 tons which wa s valued ₹ 44297.13 lakh. The analysis of EXIM data revealed that the exported quantity of rosin decreased to 45.5% in 2015-16 in comparison to 2014-15 while imported quantity of rosin was increased by 16.2% in 2015-16 in comparison to 2014-15. During this year in international market, export price of rosin increased by 7.6 %, while import price decreased by 15.5 %. Demand of resin was higher than the quantity processed in India. Hence, India depends on Indonesia, Brazil, Nepal, Vietnam and China to meet its domestic demand. About 99 % of gum rosin was supplied from these four neighboring countries of Asia and rest of this came from Europe and America. Similarly, about 93 % gum rosin was demanded by the major buyers like Nigeria, Kenya, Egypt A Rp, and U Arab Emts.

The destination and source wise demand and supply of the rosin is depicted in the Figure 20 and Figure 21. Rosin is an important natural resin in the basket of import items. Its import had a very high stability with growth rates in import quantity (10.65 %) as well as value (26.41 %).

**Table 18. Destination and source wise EXIM scenario of rosin during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000 US $) | Share (%) |
|---|---|---|---|---|
| **A. Export** | | | | |
| Nigeria | 140.00 | 213.54 | 330.00 | 70.40 |
| Kenya | 17.44 | 31.36 | 50.00 | 10.34 |
| Egypt A Rp | 16.00 | 19.22 | 30.00 | 6.34 |
| U Arab Emts | 8.88 | 17.33 | 30.00 | 5.71 |
| Sri Lanka Dsr | 6.00 | 9.47 | 10.00 | 3.12 |
| Uganda | 1.92 | 5.24 | 10.00 | 1.73 |
| Zambia | 13.00 | 3.69 | 10.00 | 1.22 |
| Indonesia | 1.00 | 3.32 | 10.00 | 1.09 |
| Nepal | 0.03 | 0.08 | 0.12 | 0.03 |
| Malaysia | 0.05 | 0.05 | 0.08 | 0.02 |
| **Total** | **204.32** | **303.30** | **470.00** | **100.00** |
| **B. Import** | | | | |
| Indonesia | 13842.32 | 14341.37 | 21810.00 | 32.38 |
| Brazil | 9804.00 | 11121.37 | 17020.00 | 25.11 |
| Nepal | 10031.20 | 8652.15 | 13250.00 | 19.53 |
| Vietnam Soc Rep | 4811.63 | 4930.65 | 7570.00 | 11.13 |
| China P Rp | 3313.05 | 4857.65 | 7520.00 | 10.97 |
| Japan | 64.03 | 164.57 | 250.00 | 0.37 |
| Belgium | 53.13 | 152.73 | 230.00 | 0.34 |
| Taiwan | 20.00 | 33.17 | 50.00 | 0.07 |
| Portugal | 12.00 | 26.61 | 40.00 | 0.06 |
| Germany | 2.83 | 7.16 | 10.00 | 0.02 |
| U S A | 5.59 | 4.81 | 10.00 | 0.01 |
| Italy & Netherland | 2.80 | 4.89 | 11.85 | 0.01 |
| **Total** | **41962.58** | **44297.13** | **67771.85** | **100.00** |

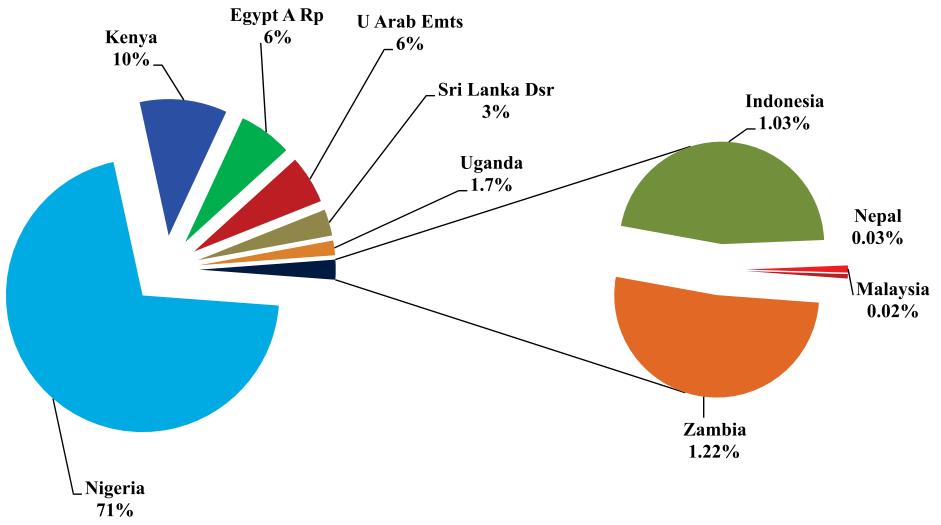



Figure 20. Destination wise scenario of rosin from India during 2015-16

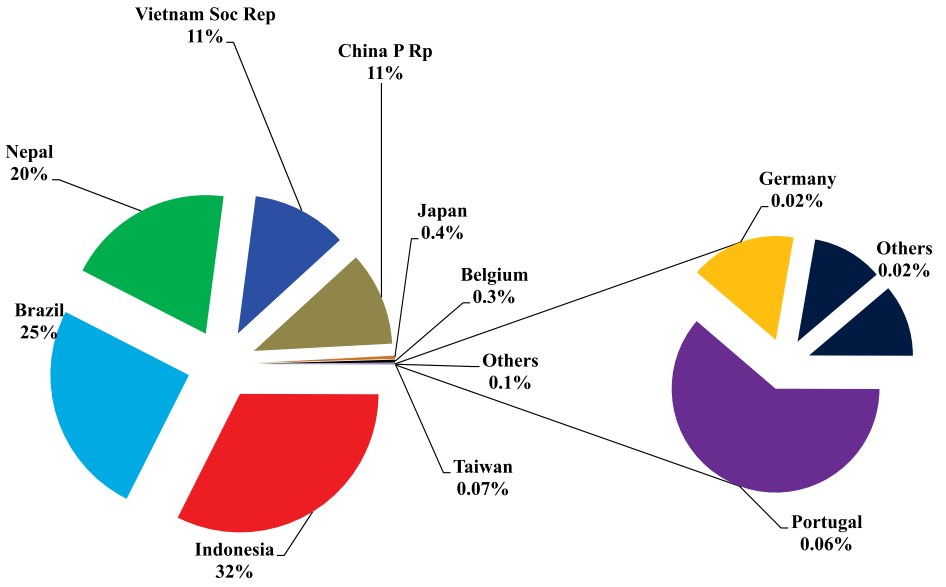



Figure 21. Source wise scenario of rosin in India during 2015-16

## Copal

Copals are derived from species of *Bursera*, *Protium* (*Burseraceae*), and *Hymenaea*. The adhesive property of copal makes it a potential coating material. It is evident from Table 19 that more than 99% of the copal in India was supplied from Indonesia (96.83%) and Philippines (3.06%) during

2015-16. A very little portion of the total imported quantity was exported to Thailand (90.21%) and Canada (9.79%) during 2014-15, but it was not exported in 2015-16.

**Table 19. Source wise import of copal during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000' US dollars) | % share |
|---|---|---|---|---|
| Indonesia | 2026.30 | 1080.38 | 1650.00 | 96.83 |
| Philippines | 51.80 | 34.17 | 50.00 | 3.06 |
| Italy | 0.10 | 1.23 | 1.88 | 0.11 |
| **Total** | 2078.10 | 1115.79 | 1115.79 | 100.00 |

## Dammar batu

Damar is tapped from the *sal* tree *(Shorea robusta)*, although some is still collected from the ground in fossilized form. Annual production of Dammar batu in India is about 80-100 tons. It is used as painting and incense material. Destination wise EXIM scenario of dammar batu during 2015-16 is given in Table 20. During 2015-16, more than 95% of this resin in India was supplied from Indonesia (58.60%) and Thailand (38.77%). A very little portion of the total imported quantity was exported to Japan (58.05%), Jordan (33.74%), Canada (7.29%) and Vietnam SOC Rep (0.46%).

**Table 20. Destination and source wise EXIM scenario of dammar batu during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000' US dollars) | Share (%) |
|---|---|---|---|---|
| **Export** | | | | |
| Japan | 0.50 | 3.82 | 5.83 | 58.05 |
| Jordan | 1.97 | 2.22 | 3.39 | 33.74 |
| Canada | 0.05 | 0.48 | 0.73 | 7.29 |
| Vietnam Soc Rep | 0.20 | 0.03 | 0.05 | 0.46 |
| Spain | 0.08 | 0.01 | 0.02 | 0.15 |
| Taiwan | 0.05 | 0.01 | 0.02 | 0.15 |
| Italy | 0.00 | 0.01 | 0.02 | 0.15 |
| **Total** | **2.85** | **6.58** | **10.05** | **100.00** |
| **Import** | | | | |
| Indonesia | 12321.87 | 3325.07 | 5076.44 | 58.60 |
| Thailand | 3784.24 | 2199.61 | 3358.18 | 38.77 |
| Vietnam Soc Rep | 220.73 | 83.22 | 127.05 | 1.47 |
| Lao Pd Rp | 42.00 | 34.90 | 53.28 | 0.62 |
| Singapore | 105.00 | 14.36 | 21.92 | 0.25 |
| China p Rp | 38.18 | 13.66 | 20.85 | 0.24 |
| Malaysia | 13.82 | 3.33 | 5.08 | 0.06 |
| **Total** | **16525.84** | **5674.15** | **8662.82** | **100.00** |

## Mastic gum

The traditional Greek product Chios Mastic gum or Mastiha, primarily known for its distinctive flavor. This is the dried exudate of the shrub-like tree *Pistacia lentiscus L. var. chia* of the *Anacardiaceae* family, cultivated exclusively in the south of the Greek island Chios.

**Table 21. Destination and source wise EXIM scenario of mastic gum during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000' US dollars) | Share (%) |
|---|---|---|---|---|
| **Export** | | | | |
| Pakistan Ir | 0.30 | 2.53 | 3.86 | 97.68 |
| Kenya | 0.08 | 0.05 | 0.08 | 1.93 |
| Sri Lanka Dsr | 0.02 | 0.01 | 0.02 | 0.39 |
| **Total** | **0.40** | **2.59** | **3.95** | **100.00** |
| **Import** | | | | |
| Greece | 4.39 | 23.24 | 35.48 | 99.96 |
| U Arab Emts | 0.01 | 0.01 | 0.02 | 0.04 |
| **Total** | **4.40** | **23.25** | **35.50** | **100.00** |

Mastic gum is shown to exert beneficial effects on a wide range of human disorders. The most comprehensive data so far has indicated that mastic gum provides protection against gastrointestinal malfunctions and bacterial infections. Substantial evidence has also suggested that mastic gum exhibits hepatoprotective and cardioprotective, antiinflammatory/antioxidant, and antiatherogenic properties. In India, it is imported from Greece (99.96%) and UAE (0.04%).

## Plant exudate based natural gums

Data and information was collected from the major collection centers like Nagpur, Bilaspur, Surat, Ajmer and Delhi. Procurement data available with Odisha Forest Development Corporation Ltd, Rajasthan Tribal Areas Development Coop Federation Ltd, Maharashtra State Cooperative Tribal Development Corporation Ltd, Jharkhand State Minor Forest Produce Cooperative Development Marketing Federation Ltd, Girijan Co-operative Corporation Limited, Visakhapatnam, Andhra Pradesh, Chhattisgarh M.F.P. (T&D) Fed. Ltd., Raipur, Chhattisgarh, Gujarat State Forest Development Corporation Limited (GSFDCL), Vadodara was also recorded and compiled. The gum tapping is mainly done in the schedule areas where tribal populations exist. The collection charges to the collectors at collection centres are paid by the purchaser at the rate fixed by the Govt. After making the payment of the collection charges to the collectors and the difference amount of sale rate and collection rate in the District Union, the purchasers are allowed to transport the collected gums wherever they desire. In India, mainly *karaya* gum (*S. urens*), *dhawada* gum (*A. latifolia*), *prosopis* gum (*P. juliflora*), *khair* gum (*Acacia catechu*), *babool*/*babul* gum (*A. nilotica*), *Jhingan* (*Lannea coromandelica*), *palas* (*B. monosperma*), *char* (*B. lanzan*) and *guggul* gum (*C. wightii*) are produced.

Exudate gums possess a unique combination of functionalities and properties that can never be matched by any other alternative synthetic polymers, which makes their complete substitution impossible. Importantly, these biopolymers are eco-friendly as they are biodegradable. The state wise gum production during 2009-10 to 2015-16 is shown in Table 22 and graphical illustration is shown in Figure 22.



Photo 3. Natural gum: Gum ghatti (Grade –I)

About 81 % of gum production in the country is contributed by Maharashtra (26.9 %), Madhya Pradesh (18.8 %), Jharkhand (13.5 %), Telengana (12.8 %), and Chhattisgarh (8.9 %). Rest of the 19 % comes from Gujarat (4.4 %), Andhra Pradesh (4.2%) and other minor gum producing states (10.5%). District wise gum producing areas in the country are presented in *Annexure III*. In parts of the Jaipur, Ajmer and Jodhpur districts *A. senegal* is common. *A. catechu* forests are common in the south-eastern regions. *e.g.* Baran, Jhalawar, Kota, Tonk , Chittorgarh and Alwar.

**Table 22. Gum production in major gum producing states of India (in tons)**

| State /Year | 2009-10 | 2010-11 | 2011-12 | 2012-13 | 2013-14 | 2014-15 | 2015-16 | Average |
|---|---|---|---|---|---|---|---|---|
| Andhra Pradesh | 35.05 | 23.97 | 34.49 | 29.87 | 42.53 | 26.17 | 19.84 | 35.88 |
| Telangana | 175.59 | 99.67 | 159.55 | 102.19 | 65.02 | 42.46 | 33.64 | 109.80 |
| Chhattisgarh | 236.94 | 41.64 | 20.30 | 23.61 | 2.54 | 4.03 | 139.00 | 76.31 |
| Gujarat | 47.20 | 27.85 | 33.57 | 42.62 | 54.72 | 51.50 | 16.05 | 37.69 |
| Jharkhand | 0.00 | 240.60 | 270.40 | 207.30 | 90.00 | 56.00 | 56.30 | 115.08 |
| Madhya Pradesh | 8.74 | 286.48 | 292.28 | 232.39 | 207.50 | 120.00 | 113.00 | 160.45 |
| Maharashtra | 0.00 | 200.00 | 203.40 | 350.77 | 539.11 | 323.79 | 221.00 | 229.76 |
| Others | 6.40 | 103.00 | 134.60 | 155.20 | 143.00 | 87.00 | 85.00 | 89.90 |
| **Total** | **509.92** | **1023.22** | **1148.59** | **1143.95** | **1144.43** | **710.94** | **683.83** | **854.86** |

*Note: These are the revised dataset and may differ with previous estimates due to validation and updates*




Figure 22. Year wise production scenario of natural gums in India

## Gum *karaya* (*Sterculia urens*)

**Vernacular names:-***Kullu, Kadaya, Kadu, Galgala, Genduli, Tapsi, Panerukh, Kandol, Salad*

Gum *karaya* is the dry exudate of *S. urens* and *S. villosa*. It is also collected from *S. urceolata* and *S. foetida* in Indonesia, *S. setigera* in Africa and from *S. caudata* in Australia (Gautami and Bhat 1992). It is also known by the name Indian tragacanth, as it resembles gum tragacanth produced by *Astragalus spp.* Gum *karaya* is one of the least soluble gums used for many industries as given below:-

- Pharmaceutical, food, paper, textiles, cosmetic industry
- Superior grades in ice-creams
- Inks, rubber, linoleum, oil clothes, paper coatings, polishes, lower grades in varnishes, *etc*,
- Engraving processes and in oil drilling operations
- In dental compounds and colostomy rings.
- Acting as mucilage it is also used as a bulk laxative
- As a binder, emulsifier and stabilizer in food industry.

## *Karaya* gum production in India

Overall production has decreased from 6838 tons in 1975–1976 to 100.35 tons in 2015-16. During this period the price increased from ₹7.4/kg to ₹110/kg. The gum production across various states has been presented in Table 23 and Figure 23.

**Table 23. Gum *Karaya* production in major gum producing states of India (in tons)**

| State /Year | 2009-10 | 2010-11 | 2011-12 | 2012-13 | 2013-14 | 2014-15 | 2015-16 | Average |
|---|---|---|---|---|---|---|---|---|
| **Andhra Pradesh** | 22.94 | 16.59 | 24.76 | 16.33 | 27.94 | 16.05 | 9.91 | 19.22 |
| **Telangana** | 173.59 | 98.51 | 157.31 | 101.59 | 64.38 | 38.95 | 30.09 | 94.92 |
| **Chhattisgarh** | 175.01 | 38.97 | 13.85 | 19.09 | 1.84 | 4.03 | 41.00 | 41.97 |
| **Gujarat** | 0.10 | 0.06 | 0.20 | 0.20 | 0.10 | 0.20 | 0.05 | 0.13 |
| **Jharkhand** | 0.00 | 0.00 | 49.40 | 51.50 | 4.40 | 4.00 | 2.30 | 15.94 |
| **Madhya Pradesh** | 8.74 | 6.48 | 12.18 | 8.19 | 12.10 | 10.00 | 5.00 | 8.96 |
| **Maharashtra** | 0.00 | 0.00 | 23.20 | 10.00 | 13.00 | 5.00 | 7.00 | 8.31 |
| **Others** | 6.40 | 3.00 | 4.00 | 5.00 | 5.00 | 5.00 | 5.00 | 4.77 |
| **Total** | **386.79** | **163.61** | **284.90** | **211.90** | **128.76** | **83.23** | **100.35** | **194.22** |

The gum producing forest divisions in the Chhattisgarh are Bilaspur (Mugeli, Dindori, Ratanpur, Takhatpur, Lormi), Raipur, E. Surguja (Balarampur), Marvahi (Kota), S. Surguja, Raigarh (Khamariya,), Dharmajaygarh, Rajnandgaon, Mahasamund, Dhamtari, Korea, Sukma, Bijapur, Dantewada and W. Bhanupratapur. In Jharkhand, the *karaya* gum is produced in the Latehar (Garu, Mahuadar, Herhanj, Balumath, Barwadih, Lesliganj, Chhipadohar and Richughutu), Chatra (Lawalang, Pratappur and Kanti), Garhwa (Ramkanda and Bhandaria), Daltonganj (Panki and Chhatarpur) and West Singhbhum (Chakradharpur). Annual average production of *karaya* gum during last seven years in India is illustrated in Figure 24.

## Collection and grading of *karaya* gum

During 2015-16, a Minimum Support Price (MSP) of ₹10800 per qt for *karaya* gum was declared by Pricing Cell, TRIFED, Ministry of tribal affairs, Govt. of India.

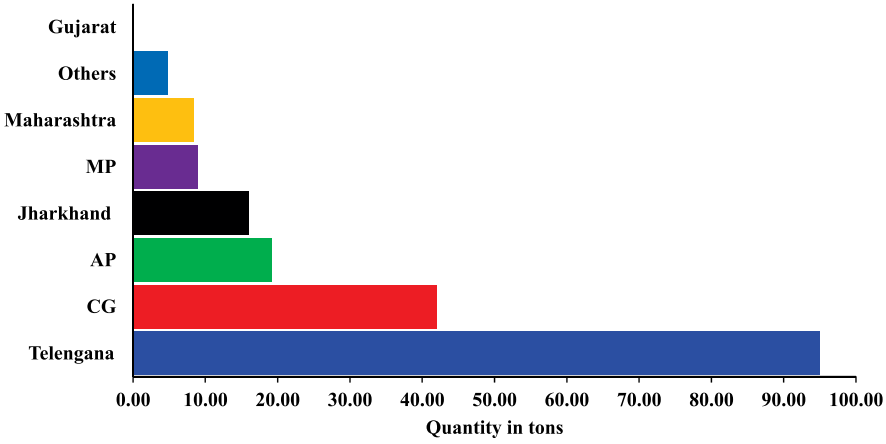



Figure 23. State-wise average production of gum *karaya*

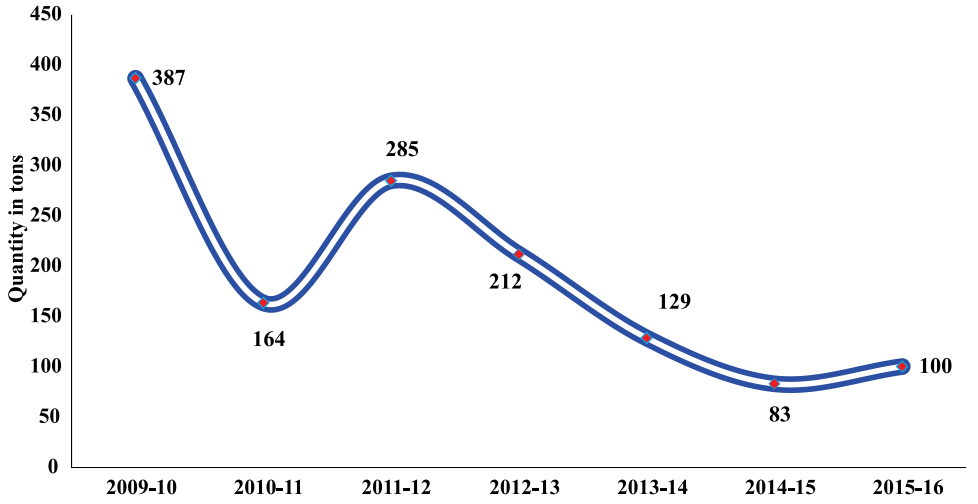



Figure 24. Gum *karaya* production in India

However, prevailing market price of gum *karaya* is ranged from ₹250 (Delhi) to ₹300 (Tamil Nadu). The scheme is initially being implemented in the states having scheduled areas and scheduled tribes in the fifth schedule of the Constitution of India (except Himachal Pradesh) namely Odisha, Jharkhand, Chhattisgarh, Maharashtra, Madhya Pradesh, Gujarat, Andhra Pradesh and Rajasthan. Now, it is extended over the country.

## International trade of *karaya* gum

Export of gum *karaya* from India during previous years have been depicted in Figure 25 while domestic demand from and overseas supply in India have been given in Table 24.

**Table 24. Destination and source wise EXIM scenario of gum *karaya* during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000' US dollars) | Share (%) |
|---|---|---|---|---|
| **A. Export** | | | | |
| **Japan** | 49.10 | 258.42 | 397.57 | 33.49 |
| **France** | 73.00 | 184.57 | 283.95 | 23.92 |
| **U S A** | 19.90 | 94.81 | 145.86 | 12.29 |
| **Pakistan Ir** | 40.46 | 80.79 | 124.29 | 10.47 |
| **Malaysia** | 10.01 | 36.19 | 55.68 | 4.69 |
| **Spain** | 8.00 | 34.21 | 52.63 | 4.43 |
| **Singapore** | 5.00 | 24.56 | 37.78 | 3.18 |
| **Syria** | 3.00 | 11.47 | 17.65 | 1.49 |
| **Germany** | 3.00 | 8.33 | 12.82 | 1.08 |
| **Others** | 19.30 | 38.24 | 58.83 | 4.96 |
| **Total** | 230.77 | 771.59 | 1187.06 | 100.00 |

| Country | Quantity (tons) | Value (₹ lakh) | Value (000' US dollars) | Share (%) |
|---|---|---|---|---|
| **B. Import** | | | | |
| **Mali** | 715.85 | 507.05 | 780.08 | 79.48 |
| **Senegal** | 49.90 | 71.71 | 110.32 | 11.24 |
| **U S A** | 3.81 | 58.02 | 89.26 | 9.09 |
| **Nigeria** | 2.00 | 1.19 | 1.83 | 0.19 |
| **Total** | 771.56 | 637.97 | 981.49 | 100.00 |

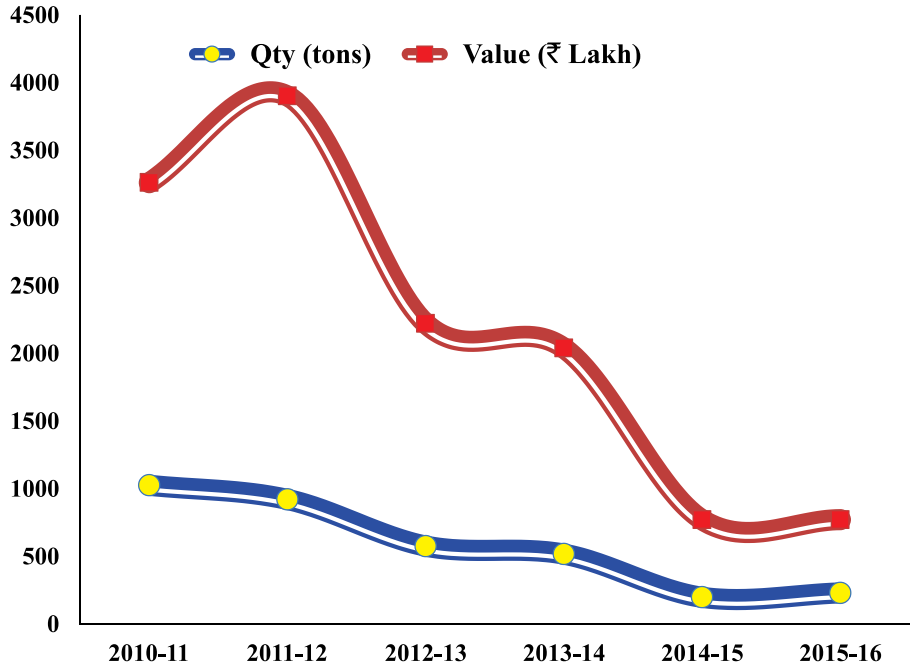



Figure 25. Export of gum *karaya* from India

During 2015-16, gum *karaya* was exported across the globe in 44 countries. There is high demand of gum *karaya* from India. Japan (33.49 %), France (23.92 %), USA (12.29%), Pakistan (10.47 %), Malaysia (4.69%), Spain (4.43%), Singapore (3.18%), Syria (1.49%) and Germany (1.08%) remained as major export destinations during 2015-16 and about 95 % gum *karaya* supplied to these top 9 destinations. Mali and Senegal had supplied 90 % raw gum *karaya* and about 9 % supply of good quality *karaya* gum was from USA. Source wise import figures are presented in Figure 27.

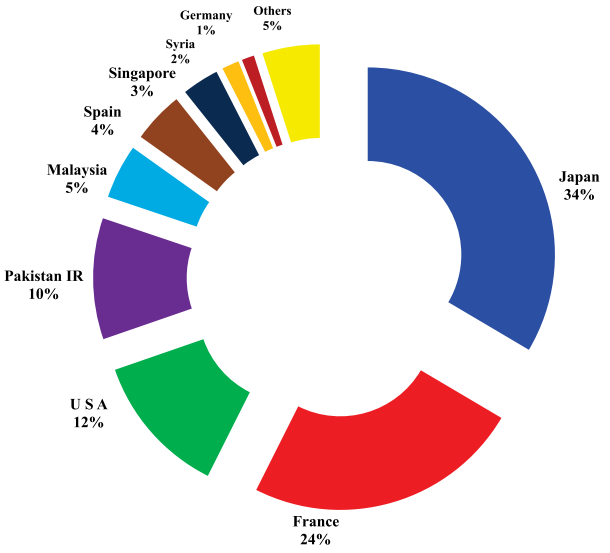



Figure 26. Destination wise scenario of gum *karaya from India* during 2015-16

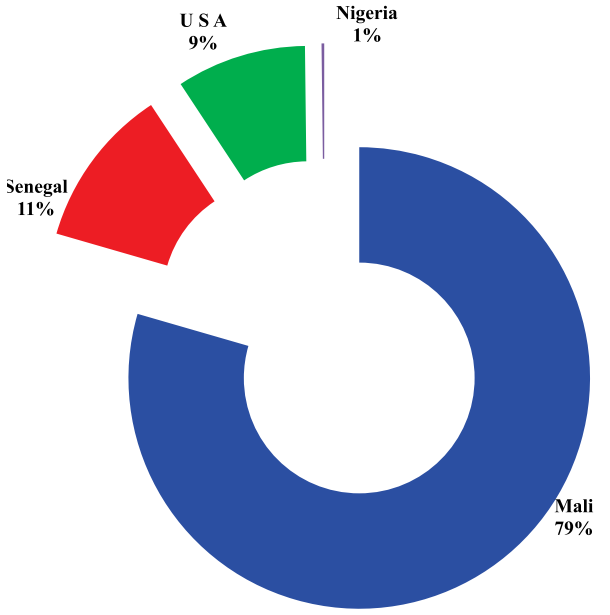



Figure 27. Source wise scenario of gum *karaya* in India during 2015-16

### *Dhawda* gum/ *Gum Ghatti* (*A. latifolia*)

*Dhawda* gum is the dry exudate of A. latifolia. It has a glassy fracture and occurs in rounded tears which are normally less than 1 cm in diameter. It often occurs in larger vermiform masses. *Dhawda*

gum is used as an emulsifier and stabilizer in beverages and butter containing Table syrups; flavour fixative for specific applications; to prepare powdered, sTable, oil-soluble vitamins; as a binder in long-fibered light weight papers; as an emulsifier of petroleum and non petroleum waxes to form liquid and wax paste emulsions; to prepare uniform and discrete prills of cross-linked polystyrene; used as drilling mud conditioner and the acidizing of oil wells and also used in powdered explosives to improve resistance to water damage.

## *Dhawda* gum production in India

*Dhawda* gum is produced in the states of Chhattisgarh, Gujarat, Jharkhand, Madhya Pradesh, Maharashtra and Andhra Pradesh. State-wise production data have been shown in Table 25. It is evident from the Figure 28 that total production of *dhawda* gum increased from 2.7 tons in 2011-12 to 194.00 tons in 2015-16.

**Table 25. *Dhawda* gum production in major gum producing states of India (in tons)**

| State / Year | 2009-10 | 2010-11 | 2011-12 | 2012-13 | 2013-14 | 2014-15 | 2015-16 | Average |
|---|---|---|---|---|---|---|---|---|
| **Andhra Pradesh** | 0.30 | 0.80 | 0.50 | 0.60 | 0.80 | 1.00 | 1.00 | 0.71 |
| **Chhattisgarh** | 42.50 | 1.60 | 3.01 | 2.62 | 0.20 | 0.00 | 6.00 | 7.99 |
| **Gujarat** | 0.50 | 0.30 | 2.33 | 1.30 | 1.40 | 1.30 | 1.00 | 1.16 |
| **Jharkhand** | 0.00 | 0.00 | 20.50 | 30.20 | 25.40 | 12.00 | 4.00 | 13.16 |
| **Madhya Pradesh** | 0.00 | 0.00 | 30.10 | 24.20 | 20.80 | 10.00 | 8.00 | 13.30 |
| **Maharashtra** | 0.00 | 0.00 | 40.20 | 240.77 | 334.11 | 230.79 | 134.00 | 139.98 |
| **Others** | 0.00 | 0.00 | 50.60 | 80.20 | 65.70 | 40.00 | 40.00 | 39.50 |
| **Total** | **43.30** | **2.70** | **147.24** | **379.89** | **448.41** | **295.09** | **194.00** | **215.80** |

## Collection and grading of *dhawda* gum

The collection rate for *dhawada* gum for the year 2015-16 was around ₹600-800/kg for Grade-I, ₹200-300/kg for Grade-II, ₹100-150/kg for Grade-III.

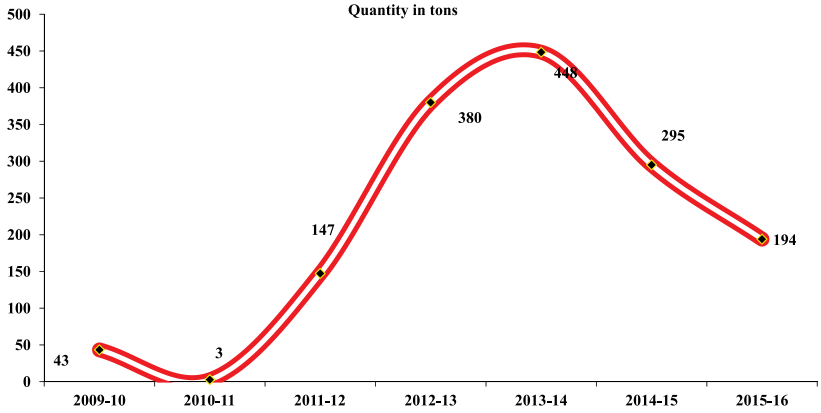



Figure 28. Year wise production scenario of gum ghatti in India

## Gum *arabic* (*Acacia senegal*)

Gum arabic, also known as Acacia gum, is a natural gum made of the hardened sap of various species of the Acacia tree. Gum *arabic* is predominantly collected from two species of Acacia, namely *A. senegal* and *Vachellia (Acacia) seyal*. Producers harvest the gum commercially from wild trees, mostly in Sudan (80%). The uses of gum *arabic* are linked to its two main characteristics *i.e.* its high solubility in water and its low viscosity. This gives gum *arabic* eminent qualities as an emulsifier, stabilizer, thickener or adhesive of a non-toxic nature and its low calorific value and high soluble fibre, which are of major importance in nutrition and dietary applications. Pharmaceutical drugs and cosmetics also use the gum as a binder, emulsifying agent, and a suspending or viscosity increasing agent. *A. senegal*, commonly known as to produce gum acacia is a deciduous shrub, growing upto 15 m tall and usually branching out from the ground. Gum production is excellent on poor soils and higher in stressed trees .Gum exudes from the duct of the inner bark and is tapped in the hot season (May-June) when the trees are under stress and annual gum production in India is about 5-10 tons. Tapping begins when trees are 4-5 years old. It commences after leaf fall and ceases during the colder months of the dry season. Annual yields stand at 188-2856 g for young trees and 379-6754 g for older trees (7-15 years). Production of gum *arabic* is concentrated within a 520,000 $Km^2$ radius of central Sudan, commonly known as the gum belt (FAO, 2014).

## International trade of gum *arabic*

The three main producing countries are Sudan, Nigeria, Cameroon and Chad which cover about 82% of the global export market of the gum *arabic*.

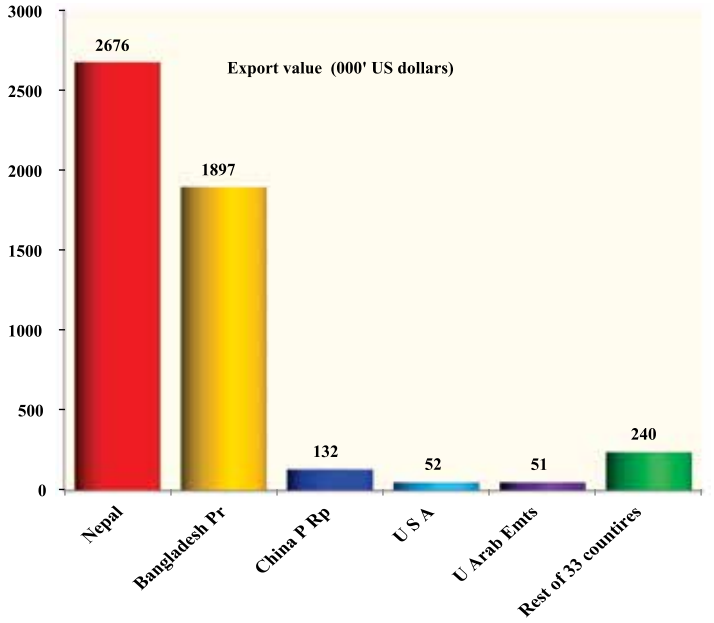



Figure 29. Destination wise scenario of gum arabic from India during 2015-16

Destination wise export and source wise import share is depicted in Table 26. There is high demand of gum *arabic* from and in India also. About 4000 tons of gum *arabic* exported to Nepal (53.02 %), Bangladesh Pr (37.58 %) and China P Rp (2.61 %) remained as major export destinations during 2015-16. About 93 % gum *arabic* supplied to these top 3 destinations (Figure 29). It was noted that Nepal emerged as major importer for gum Arabic in 2015-16. Due to this demand, the quantity exported was increased tenfold in comparison to previous year. However, there was only three time increase in the export value as the exported quantity in Nepal (88.52%) was of poor quality (grade-III). In the international market, price of gum *arabic* ranged from ₹500-600/kg for Grade-I, ₹100-300/kg for Grade-II, ₹50-100/kg for Grade-III.

**Table 26. Destination and source wise EXIM scenario of gum Arabic during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000' US dollars) | Share (%) |
|---|---|---|---|---|
| **Export** | | | | |
| Nepal | 3661.68 | 1739.56 | 2676.25 | 53.02 |
| Bangladesh Pr | 347.60 | 1233.01 | 1896.94 | 37.58 |
| China P Rp | 41.00 | 85.66 | 131.78 | 2.61 |
| U S A | 14.53 | 33.54 | 51.60 | 1.02 |
| U Arab Emts | 15.94 | 33.12 | 50.95 | 1.01 |
| U K | 0.75 | 28.48 | 43.82 | 0.87 |
| Oman | 7.52 | 25.57 | 39.34 | 0.78 |
| Japan | 7.15 | 23.74 | 36.52 | 0.72 |
| Pakistan Ir | 5.00 | 12.85 | 19.77 | 0.39 |
| Indonesia | 2.00 | 10.30 | 15.85 | 0.31 |
| Saudi Arab | 8.61 | 9.07 | 13.95 | 0.28 |
| Guatemala | 3.00 | 6.70 | 10.31 | 0.20 |
| Others | 21.81 | 39.56 | 60.86 | 1.21 |
| **Total** | **4136.59** | **3281.16** | **5047.94** | **100.00** |
| **Import** | | | | |
| Sudan | 17777.73 | 12624.84 | 19422.83 | 61.59 |
| Nigeria | 4560.90 | 1876.92 | 2887.57 | 9.16 |
| Cameroon | 2598.00 | 1310.45 | 2016.08 | 6.39 |
| Chad | 1441.00 | 927.82 | 1427.42 | 4.53 |
| Mali | 1838.60 | 790.60 | 1216.31 | 3.86 |
| France | 273.78 | 775.98 | 1193.82 | 3.79 |
| Senegal | 1121.67 | 586.31 | 902.02 | 2.86 |
| Benin | 266.00 | 367.40 | 565.23 | 1.79 |
| U Arab Emts | 796.00 | 364.55 | 560.85 | 1.78 |
| Ethiopia | 512.00 | 337.87 | 519.80 | 1.65 |
| Tanzania Rep | 377.90 | 227.04 | 349.29 | 1.10 |
| Others | 353.22 | 307.17 | 472.57 | 1.50 |
| **Total** | **31916.80** | **20496.95** | **31533.77** | **100.00** |

The international market for gum *arabic* is subject to different trends and fluctuations, determined by an increasing demand, variable capacity of African producer countries to stabilize the supply, variability of quality and prices and use of substitutes by importing countries, which can negatively affect the gum *arabic* market demand. During 2015-16, gum *arabic* is exported across the globe in 38 countries. Sudan, Nigeria, Cameroon and Chad have supplied 82 % raw gum *arabic* and rest 18 % was supplied from Mali, France, Senegal, Benin, U Arab Emts, Ethiopia, Tanzania Rep and others (Figure 30). India is second only to the US in consumption of Gum Arabic (NGARA, 2004). Re-export of processed and value added gum *arabic* (solubilized, atomized, kibbled) is a very common trade practice with France and United Kingdom leading the countries engaged in such re-export.

## International trade of *asian* gum

There is high demand of *asian* gum from and in India also. Destination wise export and source wise import share of *asian* gum is depicted in Table 27 and graphical presentation of direction of trade is illustrated in Figure 31.

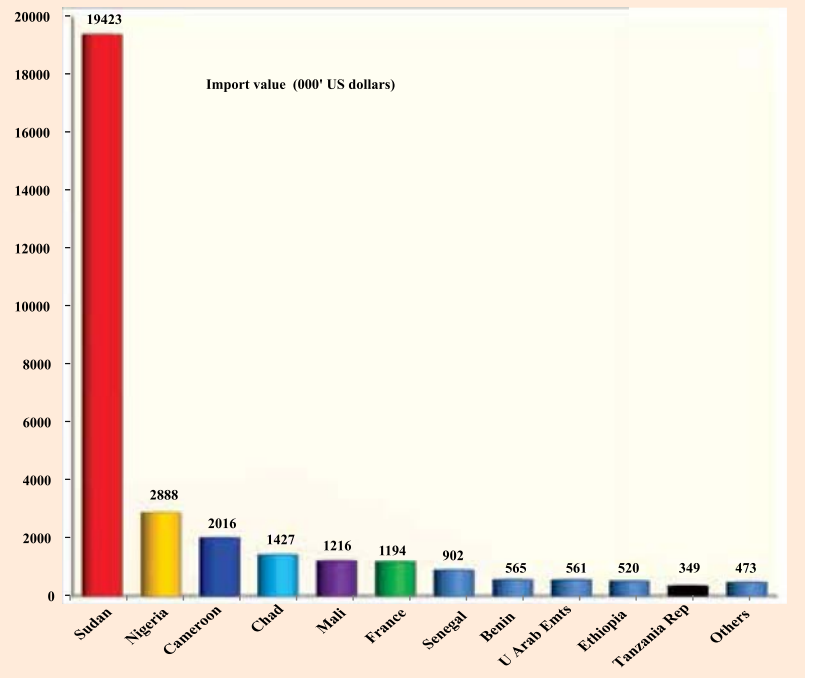



Figure 30. Source wise scenario of gum arabic in India during 2015-16

During 2015-16, about 440 tons of *asian* gum was exported from the country. Germany (29.89 %), Bangladesh Pr (14.11 %), Malaysia (10.21 %), Belgium (6.75 %), Lithuania (6.74 %), Japan (6.38 %), South Africa (4.36%) and Netherland (4.33 %) remained as major export destinations. About 83 % gum Asian supplied to these top 8 destinations. During 2015-16, Asian gum was exported across the globe in 14 countries and Singapore alone had supplied raw *asian* gum for India.

**Table 27. Destination and source wise scenario of *asian* gum during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000' US dollars) | Share (%) |
|---|---|---|---|---|
| **Export** | | | | |
| **Germany** | 126.00 | 517.22 | 789.65 | 29.89 |
| **Bangladesh Pr** | 72.73 | 244.39 | 373.11 | 14.11 |
| **Malaysia** | 40.23 | 176.71 | 269.79 | 10.21 |
| **Belgium** | 28.00 | 116.83 | 178.37 | 6.75 |
| **Lithuania** | 36.00 | 116.73 | 178.21 | 6.74 |
| **Japan** | 14.70 | 110.40 | 168.55 | 6.38 |
| **South Africa** | 17.76 | 75.38 | 115.08 | 4.36 |
| **Netherland** | 16.50 | 74.89 | 114.34 | 4.33 |
| **Korea Rp** | 7.45 | 55.72 | 85.07 | 3.22 |
| **France** | 20.00 | 35.42 | 54.08 | 2.05 |
| **U S A** | 9.37 | 31.49 | 48.08 | 1.82 |
| **Russia** | 5.00 | 28.38 | 43.33 | 1.64 |
| **Spain** | 4.08 | 26.32 | 40.18 | 1.52 |
| **U K** | 11.02 | 22.44 | 34.26 | 1.30 |
| **Turkey** | 4.06 | 20.36 | 31.08 | 1.18 |
| **Hungary** | 6.00 | 19.07 | 29.11 | 1.10 |
| **Australia** | 5.36 | 18.71 | 28.56 | 1.08 |
| **Others** | 16.58 | 40.16 | 61.31 | 2.32 |
| **Total** | **440.84** | **1730.62** | **2642.17** | **100.00** |
| **Import** | | | | |
| **Singapore** | 29.89 | 15.34 | 20.05 | 100.00 |
| **Total** | **29.89** | **15.34** | **20.05** | **100.00** |

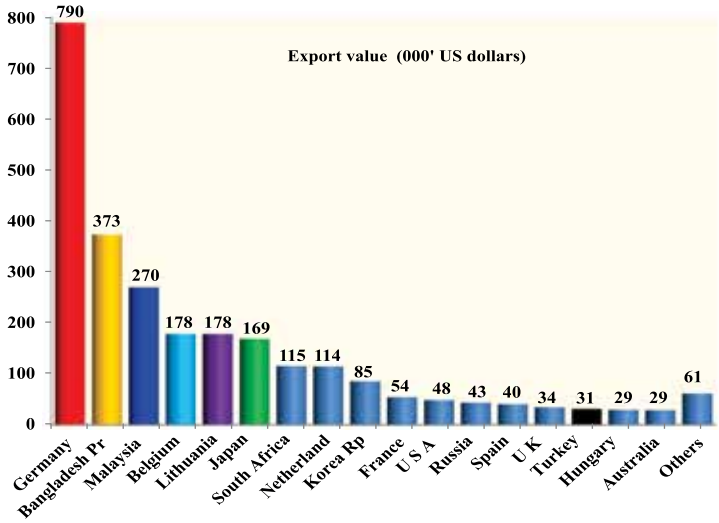



Figure 31. Destination wise scenario of *asian* gum from India during 2015-16

## International trade of *african* gum

In India, *african* gum is supplied from African countries for domestic consumption and rest is re-exported. It is evident from the Table 28 that Kenya was only the major export destination of *african* gum.

**Table 28. Export of *african* gum from India during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000' US dollars) | Share (%) |
|---|---|---|---|---|
| **Export** | | | | |
| **Kenya** | 0.6 | 1.11 | 1.71 | 100.00 |
| **Total** | **0.6** | **1.11** | **1.71** | **100.00** |

## *Jhingan* gum (*Lannea coromandelica*)

*L. coromandelica* (*Jhingan*, Indian Ash Tree, *Moi* tree) is a deciduous tree which grows upto 14 m height. *Jhingan* gum is mostly obtained from natural exudation and sometimes by tapping and annual gum production is about 60-80 tons in India. It is used as an alternative to gum *arabic* in food and other purposes. It was found that the emulsion containing *jhingan* gum produced comparatively more sTable emulsion. Production and export performance of *agarbathi* industry depends on availability of forest based raw materials.

## *Saja (Terminalia tomentosa)*

*T. tomentosa* (*Saja, Saj*) is a tree growing upto 30 m tall, with a trunk upto diameter of 1 m. A copious transparent gum exudes in large globular tears from the trunk. In India, the annual gum production is about 30-40 tons. It is used as an incense and cosmetic. *T. tomentosa* has a remarkable attribute as some members of the species store water in the dry season.

## Tragacanth (Adracanth) *(Astragalus gummifer)*

Tragacanth gum (TG) is a natural and acidic polysaccharide that secretes spontaneously or with a scratch on the different species of *Astragalus* plant.

**Table 29. Destination and source wise scenario of Tragacanth (Adracanth) during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000 ₹ US dollars) | Share (%) |
|---|---|---|---|---|
| **Export** | | | | |
| Philippines | 0.06 | 0.21 | 0.32 | 16.67 |
| Sri Lanka DSR | 0.10 | 0.67 | 1.03 | 53.17 |
| U S A | 0.20 | 0.38 | 0.58 | 30.16 |
| **Total** | **0.36** | **1.26** | **1.94** | **100.00** |
| **Import** | | | | |
| Czech Republic | 0.01 | 1.45 | 2.23 | 95.39 |
| Italy | 0.00 | 0.07 | 0.11 | 4.61 |
| **Total** | **0.01** | **1.52** | **2.34** | **100.00** |

*A. gummifer* is an important species of this plant. TG obtained in two different forms: ribbon and flake. The name tragacanth is originated from two Greek words of tragos (goat) and akantha (thorn) showing its curved shape. EXIM scenario of Tragacanth (Adracanth) during 2015-16 is given in Table 29.

### *Xanthan* gum

*Xanthan* gum is a polysaccharide used as a food additive and rheology modifier, commonly used as a food thickening agent. *Xanthan* gum was discovered by Allene Rosalind Jeanes. It was brought into commercial production under the trade name Kelzan in the early 1960s. It was approved for use in foods after animal testing in 1968. It is accepted as a safe food additive with E number E415. *Xanthan* gum derives its name from the strain of bacteria used during the fermentation process, *X. campestris*. *X. campestris* is the same bacterium responsible for causing black rot to form on broccoli, cauliflower and other leafy vegeTables. The bacterium forms a slimy substance that acts as a natural stabilizer or thickener. In foods, *xanthan* gum is most often found in salad dressings and sauces. It helps to prevent oil separation by stabilizing the emulsion, although it is not an emulsifier. Toothpaste often contains xanthan gum, wherein it serves as a binder to keep the product uniform. EXIM scenario of *xanthan* gum during 2015-16 is given in Table 30.

**Table 30. Destination and source wise scenario of *xanthun* gum during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000' US dollars) | Share (%) |
|---|---|---|---|---|
| **Export** | 0.58 | 0.48 | 7.83 | 61.61 |
| U Arab Emts | 2.17 | 12.57 | 16.54 | 54.96 |
| Nigeria | 1.00 | 5.06 | 6.66 | 22.13 |
| Bangladesh Pr | 0.30 | 1.56 | 2.05 | 6.82 |
| Ghana | 0.25 | 1.20 | 1.58 | 5.25 |
| Bhutan | 0.75 | 1.11 | 1.46 | 4.85 |
| Ethiopia | 0.10 | 1.04 | 1.37 | 4.55 |
| Nepal | 0.09 | 0.17 | 0.22 | 0.74 |
| Egypt A Rp | 0.03 | 0.16 | 0.21 | 0.70 |
| **Total** | **4.69** | **22.87** | **30.09** | **100.00** |
| **Import** | | | | |
| China P Rp | 24.00 | 46.32 | 75.93 | 94.96 |
| Poland | 0.25 | 2.27 | 3.72 | 4.65 |
| U S A | 0.05 | 0.18 | 0.30 | 0.37 |
| Denmark | 0.01 | 0.01 | 0.02 | 0.02 |
| **Total** | **24.31** | **48.78** | **79.97** | **100.00** |

### Plant Exudate: Other gums

The other important gums are gum *arabic* (*A. senegal)*, Gum kondagogu (*Cochlospermum gossypium), saja* (*T. tomentosa*), *Jhingan* (*L. coromandelica), babool* (*A. nilotica*), *palas* (*B. monosperma*), *khair* (*A. catechu*) and *char* (*B. lanzan* **Spreng**). Gujarat is major *guggal* gum and

*prosopis* gum/*babool* gum producing state. Gum kondagogu is a naturally occurring nontoxic polysaccharide derived as an exudate from the bark of *C. gossypium* (*Bixaceae* family), a native tree of India. The total production of other gums across various states was around 389.5 tons during 2015-2016 (Table 31). It has been demonstrated graphically in Figure 32.

**Table 31. Production of other gums in India (in tons)**

| State /Year | 2009-10 | 2010-11 | 2011-12 | 2012-13 | 2013-14 | 2014-15 | 2015-16 | Average |
|---|---|---|---|---|---|---|---|---|
| Andhra Pradesh | 11.81 | 6.59 | 9.23 | 12.94 | 13.79 | 9.12 | 8.93 | 10.34 |
| Telangana | 2.00 | 1.16 | 2.24 | 0.60 | 0.65 | 3.50 | 3.55 | 1.96 |
| Chhattisgarh | 19.43 | 1.07 | 3.44 | 1.90 | 0.50 | 0.00 | 92.00 | 16.91 |
| Gujarat | 46.60 | 27.49 | 31.04 | 41.12 | 53.22 | 50.00 | 15.00 | 37.78 |
| Jharkhand | 0.00 | 240.60 | 200.50 | 125.60 | 60.20 | 40.00 | 50.00 | 102.41 |
| Madhya Pradesh | 0.00 | 280.00 | 250.00 | 200.00 | 174.60 | 100.00 | 100.00 | 157.80 |
| Maharashtra | 0.00 | 200.00 | 140.00 | 100.00 | 192.00 | 88.00 | 80.00 | 114.29 |
| Others | 0.00 | 100.00 | 80.00 | 70.00 | 72.30 | 42.00 | 40.00 | 57.76 |
| **Total** | **79.83** | **856.91** | **716.45** | **552.16** | **567.26** | **332.62** | **389.48** | **499.24** |

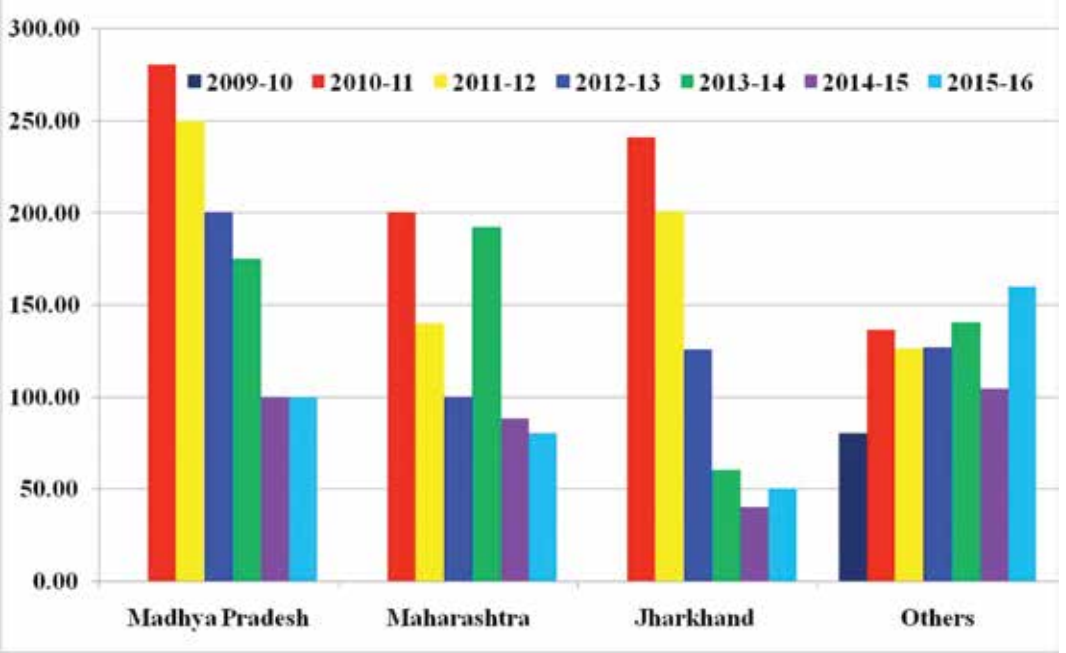



Figure 32. Total production of other gums across various states in India

## Cassia tora seeds

Cassia tora seeds, traded under ITC (HS) Code 09109915, have remained another commodity in high exports over the ten year period from 2005-06 to 2014-15 with its export volume increasing over ten times from 2083 tons during 2005-06 to 28190 tons during 2014-15, with a corresponding increase in export value from ₹ 3.57 crore in 2005-06 to ₹123.99 crore in 2014-15. However, its export has declined to 19741 tons with an export value of ₹138 crore in 2015-16. The export price per kg has also increased from just about ₹ 17 per kg in 2005-06 to ₹ 44 per kg in 2014-15 and continue declining.

## Collection price and grading

The collection rate of *jhingan*, *khair* and *babool* gum ranged from ₹ 2000 to 5000/qt. Retail packing price of the *babool* gum and *salai dhoop* in Gujarat increased from ₹ 170/kg to ₹ 200/kg and ₹ 250/kg to ₹ 400/kg, respectively. Acacia gum has been harvested in Arabia, Sudan, and West Asia since antiquity, sub-Saharan acacia gum has a long history as a prized export and get lesser value due to lac of primary processing. Currently, the main populations of gum-producing Acacia species are harvested in Mauritania, Senegal, Mali, Burkina Faso, Niger, Nigeria, Chad, Cameroon, Sudan, Erita, Somalia, Ethiopia, Kenya and Tanzania. Grade-wise collection price of other gums in India is depicted in Table 32.

**Table 32. Collection price of other gums in India (grade wise)**

| Natural gum | Scientific name | Grade | Price (₹/Kg) |
|---|---|---|---|
| *Guggul* gum | *C. wightii* | I | 600-650 |
| *Guggul* gum | *C. wightii* | II | 250-260 |
| *Guggul* gum | *C. wightii* | III | 150-175 |
| *Salai* gum | *B. serrata* | I | 125-150 |
| *Khair* gum | *A. catechu* | I | 50-60 |
| *Babool* gum | *A. nilotica* | I | 40 |
| *Babool* gum | *A. nilotica* | II | 30 |
| *Babool* gum | *A. nilotica* | III | 27 |
| Other minor gums | | | 30-40 |

## Regulations and governance

On the basis of regulations and governance issues for trading of NTFPs including natural resins and gums, the Government has classified as nationalized items, monopoly items and non-nationalized items. State Government classifies NTFPs as nationalized items, which can only be traded / marketed by Forest Department and it may vary from state to state. Four major gums namely *karaya/kullu* gum (*S. urens*), gum ghatti/ *dhawada* gum (*A. latifolia*), *babool* gum (*A. nilotica*) and *khair* gum (*A. catechu*) are categorized under nationalized gums in some states. A trade wing of the Forest Department's trade division deals with its marketing. The forest Department gives the monopoly rights for value added processes and marketing of NTFPs to some agency under lease agreement, based on sustainable harvesting and economic value. Apart from nationalized and monopoly items, remaining NTFPs are open to all. Primary collectors or their associations must get permits from the Divisional Forest Officer (DFO) to transport and market the produce. The state-wise list of gums producing trees and respective categories is presented in *Annexure IV*.

## Seed based natural gum-*Guar* gum



Photo 4. Natural seed gum : *Guar* gum

India is a native of *guar* or cluster bean *(Cyamopsis tetragonoloba)* grown in dry regions. It is used as a vegeTable by households, as cattle food (whole green plant), as cattle feed (*churi* and *korma)* and as a green manure crop in agriculture. The growing season of *guar* is 14 to 16 weeks and requires reasonably warm weather and moderate flashing rainfall with plenty of sunshine. Too much rain can cause the plant to become more 'leafy' resulting thereby reducing the number of pods or the number of seeds per pod which affects the size and yield of seeds. The *guar* is a naturally rainfed crop. Depending on the monsoon rainfall the total size of *guar* crop varies from year to year. After harvesting, when the pods become dry through sunlight, they are beaten off and during this process, the seeds come out of the pods. Peak arrival of the produce in the market is seen in October to December while lean arrival in the months of January to May. In major *guar* producing areas like Rajasthan, Haryana and Gujarat, *guar* crop is grown after the first shower of monsoon in July to early August and is harvested in late October to early November. It is grown as the rainfed crop and its production totally depends on the performance of monsoon. The decline in production due to deficient rainfall during 2009-10 also confirmed this fact. The price hit had been all time high in the spot and future commodity markets during this period.

*Guar* gum comes from the endosperm (30%) of the seed. There are various grades of *guar* gums pure and its derivatives. *Guar* gum is a white to creamy coloured, free flowing powder without any extraneous matter. Its ability to bind water, control the viscosity of aqueous solutions, form strong and tough films have accounted for its wider application and consequently the rapid growth.

## *Guar* seed production in India

India contributes about 80% to the world *guar* production and Pakistan is the second largest producer. All India area, production and yield of *guar* seed has been presented in Table 33. State wise area, production and yield of *guar* seed has been presented in Table 34.

**Table 33. All India area, production and yield of *guar* seed and potential *of guar* gum production** *(Area- thousand ha, Production- thousand tons, Yield- kg/ha,)*

| Year | Area | Production | Yield | Potential of *guar* gum production |
|---|---|---|---|---|
| 2009-10 | 2995.2 | 594.7 | 199 | 198.2 |
| 2010-11 | 3382.2 | 1965.3 | 581 | 655.1 |
| 2011-12 | 3444.2 | 2217.6 | 644 | 744.3 |
| 2012-13 | 5151.7 | 2460.7 | 478 | 819.0 |
| 2013-14 | 5887.7 | 3388.4 | 576 | 1129.1 |
| 2014-15 | 5359.8 | 3284.0 | 613 | 1094.6 |
| 2015-16 | 5484.01 | 2718.21 | 496 | 817.0 |

*Source: Directorate of Economics and Statistics, MoA, GoI; www.rsamb.rajasthan.gov.in*

**Table 34. State wise area, production and yield of *guar* seed during previous seven years**

| State | 2009-10 | 2010-11 | 2011-12 | 2012-13 | 2013-14 | 2014-15 | 2015-16 | Average |
|---|---|---|---|---|---|---|---|---|
| State wise area of *guar* seed during previous seven years ( thousand ha) | | | | | | | | |
| Andhra Pradesh | - | - | - | - | 3.03 | 1.06 | 1.20 | 1.70 |
| Gujarat | 132.70 | 124.60 | 128.30 | 207.10 | 278.00 | 278.00 | 297.00 | 206.53 |
| Haryana | 252.00 | 256.00 | 215.00 | 388.00 | 481.00 | 406.00 | 367.00 | 337.86 |
| Punjab | 21.80 | 24.00 | 25.00 | 26.00 | 28.20 | 48.60 | 30.00 | 29.09 |
| Rajasthan | 2586.79 | 2980.95 | 3094.16 | 4533.27 | 5070.92 | 4625.21 | 4787.01 | 3954.04 |
| Uttar Pradesh | 2.54 | 2.40 | 2.50 | 2.50 | 2.50 | 2.00 | 2.00 | 2.35 |
| **All India** | **2995.83** | **3387.95** | **3464.96** | **5156.87** | **5863.66** | **5360.87** | **5484.01** | **4530.59** |
| State wise production of *guar* seed during previous seven years (thousand MT) | | | | | | | | |
| Andhra Pradesh | - | - | - | - | 3.00 | 1.00 | 0.90 | 1.63 |
| Gujarat | 44.70 | 73.00 | 74.50 | 128.50 | 133.60 | 167.00 | 179.00 | 114.33 |
| Haryana | 329.00 | 333.00 | 290.00 | 278.00 | 369.00 | 338.00 | 283.00 | 317.14 |
| Punjab | 16.80 | 18.00 | 20.00 | 22.00 | 21.20 | 33.40 | 30.00 | 23.06 |
| Rajasthan | 201.00 | 1540.54 | 1846.61 | 2026.67 | 2861.94 | 2743.97 | 2223.71 | 1920.64 |
| Uttar Pradesh | 2.10 | 2.00 | 2.00 | 2.00 | 2.00 | 1.60 | 1.60 | 1.90 |
| **All India** | **593.60** | **1966.54** | **2233.11** | **2457.17** | **3390.74** | **3284.97** | **2718.21** | **2377.76** |
| State wise yield of *guar* seed during previous seven years (Kg/ hectare) | | | | | | | | |
| Andhra Pradesh | - | - | - | - | 987 | 937 | 900 | 941 |
| Gujarat | 337 | 586 | 581 | 620 | 434 | 602 | 602 | 537 |
| Haryana | 1305 | 1301 | 1350 | 718 | 767 | 832 | 772 | 1006 |
| Punjab | 771 | 750 | 800 | 846 | 753 | 688 | 1000 | 801 |
| Rajasthan | 78 | 517 | 597 | 447 | 564 | 593 | 465 | 466 |
| Uttar Pradesh | 835 | 800 | 800 | 800 | 800 | 800 | 800 | 805 |
| **All India** | **198** | **580** | **644** | **476** | **578** | **613** | **496** | **512** |

*Source: Directorate of Economics and Statistics, MoA, GoI*

Rajasthan has been a leading state in production as well as processing of *guar* seed in India followed by Haryana and Gujarat. In Rajasthan, Churu, Bikaner, Jaisalmer, Barmer, Nagaur, Hanumangarh, Jodhpur, Ganganagar, Jaipur, Sirohi, Dausa, Jhunjhunu and Sikar are the districts where *guar* crop cultivated. Gurgaon, Mahendragrh and Rewari districts of Haryana and few districts like Kutch, Banaskantha, Mehsana, Sabarkantha, Vadodara and Ahmadabad of Gujarat are also *guar* producing areas in the country. Cultivation of *guar* crop has been started in newer areas like Andhra Pradesh including Telengana and other southern states also.

## *Guar* seed processing and value addition in India

India is leading in *guar* gum production due to its well established *guar* gum industry in Jodhpur, Rajasthan. Other states like Gujarat, Haryana and Maharashtra has *guar* seed collection centers and *guar* gum processing units. Recently, more units are also being established in Bikaner, Rajasthan with high capacity. Sri Ganganagar, Hissar, Alwar, Sirsa, Jodhpur, Bikaner, Jaipur are the major markets for *guar* seed. Mainly *guar* gum is used in food, paper, textile, oil drilling, mining, explosives, ore flotation and other various industrial applications. Out of the decadal average potential production of *guar* gum around 6.5 lakh tons in the country, around 3.0 lakh tons is consumed in the domestic market as raw as well as processed and around 3.5 lakh tons is exported.

Processing of *guar* gum are intended to impart desired chemical properties in the galactomannan like increased solubility in water, clarity of solution, increased shelf life, ionic character according to the needs of the particular final application. By these modifications lot of opportunities are opened, where chemically modified *guar* derivatives can be used (Table 35).

**Table 35. Various varieties of processed products of *guar* gum for industrial applications**

| Varieties | Applications | Varieties | Applications |
|---|---|---|---|
| Hydroxy alkylated *guar* gum | Oil well drilling | Sulphated *guar* gum | Mining |
| Carboxy methylated *guar* gum | Textile printing, Tobacco industry, water based paints | *Guar* gum formate | Food-Human and animal, Miscellaneous |
| Oxidised *guar* gum | Paper industry Food, textile printing applications | *Guar* gum acryl amide | Pharmaceutical industry |
| Acetates of *guar* gum | Food-Human and animal | Borate cross linked *guar* gum | Photography, Oil well drilling, carpet printing |
| Cationic derivatives of *guar* gum | Cosmetic and paper industries | Reticulated *guar* gum | Stick explosives, blasting slurries |

## Movement in price of *guar* gum

The seasonal demand for *guar* gum is at its peak during the months of Nov-March around the world.

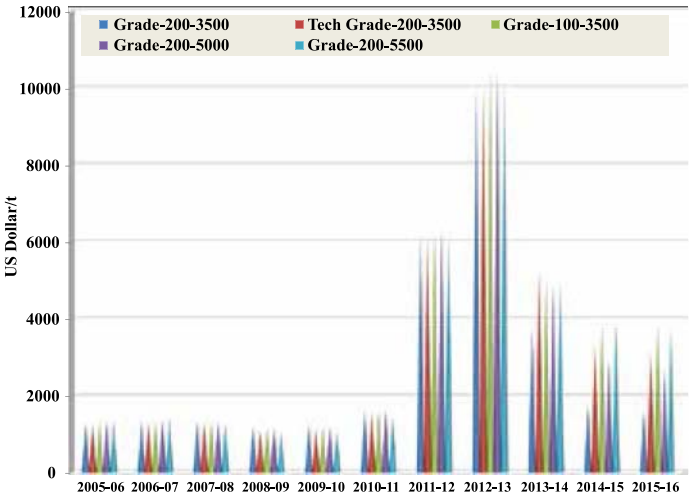



Figure 33. International price level of different grades of *guar* gum

As the reason being quite simple, the majority of the drilling activities of crude oils take place during these months. *Guar* seed prices in near future in domestic market may remain sTable as the normal production during 2015-16. The international price scenario of different grades of *guar* gum during 2005-06 to 2015-16 is shown in Figure 33.

## International trade of *guar* gum

The share of agricultural exports in India's total exports decreased from 13.79% in 2013-14 to 12.55% in 2015-16. Agricultural exports decreased from ₹ 2627.78 billion in 2013-14 to ₹ 2153.95 billion in financial year 2015-16 registers a decline of nearly 18.03%. Decrease in the value of agricultural exports during 2015-16 was primarily on account of lower exports of wheat, oil meals, *guar* gum meals, *basmati* and non-*basmati* rice. India's top exports of processed agricultural commodities in terms of quantity and value for three years have been given in the Table 36. Decrease in value of agricultural exports during 2015-16 was primarily on account of lower exports of *guar* gum (62 %), wheat (79%), basmati rice (18%) and Buffalo meat (9%). *Guar* gum has been the topmost export commodity in agri-export during previous years like in 2012-13. But, in 2014-15 it ranked on 4$^{th}$ position after buffalo meat, *basmati* and non-*basmati* rice and subsequently in 2015-16 it ranked on 6$^{th}$ position after buffalo meat, *basmati*, non-*basmati* rice, groundnuts and cereal preparations.

**Table 36. Top agri-export commodities during previous three years**

| Particulars | Export Value (₹ in crores) | | | % share in total agri-export | | |
|---|---|---|---|---|---|---|
| | 2013-14 | 2014-15 | 2015-16 | 2013-14 | 2014-15 | 2015-16 |
| 1. *Guar* gum | 11730 | 9480 | 3613 | 8.7 | 7.3 | 3.0 |
| 2. *Basmati* rice | 29300 | 27599 | 22714 | 21.7 | 21.2 | 21.2 |
| 3. Buffalo meat | 27250 | 29283 | 26682 | 20.2 | 22.4 | 24.8 |
| 4. Non-*basmati* rice | 17490 | 20336 | 15086 | 13.0 | 15.6 | 14.0 |
| 5. Other cereals | 7130 | 5258 | 5405 | 5.3 | 4.0 | 5.0 |
| 6. Wheat | 9260 | 4975 | 1062 | 6.9 | 3.8 | 1.0 |
| Total (S.No.1-6) | 102160 | 96931 | 74562 | 75.8 | 74.3 | 69.0 |
| **Total agri-export (All items)** | **134940** | **130460** | **107430** | **100.0** | **100.0** | **100.0** |

*Source: DGCIS & APEDA Annual Report.*

Export of *guar* gum during X and XI plan period, export and production capacity during 2009-10 to 2015-16 is given in Table 37 and Figure 34. India is the major exporter of *guar* gum to the world. It exports various forms of *guar* products to a large number of countries. The country has exported 325.25 thousand tons of *guar* gum to the world for the worth of ₹ 323.38 crores during the year 2015-16. India is also the leading net exporter of *guar* seeds. The country exports over 3.25 lakh tons of *guar* and its derivatives, which is comprised by 45,667 tons of refined split *guar* gum (14.04 %), 2.11 lakh tons of treated and pulverized *guar* gum (64.88 %), 68,573 tons of *guar* meal (21.08 %) and traces of *guar* seed. Composition of *guar* gum export from India is given in Figure 35. Destination wise overseas demand and source wise supply of *guar* meal, *guar* gum refined split (GGRS), *guar* gum treated and pulverized (GGTP) and in India from the overseas countries is illustrated through pie charts in Figure 36, Figure 37, Figure 38 and Figure 39, respectively.

**Table 37. Export of *guar* gum from India during previous years**

| Year | Quantity (in '000 MT) | Value (in ₹ crore) | Value (in US million dollar) |
|---|---|---|---|
| 2009-10 | 217.41 | 1133.30 | 240.70 |
| 2010-11 | 421.43 | 2938.70 | 646.08 |
| 2011-12 | 702.73 | 16523.90 | 3354.82 |
| 2012-13 | 406.32 | 21287.00 | 3919.23 |
| 2013-14 | 601.97 | 11735.40 | 1979.70 |
| 2014-15 | 665.23 | 9479.90 | 1552.14 |
| 2015-16 | 325.25 | 3233.87 | 496.57 |

*Source: Directorate of Economics and Statistics, MoA, GoI; APEDA*

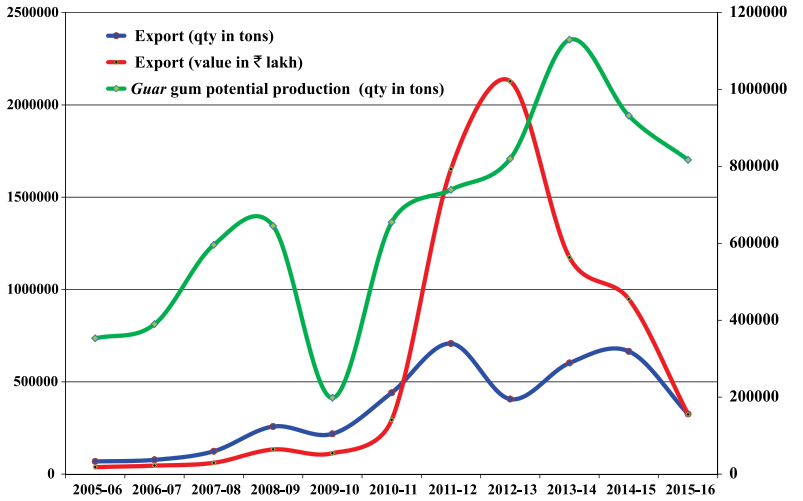



Figure 34. Trends in production potential and export of *guar* gum from India

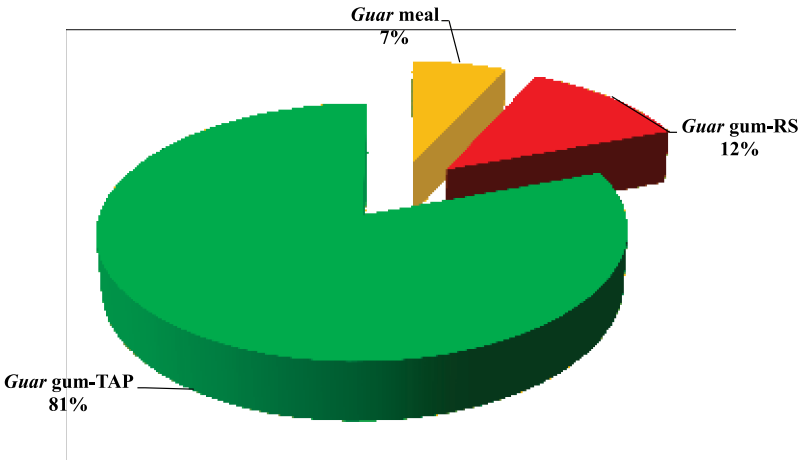



Figure 35. Composition of *guar* gum export from India

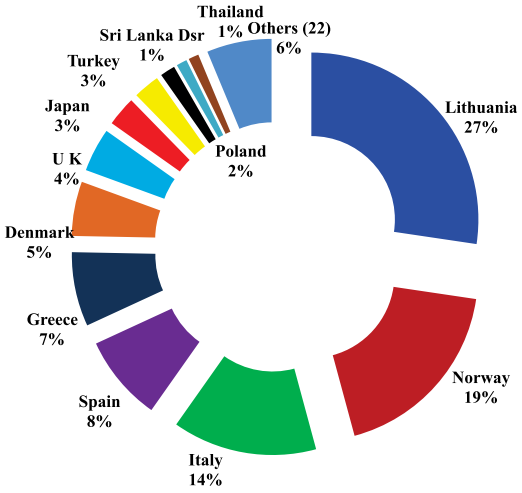



Figure 36. Destination wise scenario of *guar* meal during 2015-16

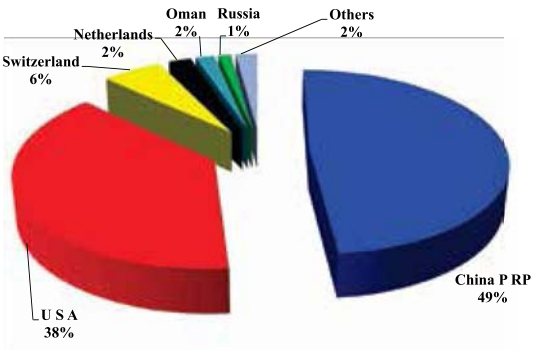



Figure 37. Destination wise scenario of *guar* gum refined split during 2015-16

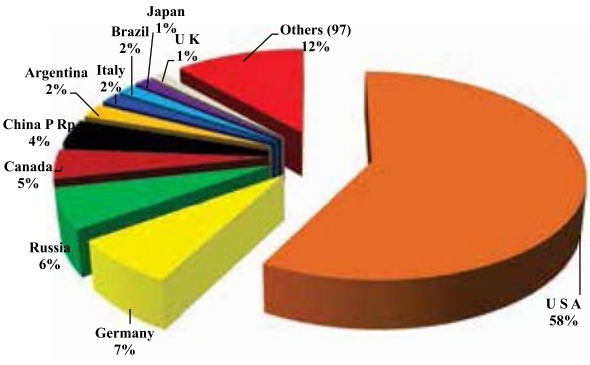



Figure 38. Destination wise scenario of *guar* gum treated and pulverized during 2015-16

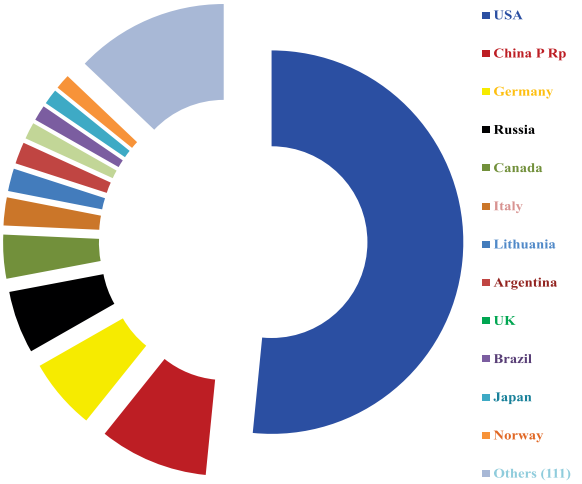



Figure 39. Destination-wise quantitative share of the *guar* gum exported from India (Overall)

United States, China, Germany, Russia, Canada, Italy, Lithuania, Argentina, UK, Brazil, Japan and Norway remained as major export destinations in 2015-16. During 2015-16, *guar* gum was exported across the globe to 123 countries. About 87 % was exported to top 12 countries and rest 13 % was exported to 111 countries around the world. Other countries are Switzerland, Spain, Australia, United Arab Emirates, Indonesia, Netherlands, Denmark, Belgium, Turkey, Greece, Thailand, France, Mexico, South Africa, Korea Republic, Poland, Chile, Iran, Malaysia, Philippines, Oman, Egypt Arab Republic, Ukraine, Taiwan, Israel, Colombia, Saudi Arabia, Iraq, Sri Lanka, *etc*. Destination wise export share is depicted in Figure 20.

## Natural Gum Resins

Under this category, *asafoetida*, *salai* gum and *myrrh* are the major gum resins. About 1100 tons of the gum resins exported from the country and ₹657 million earned as a foreign exchange.

## Asafoetida (*Hing*)

Asafoetida (*Ferula anthrax* and *F. foetida*) is a popular spice used in daily food by Indians. It was found to contain mainly ferulic acid, umbellic acid and ketonic substance known as umbelliferone. Powder of Asafoetida is used as carminative and also used in fainting, flatulent colic and chronic bronchitis as well as it is used to treat asthma in adults (Kokate *et al* 2002). New pharmacological investigations indicate possible anti-inflammatory, anti-diabetic and anti-bacterial effects (Kandziora *et al.*, 2015).

## International trade of Asafetida

In India, Asafetida is supplied from Afghanistan (90.44%) and Uzbekistan (7.32%). On other hand, UAE (27.25%), USA (14.78%), Saudi Arab (7.33%) and Singapore (7.01%) are the major export destination of Asafetida (Table 38).

**Table 38. Destination and source wise scenario of asafetida during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (US million dollars) | Share (%) |
|---|---|---|---|---|
| **Export** | | | | |
| U Arab Emts | 243.67 | 1260.31 | 1.92 | 27.25 |
| U S A | 106.75 | 683.69 | 1.04 | 14.78 |
| Saudi Arab | 69.61 | 338.94 | 0.52 | 7.33 |
| Singapore | 48.78 | 324.09 | 0.49 | 7.01 |
| U K | 53.30 | 253.09 | 0.39 | 5.47 |
| Malaysia | 45.06 | 221.09 | 0.34 | 4.78 |
| Myanmar | 54.10 | 209.61 | 0.32 | 4.53 |
| Thailand | 41.00 | 202.81 | 0.31 | 4.38 |
| Kuwait | 26.60 | 147.00 | 0.22 | 3.18 |
| Australia | 29.10 | 141.37 | 0.22 | 3.06 |
| Oman | 19.66 | 120.93 | 0.19 | 2.61 |
| Qatar | 15.08 | 87.24 | 0.13 | 1.89 |
| Baharain Is | 12.87 | 75.96 | 0.12 | 1.64 |
| Canada | 11.22 | 72.32 | 0.11 | 1.56 |
| Bangladesh Pr | 9.00 | 68.46 | 0.10 | 1.48 |
| South Africa | 17.19 | 60.42 | 0.09 | 1.31 |
| Others | 82.42 | 358.18 | 0.55 | 7.74 |
| **Total** | **885.41** | **4625.51** | **7.06** | **100.00** |
| **Import** | | | | |
| Afghanistan | 912.65 | 47701.89 | 72.50 | 90.44 |
| Uzbekistan | 148.54 | 3859.47 | 5.87 | 7.32 |
| Iran | 132.03 | 1023.04 | 1.55 | 1.94 |
| Kazakhstan | 1.00 | 27.18 | 0.04 | 0.05 |
| Tajikistan | 4.89 | 130.55 | 0.20 | 0.25 |
| **Total** | **1199.11** | **52742.14** | **80.16** | **100.00** |

## *Sal*ai gum

*Salai* gum (*Boswellia serrata*) *Frankincense*, also known as *Olibanum*, is an aromatic oleogum resin obtained as pale yellow to red tears from the bark of trees belonging to the genera *Boswellia* of the *Burseraceae* family thriving in arid regions in the horn of Africa and southern Arabia. There are 43 different reported species in India, Arabian Peninsula and North Africa. Olibanum is produced in a restricted geographical area from uncultivated trees, and usually collected by small nomadic groups; the name is derived from the Arab word “al Luban”, which means milk and is references to the milky sap that, exudes from the tree upon incision. Today frankincense is widely employed in aromatherapy, Catholic Christian ceremonies as well as other religious and secular traditions

(Camarda *et al.*, 2016). It is also an important component in cosmetic industry and it is widely marketed as a food supplement (Khan, *et al*., 2009). The smoke is also a powerful insect deterrent and thus served as a prophylactic to prevent the bites of malaria carrying mosquitoes. The biological activity of frankincense resins is due to the pentacyclic triterpenic acids, α- and β-boswellic acids and their derivatives, which showed a well documented anti-inflammatory and immunomodulatory activities. The main component of frankincense is oil (60%). It contains mono- (13%) and diterpenes (40%) as well as ethyl acetate (21.4%), octyl acetate (13.4%) and methylanisole (7.6%).

## Production of Salai gum / olibanum

*B. serrata* (Family: Burseraceae) is a deciduous middle sized tree, which is mostly concentrated in tropical areas; parts of Asia and Africa. In India, it occurs in dry hilly forests of Rajasthan, Madhya Pradesh, Gujarat, Bihar, Assam, Odisha as well as central penisular regions of Andhra Pradesh, Assam *etc*. This gum is tapped from the incision made on the trunk of the tree which is then stored in specially made bamboo basket and converted into different grades of material according to flavor, color, shape and size. Harvesting frankincense is a time consuming process that begins in December, reaching a peak from March to May (Marshall 2003). The trees start producing resin when they are about 8 to 10 years old (Michie *et al*., 1991). In India, *S. robusta*, *A. latifolia*, *T. tomentosa*, *B. serrata, B. lanzan, A. catechu*, *etc*. are reported as the important tree species by Champion and Seth (1968) and these species exhibit local dominance. According Sagar *et al* (2003) half of these species changed dispersion behaviour as a result of disturbance. Species changing from clumped distribution to uniform distribution included *B. serrata, H. antidysenterica* and *L. coromandelica.*

## International trade of olibanum

In India, olibanum is supplied from U Arab Emts (57.53%), Somalia 23.43%), Yemen Republic (12.43%) and Ethiopia (5.53%). Trinidad (26.66%), Germany (12.69%), Mauritius (7.63%), Morocco (7.42%), Mexico (7.35%) and U S A (6.61%) are the major export destinations (Table 39).

**Table 39. Destination and source wise scenario of olibanum during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000' US dollars) | Share (%) |
|---|---|---|---|---|
| **Export** | | | | |
| **Trinidad** | 39 | 38.34 | 0.06 | 26.66 |
| **Germany** | 5 | 18.24 | 0.03 | 12.69 |
| **Mauritius** | 7.6 | 10.97 | 0.02 | 7.63 |
| **Morocco** | 7 | 10.67 | 0.02 | 7.42 |
| **Mexico** | 27 | 10.57 | 0.02 | 7.35 |
| **U S A** | 3.03 | 9.51 | 0.01 | 6.61 |
| **Argentina** | 2 | 8.65 | 0.01 | 6.02 |
| **Oman** | 1.83 | 8.65 | 0.01 | 6.02 |
| **Australia** | 0.56 | 7.23 | 0.01 | 5.03 |
| **Netherland** | 0.2 | 6.24 | 0.01 | 4.34 |
| **Others (7)** | 9.58 | 14.72 | 0.03 | 10.23 |
| **Total** | **102.8** | **143.79** | **0.23** | **100.00** |

| Import | | | | |
|---|---|---|---|---|
| **U Arab Emts** | 37 | 276.12 | 0.42 | 57.53 |
| **Somalia** | 11 | 112.45 | 0.17 | 23.43 |
| **Yemen Republc** | 11.7 | 59.66 | 0.1 | 12.43 |
| **Ethiopia** | 14 | 26.57 | 0.04 | 5.53 |
| **Others (2)** | 0.76 | 5.21 | 0.01 | 1.08 |
| **Total** | **74.46** | **480.01** | **0.74** | **100.00** |

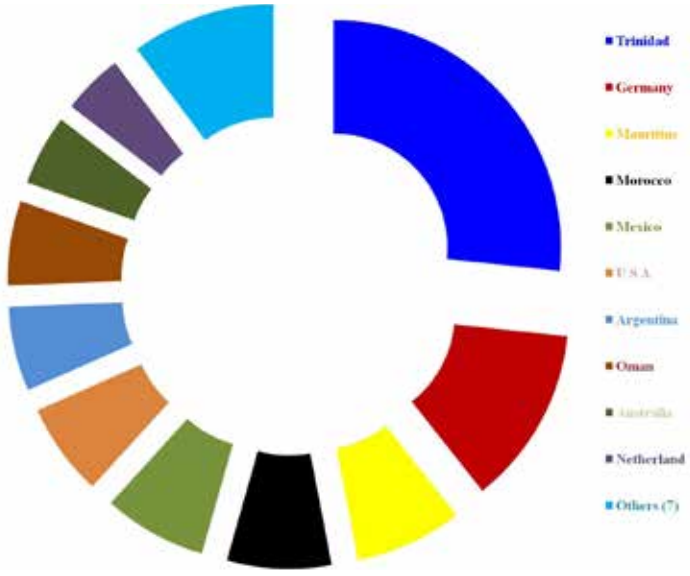



Figure 40. Destination-wise quantitative share of the olibanum exported from India

### *Myrrh*

*Myrrh* comes from a small, thorny tree. *Commiphora myrrha* was cultivated in ancient times in the Arabian Peninsula. The growers make a small cut in the bark, from where the resin would leak out. It is then collected and stored for about three months until it hardens into fragrant globules. *Myrrh* is used raw or crushed and mixed with oil to make perfume. Myrrh oil, which is steam distilled directly from the *myrrh* resin, has an aroma that is woody, earthy and a bit balsamic. Also, *myrrh* is occasionally used as a flavoring agent. Somalia and Ethiopia are by far the largest producers of the substance.

### International trade of *myrrh*

In India, *myrrh* is supplied from U Arab Emts (61.48%), Somalia (24.07%) and Kenya (6.96%). Turkey (47.21%), Iraq (47.21%) and UK (3.62%) are the major export destinations of *myrrh* (Table 40).

**Table 40. Destination and source wise scenario of myrrh during 2015-16**

| Country | Quantity (tons) | Value (₹ lakh) | Value (000' US dollars) | Share (%) |
|---|---|---|---|---|
| **Export** | | | | |
| **Turkey** | 0.30 | 4.83 | 7.58 | 47.21 |
| **Iraq** | 0.30 | 4.83 | 7.58 | 47.21 |
| **U K** | 0.02 | 0.37 | 0.58 | 3.62 |
| **Israel** | 0.01 | 0.13 | 0.20 | 1.27 |
| **Hungary** | 0.004 | 0.07 | 0.11 | 0.68 |
| **Total** | **0.64** | **10.23** | **16.05** | **100.00** |
| **Import** | | | | |
| **U Arab Emts** | 19.60 | 221.26 | 0.33 | 61.48 |
| **Somalia** | 14.70 | 86.64 | 0.13 | 24.07 |
| **Kenya** | 32.03 | 25.03 | 0.04 | 6.96 |
| **Germany** | 1.50 | 10.44 | 0.02 | 2.90 |
| **Ethiopia** | 2.00 | 10.23 | 0.02 | 2.84 |
| **Brazil** | 0.18 | 3.45 | 0.01 | 0.96 |
| **Italy** | 0.15 | 2.03 | 3.18 | 0.56 |
| **Yemen Republic** | 1.00 | 0.80 | 1.26 | 0.22 |
| **Total** | **71.16** | **359.88** | **4.99** | **100.00** |

### *Dikamali* (*Gardenia gummiferra*)

*G.gummiferra* (*Dikamali*) family *Rubiaceae* is the gum resin obtained from the leaf buds of a shrubby plant by making a cut on the stem or branches. It is geographically distributed in all districts of south India, Burma, Bangladesh, Konkan region, North Kanara, and Malabar Coast. Other names are Dikamali, Gandharaj, Hingunadika, Nadihingu, pindava, *etc.* It contains 89.9% resin and 0.1% oil and gardenin, a coloring agent and annual production is 10-15 tons. Producers' price in the market is ₹100-120 and retail price of processed product is about ₹1200-1400.

## Natural Resins and Gums

NRG production level during 2015-16 was estimated to be comparatively lower (~843178 tons) than previous year (Table 41). Other resins and gum production have declined during the current year. Production and trade of NRGs including *guar* gum, lac, pine resin, *karaya* gum, dhawda gum and other natural resins and total production figure of all the NRGs is tabulated as under.

**Table 41. Total NRG production (quantity in tons)**

| Name of product | 2011-12 | 2012-13 | 2013-14 | 2014-15 | 2015-16 | Average | % share |
|---|---|---|---|---|---|---|---|
| *Guar* gum* | 744295 | 818975 | 1129134 | 1094547 | 817042 | 920799 | 97.05 |
| Lac | 17900 | 19577 | 21008 | 16978 | 18746 | 18842 | 1.99 |
| Pine resin | 8514 | 8361 | 6875 | 6699 | 5726 | 7235 | 0.76 |
| Karaya gum | 285 | 212 | 129 | 83 | 100 | 162 | 0.02 |
| *Dhawda* gum | 147 | 380 | 448 | 295 | 194 | 293 | 0.03 |
| Other gums | 716 | 552 | 567 | 333 | 389 | 512 | 0.05 |
| Other resins and gums | 691 | 1083 | 975 | 1190 | 980 | 984 | 0.10 |
| **Grand Total** | **772549** | **849140** | **1159137** | **1120125** | **843178** | **948826** | **100.00** |

**Estimation of guar gum is based on the conversion of total guar seed production with a coefficient of conversion*

**Export and import of Natural Resins and Gums during 2015-16**

The total export of natural resins, gums and gum-resins during the year 2015-16 was 272462.21 tons valued ₹ 3439.95 crores and total import was 96387.62 tons valued ₹ 1291.69 crores. A comparative decrease of 49.88% was observed in the volume of total NRG export during 2015-16. Due to deceleration in price, particularly for *guar* gum, the total value of NRG export was less than previous year. Details of export and import of natural resins, gums and gum-resins is presented in the Table 42.

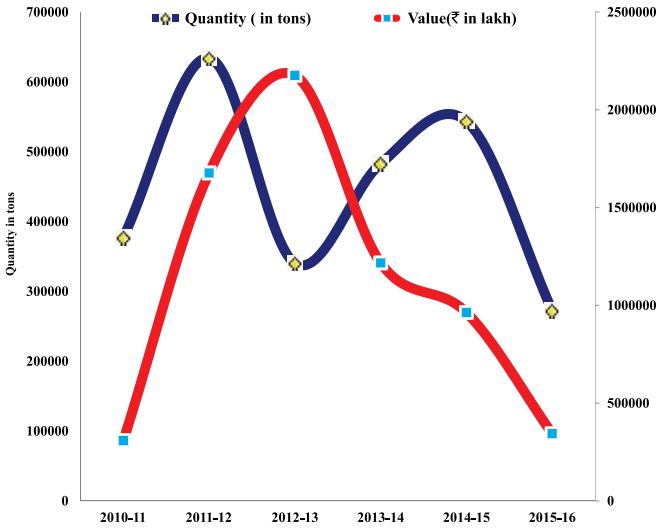



Figure 41. Trend in export of natural resins and gums from India during last five years

**Table 42.Total NRG export from and import in India during 2015-16**

| Sl. No. | Name of product | Export | | Import | |
|---|---|---|---|---|---|
| | | Quantity (tons) | Value (` Lakh) | Quantity (tons) | Value (` Lakh) |
| **A. Natural resins** | | | | | |
| 1. | Copal | 0.00 | 0.00 | 2078.20 | 1115.79 |
| 2. | Dammar batu | 2.84 | 6.58 | 16525.84 | 5674.14 |
| 3. | Gum rosin | 204.33 | 303.30 | 41962.57 | 44297.13 |
| 4. | Lac | 7668.42 | 24755.18 | NA | NA |
| 5. | Mastic gum | 0.38 | 2.59 | 4.40 | 23.25 |
| 6. | Other resins | 344.96 | 360.93 | 106.03 | 315.77 |
| | **Total** | **8220.93** | **25428.57** | **60677.03** | **51426.08** |
| **B. Gums** | | | | | |
| 7. | African gum | 0.60 | 1.11 | 0.00 | 0.00 |
| 8. | Asian gum | 440.86 | 1730.61 | 29.89 | 15.34 |
| 9. | Gum arabic | 4136.56 | 3281.17 | 31916.79 | 20496.96 |
| 10. | *Guar* gum refined split | 45667.65 | 40311.47 | 32.93 | 206.08 |
| 11. | *Guar* gum treated and pulverized | 211009.53 | 260841.87 | 600.20 | 1177.43 |
| 12. | Gum Karaya (Indian tragacanth) | 230.79 | 771.62 | 771.56 | 637.97 |
| 13. | Tragacanth (Adracanth) | 0.36 | 1.26 | 0.02 | 1.52 |
| 14. | Xanthum gum | 4.70 | 22.87 | 24.25 | 48.78 |
| 15. | Other natural gums | 1639.90 | 5033.62 | 303.77 | 549.50 |
| | **Total** | **263130.94** | **311995.60** | **33679.40** | **23133.58** |
| **C. Gum resins** | | | | | |
| 16. | Asafoetida | 885.43 | 4625.51 | 1199.12 | 52742.14 |
| 17. | Myrrh | 0.61 | 10.22 | 71.16 | 359.87 |
| 18. | Olibanum or frankincence | 102.79 | 143.79 | 74.46 | 480.00 |
| 19. | Other gum resins | 121.51 | 1792.28 | 686.46 | 1027.69 |
| | **Total** | 1110.34 | 6571.80 | 2031.19 | 54609.70 |
| | **G. Total** | **272462.21** | **343995.98** | **96387.62** | **129169.37** |

*Source: DGCIS, SHEFEXIL & APEDA Annual Reports.*

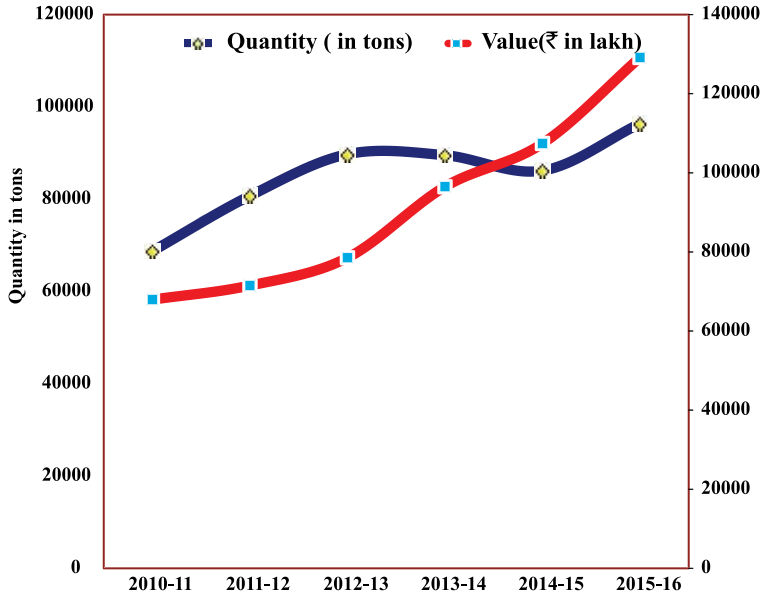



Figure 42. Trend in import of natural resins and gums in India during last five years

NRGs are important natural products in the export basket for foreign exchange reserve in India. The average annual export quantity of NRGs was 218971.29 tons valued ₹ 357155.57 lakh during the period 1998-99 to 2012-13. However, during last 15 years exported quantity has registered increasing trend with an annual compound growth rates of 10.69%. Consequently, export value of NRGs has also shown increasing trend with an annual compound growth rates of 22.04%. Similarly, the average annual import quantity of rosin was 50523.48 tons valued ₹ 29896.12 lakh during the period 1998-99 to 2012-13. Its import had a very high stability with growth rates in import quantity (13.64%) as well as value (24.95%). Trend in export of natural resins and gums from India and import in India during last five years is presented in Figure 41 and Figure 42, respectively.

# POLICY INTERVENTIONS

To provide fair prices of NWFPs which differ widely state to state and to prevent exploitation of the poor tribals'; Minimum Support Price (MSP) was introduced. The practices adopted by different states for fixation of prices for different Non Wood Forest Products (NWFPs) were on the basis of market prices by reducing all the overheads and the second methodology for arriving at procurement price was to follow cost plus method in which cost of collection was needed to be calculated for NWFPs in different states. Therefore, government has taken the initiatives under the scheme for NWFPs.

## Mechanism for marketing of Minor Forest Produce through Minimum Support Price and development of Value Chain for MFP

A Pricing Cell was constituted in pursuant to the Ministry of Tribal Affairs, Government of India and notified vide TRIFED, to recommend MSP for Minor Forest Produce under the scheme "Mechanism for marketing of Minor Forest Produce through MSP and development of Value Chain for MFP". To determine fair and remunerative MSP for lac and gum *karaya* the deliberations of the Pricing Cell with State Procurement Agencies (SPAs) and federations were conducted. Computation of cost of production of gum *karaya* and lac for MSP was furnished by ICAR-Indian Institute of Natural Resins & Gums (ICAR-IINRG), Ranchi. The Pricing Cell analysed the above information furnished by the SPAs and observed the wide variation in the data on the related parameters from state to state. Following recommendations were made in the context of price stabilizing policy for NWFPs.

## Minimum Support Price of gum *karaya* during 2014-15

Gum *karaya* is procured from gum pickers in three grades and procurement prices are also announced grade wise. Despite announcement of MSP in Gujarat hardly any quantity of gum *karaya* was procured. Similarly in Jharkhand and Maharashtra also, the quantity procured was very low. The Pricing Cell felt it appropriate to determine MSP on the basis of the cost of production as calculated by ICAR-IINRG plus 10 % increase thereon. Accordingly, the Pricing Cell recommended MSP of ₹ 108/kg for gum *karaya* Grade-III for the current procurement season. The Pricing Cell recommended that a study be immediately initiated so that relevant data to calculate cost of collection for 12 MFPs in 8 states can be collected.

## A. Minimum Support Price of lac for the year 2014-15

Major quantity of lac is available in the state of Jharkhand followed by Chhattisgarh, Madhya Pradesh and Odisha. Accordingly, based on the inputs provided by IINRG, Pricing Cell had recommended a Minimum Support Price (MSP) of ₹210 and ₹290/ kg for *rangeeni* and *kusumi* lac, respectively.

## B. Minimum Support Price of lac for the year 2015-16

The Pricing Cell recommended enhancing the current MSP by 10 %. Pricing Cell had recommended a MSP of ₹230 and ₹320/kg for *rangeeni* and *kusumi* lac, respectively.

## Minimum Support Price of lac for the year 2016-17

During the current year, the lac growers have used broodlac of July 2015 crop in their cultivation when the prices were ranging between ₹150-200/kg, hence during harvesting season in December and January cost of production of lac may be comparatively lower than previous season. Pricing Cell had recommended a Minimum Support Price (MSP) of ₹100 and ₹150/kg for *rangeeni* and *kusumi* lac, respectively for 2016-17 crops.

# CONCLUSIONS

New and young entrepreneurs have entered in lac sector with big investments for production of high value products like Aleuritic acid and isoambrettolide. Value addition in lac based products has an encouraging market scenario as the export of Aleuritic acid is increasing since last two years. To meet the domestic demand pine resin processing infrastructure is needed to be revamped. Both the units are functional at under capacity level. Production and procurement of some gums is declining during last two years. MSPs for NWFP may be key intervention for procurement, marketing and value chain establishment. It will be helpful to stabilize the price fluctuation particularly in case of lac. Although resins discussed have proved to be popular alternative or complementary medicine used in the treatment of many diseases, clinical trial evaluation of these claims using currently accepted protocols is needed. The reported resins offer huge potential as a possible pharmacological application but its necessary a further investigation to verify whether purified compounds isolated may have better therapeutic potential as compared to crude extracts. These new chemicals will serve to enhance the continous usefulness of higher plants and their products as renewable resources of chemicals.

Major gums and resins of international commerce produced and traded in and from India include *guar* gum, lac, gum *karaya*, gum *arabic* from *Acacia* species, and *myrrh*, asafetida, rosin and olibanum from *Commiphora* species. The volume of export of natural resins was increased by 14.3% and the volume of import of natural resins was also increased by 16.6% in 2015-16. However, the value of export for natural resins decreased by 23.4%. The volume of export of natural gums was decreased by 49.9% and the volume of import of natural gums was increased by 3.7% in 2015-16. But, the value of export for natural gums decreased by 66.2%. Similarly, the volume of export of natural gum-resins was decreased by 90.1% and the volume of import of natural gums was increased by 20.4% in 2015-16. However, the value of export for natural gum resins increased by 4.3%. Overall, the volume of export of NRGs was decreased by 49.9% and the volume of import of natural resins was increased by 11.8% in 2015-16. But, the value of export for NRGs decreased by 64.3%.

Enhancing the offer of Indian market products and increasing their role in the rural economy could help to reduce the costs of forest protection. A well-structured forest economy that is able to provide sTable flow of incomes can provide a fundamental set of public non-market services and social values to both local people and the whole community. Understanding the true value of natural resources, for both land users and policymakers, is an essential step for promoting their protection and sustainable use. Since markets do not reflect all values flowing from NWFPs, further research is needed to better understand the link between ecosystem functioning and the delivery of services, as well as their appropriate economic evaluation. As a final point, we are aware that synergies and trade-offs exist between market products and nonmarket services, as well as among different ecosystem services. The constraints related to export of natural resins and gums were small and scattered quantity of produce, fluctuation in domestic production and prices, adulteration in raw material, limited export promotional measures, improper international market information, lack of future trading and prior

agreement as expressed by the exporter. India has a great potential in production and export of natural resins and gums because of availability of resins and gums yielding trees, manpower, favorable climate and R&D support. The proper attention and action for intensification of effort for increasing natural resins and gums production will definitely improve the export earning of the country. Scientific inoculation and gum tapping methods would produce more NRGs than traditional method because they have been practiced successfully by innovative/progressive farmers during last few years. It would be preferable to consider scientific inoculation and gum tapping methods to get better quality and yield, and to change the economic standards of cultivators of India.

***Annexure I*. Sticklac production scenario in India during 2015-16 (in tons)**

| Name of states / Districts | Baisakhi | Jethwi | Katki | Aghani | Total |
|---|---|---|---|---|---|
| **I. Andhra Pradesh** | 60 | 5 | 100 | 7 | 172 |
| **II. Assam** | 65 | 0 | 250 | 0 | 315 |
| **III. Chhattisgarh** | | | | | |
| **1.** Bastar | 15 | 35 | 10 | 80 | 140 |
| **2.** Bilaspur | 40 | 20 | 60 | 60 | 180 |
| **3.** Dhamtari | 32 | 30 | 20 | 25 | 107 |
| **4.** Durg | 10 | 0 | 15 | 0 | 25 |
| **5.** Janjgir-Champa | 45 | 25 | 20 | 35 | 125 |
| **6.** Kanker | 25 | 265 | 65 | 350 | 705 |
| **7.** Korba | 210 | 200 | 200 | 400 | 1010 |
| **8.** Mahasamund | 40 | 10 | 50 | 15 | 115 |
| **9.** Raipur | 10 | 90 | 10 | 100 | 210 |
| **10.** Rajnandgaon | 50 | 20 | 40 | 30 | 140 |
| **11.** Surguja | 70 | 0 | 40 | 0 | 110 |
| **12.** Ambikapur | 5 | 10 | 10 | 0 | 25 |
| **13.** Raigarh | 30 | 20 | 10 | 25 | 85 |
| **14.** Others | 40 | 40 | 50 | 80 | 210 |
| **Total** | **622** | **765** | **600** | **1200** | **3187** |
| **IV. Gujarat** | 20 | 20 | 3 | 5 | 48 |
| **V. Jharkhand** | | | | | |
| **15.** Garhwa | 15 | 0 | 30 | 0 | 45 |
| **16.** Gumla | 10 | 840 | 55 | 700 | 1605 |
| **17.** Latehar | 10 | 15 | 20 | 15 | 60 |
| **18.** Palamau | 40 | 0 | 125 | 20 | 185 |
| **19.** Ranchi | 270 | 1800 | 220 | 900 | 3190 |
| **20.** Khunti | 250 | 850 | 110 | 400 | 1610 |
| **21.** Simdega | 50 | 1000 | 70 | 900 | 2020 |
| **22.** West Singhbhum | 100 | 310 | 45 | 370 | 825 |
| **23.** Others | 80 | 120 | 110 | 100 | 410 |
| **Total** | **825** | **4935** | **785** | **3405** | **9950** |
| **VI. Madhya Pradesh** | | | | | |
| **24.** Annuppur & Shahdol | 5 | 3 | 5 | 1 | 14 |

| | | | | | |
|---|---|---|---|---|---|
| **25.** Balaghat | 580 | 7 | 150 | 4 | 741 |
| **26.** Chhindwada | 10 | 8 | 10 | 5 | 33 |
| **27.** Dindori | 100 | 8 | 10 | 4 | 122 |
| **28.** Hosangabad | 10 | 15 | 10 | 5 | 40 |
| **29.** Mandla | 95 | 6 | 35 | 15 | 151 |
| **30.** Narshinghpur | 5 | 2 | 5 | 2 | 14 |
| **31.** Seoni | 740 | 10 | 140 | 10 | 900 |
| **32.** Others | 125 | 7 | 60 | 15 | 207 |
| **Total** | **1670** | **66** | **425** | **61** | **2222** |
| **VII. Maharashtra** | | | | | |
| **33.** Bhandara | 60 | 0 | 25 | 0 | 85 |
| **34.** Chandrapur | 60 | 0 | 25 | 0 | 85 |
| **35.** Garhchiroli | 140 | 0 | 55 | 0 | 195 |
| **36.** Gondia | 600 | 0 | 500 | 0 | 1100 |
| **Total** | **860** | **0** | **605** | **0** | **1465** |
| **VIII. Meghalaya (Garo hills)** | 120 | 0 | 7 | 0 | 127 |
| **IX. Odisha** | | | | | |
| **37.** Balasore | 10 | 4 | 15 | 10 | 39 |
| **38.** Koraput | 5 | 6 | 10 | 5 | 26 |
| **39.** Mayurbhanj | 5 | 6 | 10 | 100 | 121 |
| **40.** Nabarangpur | 15 | 25 | 20 | 70 | 130 |
| **41.** Sundergarh | 10 | 60 | 0 | 140 | 210 |
| **42.** Keonjhar | 0 | 5 | 0 | 10 | 15 |
| **43.** Others | 20 | 12 | 20 | 30 | 82 |
| **Total** | **65** | **118** | **75** | **365** | **623** |
| **X. Uttar Pradesh** | 15 | 0 | 40 | 0 | 55 |
| **XI. West Bengal** | | | | | |
| **44.** Bankura | 15 | 5 | 20 | 5 | 45 |
| **45.** Midnapur | 90 | 2 | 160 | 5 | 257 |
| **46.** Purulia | 25 | 40 | 25 | 100 | 190 |
| **47.** Others | 20 | 15 | 30 | 25 | 90 |
| **Total** | **150** | **62** | **235** | **135** | **582** |
| **Grand total** | **4472** | **5971** | **3125** | **5178** | **18746** |

***Annexure II*. Lac producing areas in the country**

| Sl. No. | Name of State/ District | Major lac producing areas |
|---|---|---|
| **I** | **Andhra Pradesh including Telangana** | |
| 1 | Adilabad | Utnoor |
| 2 | Vishakhapatnam | Paderu |
| **II** | **Assam** | |
| 3 | Kamrup | Boko |
| 4 | Karbi-Anglong | Amtreng, Baithalangsu |
| 5 | Marigaon | Nellei |
| 6 | Nagaon | Amsoi, Hojai |
| **III** | **Bihar** | |
| 7 | Gaya | Raniganj |
| 8 | Jehanabad | Malichak |
| **IV** | **Chhattisgarh** | |
| 9 | Balrampur | Chando, Dharmi, Wandrafnagar |
| 10 | Bastar | Keshkal |
| 11 | Bilaspur | Gaurella, Kota, Lorami, Pendra |
| 12 | Dhamtari | Gatta Silli, Nagri, Sihawa |
| 13 | Durg | Balod, Daudi, Dondi-Lohara, Kusumkasa |
| 14 | Gariyaband | Amlipadar, Chhaila, Chhura Gariyaband, Indagaon, Mainpur, Udanti |
| 15 | Janjgir-Champa | Sakti, Saragaon |
| 16 | Kanker | Antagarh, Bhanupratapur, Biragaon, Kondagaon, Korar, Narharpur, Sambalpur |
| 17 | Korba | Bhaisama Bazar, Chaitama, Chhuri, Haldi Bazar, Kartala, Katghora, Korbi, Madanpur, Pali, Pasan, Pasarkhet |
| 18 | Korea | Bhartarpur, Kusmi |
| 19 | Mahasamund | Bagbahara, Basana, Khalari, Mahasamund, Pithora, Tendukona |
| 20 | Narayanpur | Narainpur |
| 21 | Raigarh | Dharmjaigarh, Lailunga, Pathalgaon,Tamnar |
| 22 | Raipur | Abhanpur |
| 23 | Rajnandgaon | Aawadhi, Bharitola, Khardi, Manpur, Mohala-chowki |
| 24 | Sarguja | Chalgi, Mainpat, Pasta, Pratappur, Premnagar, Raghunathnagar, Ramanujganj, |
| **VI** | **Gujarat** | |
| 25 | Panchamahal | Dakor, Godhara, Ghoghamba, Nadiad and Santrampur |

| | | |
|---|---|---|
| 26 | Vadodara | Chhota Udepur, Devhant, Jambaguda, Jatpurpavi, Jhonjh, Kawant, Keori, Kundal Ghata, Tejgadh and Zoz |
| **VII** | **Jharkhand** | |
| 27 | Chatra | Tandwa |
| 28 | Garhwa | Bargarh, Garhwa, Godarmana, Rakshi, Ramganga, Ramkonda, Ranka |
| 29 | Gumla | Chainpur, Dumari, Kanshir, Palkot, Patratoli, Raidih |
| 30 | Khunti | Murhu, Soeko Torpa, Karra, Rania,Tapkara |
| 31 | Latehar | Balumath, Barwadih Brahmani, Chandwa, Garu Latehar, Manika, Sarju, Satbarwa |
| 32 | Palamu | Matalong, Panki, Raj Chaipur, |
| 33 | Ranchi | Angara, Banta, Birbanki, Bundu, Jonha, Namkum, Ormanjhi, Silli, Maranghada, Sonahatu, Tamar |
| 34 | Simdega | Bano, Hating hode, Jaldega, Kolebira, Lachragarh |
| 35 | West Singhbhum | Anandpur, Bandgaon, Baskata, Chaibasa, Chakradharpur, Goelkera, Lodai, Manoharpur, Sonuwa, Toklo |
| **VIII** | **Madhya Pradesh** | |
| 36 | Anuppur | Jaitahari, Keshwahi, Kotma, Venkatnagar |
| 37 | Balaghat | Baihar, Katangi, Lalbarra, Lamta, Langi, Parashwada, Waraseoni |
| 38 | Chhindawada | Damoa |
| 39 | Dindori | Bazak, Bhanupur, Karanjia, Ramnagar, Rampur |
| 40 | Hosangabad | Bankhedi, Babai, Daggrai, Darawpadaw, Hapa, Jonahata, Kekra, Lokamti, Pipariya |
| 41 | Mandla | Bichhia, Chabbi, Ghughari, Kalpi, Mahegaon, Mavai, Nainpur, Narainganj, Navas-Bablia |
| 42 | Narshinghpur | Chichli, Godarwara, Kalakhar, Kalyanpur, Nayakheda, Salechauka |
| 43 | Seoni | Barghat, Ghansore, Kahani, Kanewara, Keolari, Khamaria, Khari |
| 44 | Shahdol | Burhar, Jaitpur, Sohagpur |
| **IX** | **Maharashtra** | |
| 45 | Bhandara | Gobarwahi, Nakadongri |
| 46 | Chandrapur | Bandh, Navargaon |
| 47 | Garhchiroli | Allapalli, Bamragarh |
| 48 | Gondia | Amgaon, Chopa, Goregaon, Hirapur, Kampta, Kati, Kotjamura, Kurodhi, Salekasa, Tiroda |
| **X** | **Meghalaya** | |
| 49 | Garo Hills | Damra, Dodno, Nongpoh, Tura |
| 50 | Odisha | |

| | | |
|---|---|---|
| 51 | Balasore | Haldipada, Jaleshwar, Nilagiri |
| 52 | Keonjhar | Telkoi |
| 53 | Koraput | Ramgiri |
| 54 | Mayurbhanj | Jashipur, Kaptipada, Karanjia, Kusumi, Padampokhari, Sarat, Thakurmunda, Udala, |
| 55 | Nabarangpur | Chandahandi, Raighar |
| 56 | Sundergarh | Gurundia, Khandadhar, Kutra, Rajgangpur |
| **XI** | **Uttar Pradesh** | |
| 57 | Sonbhadra | Doodhi |
| 58 | Allahabad | Koraon, Meja |
| **XII** | **West Bengal** | |
| 59 | Bankura | Idpur, Khatra, Raipur, Ranibandh, |
| 60 | Midnapur | Katai, Kuti, Moyna, Panskurah, Ramnagar, Tamluk |
| 61 | Purulia | Ayodhya Pahar, Bagh Mandi, Balarampur, Jhalda, Kashipur, Kutidih, Raghunathpur, Tulin |

***Annexure III*. Gums producing districts and areas**

| Sl. No. | Name of State/ District | Major gum producing areas |
|---|---|---|
| **I** | **Andhra Pradesh including Telangana** | |
| 1 | Adilabad | Jannaram, Kamma Reddy, SK Nagar and Utnoor |
| 2 | Chittur | Chittur, Kaddapa and Nellore |
| 3 | East Godavari | Addategella, Maredumilli, Rajavomma Kangi and Rampachodavaram |
| 4 | Khammam | Bhadrachalam, Chintoor, Dammapeta and Kukunaru |
| 5 | Mahboobnagar | Mannanoele, Nanjaria and Pedadornala |
| 6 | Srikakulam | Pathpattanam and Setampeda |
| 7 | Visakhapatnam | Arku, Chintapalli, GKVidhi, G.Murugala, Kashipatnam, Koyuru, Munchingput, Paderu and Petabayallu |
| 8 | Vizianagram | Gummalaxmipuram, Parvatipuram and Salur |
| 9 | Warangal | Etrunagram, Mulug and Narshimhpeda |
| 10 | West Godavari | AR Puram |
| **II** | **Chhattisgarh** | |
| 11 | Balrampur | Chando, Dharmi |
| 12 | Bastar | Jagdalpur, Keshkal |
| 13 | Bilaspur | Dindori, Lormi, Mugeli, Ratanpur and Takhtpur |
| 14 | Dhamtari | Gatta Silli, Nagri, Sihawa |
| 15 | Gariyaband | Amlipadar, Chhaila, Chhura Gariyaband, Indagaon, Mainpur, Udanti |

| | | |
|---|---|---|
| 16 | Janjgir-Champa | Sakti, Saragaon |
| 17 | Kanker | Antagarh, Bhanupratapur, Biragaon, Kondagaon, Korar, Narharpur, Sambalpur |
| 18 | Korba | Bhaisama Bazar, Chaitama, Chhuri, Haldi Bazar, Kartala, Pali, Pasan, Pasarkhet |
| 19 | Korea | Bhartarpur |
| 20 | Mahasamund | Bagbahara, Basana, Khalari, Pithora |
| 21 | Narayanpur | Narainpur |
| 22 | Raigarh | Khamariya, Dharmjaigarh, Manendragarh Milupara, Pathal-gaon |
| 23 | Raipur | Abhanpur |
| 24 | Rajnandgaon | Khardi, Manpur, Mohala-chowki |
| 25 | Sarguja | East Sarguja, South Sarguja |
| 26 | Sukma | Sukma |
| 27 | Bijapur | Bijapur, Bairamgarh, Bhopalpatnam, Usoo |
| 28 | Dantewada | Dantewada,Geedam,Kuwakonda, Katekalyan |
| **III** | **Gujarat** | |
| 29 | Balsar | Dharampur, Malanpada and Pangarbari |
| 30 | Banskantha | Ambaji, Pata and Virampur |
| 31 | Bharuch | Jaghadiya and Jankhawav |
| 32 | Dahod | Baria and Sagtala |
| 33 | Dang | Ahwa and Waghai |
| 34 | Kuchachh | Bhachau, Bhirandiyara, Bhuj, Dwarika, Mandvi, Nakhatrana and Rapar |
| 35 | Narmada | Dediapada, Mandvi and Rajpipla |
| 36 | Navsari | Ankalachh, Bansda and Chikhali |
| 37 | Panchamahal | Dakor, Ghoghamba, Nadiad and Santrampur |
| 38 | Sabarkantha | Bhiloda, Choriwada Modasa and Posina |
| 39 | Vadodara | Chhota Udepur, Devhant, Jambaguda, Jatpurpavi, Kawant, Tejgadh and Zoz |
| **IV** | **Jharkhand** | |
| 40 | Chatra | Lawalang, Pratappur and Kanti |
| 41 | Garhwa | Ramkanda and Bhandaria |
| 42 | Gumla | Dumari, Kanshir, Palkot, Patratoli, Raidih |
| 43 | Khunti | Murhu, Soeko Rania,Tapkara |
| 44 | Latehar | Garu, Mahuadar, Herhanj, Balumath, Barwadih, Lesliganj, Chhipadohar and Richughutu |
| 45 | Palamu | Daltonganj,Panki, Chhatarpur |

| | | |
|---|---|---|
| 46 | Ranchi | Angara, Bundu, Jonha, Sonahatu, Tamar |
| 47 | West Singhbhum | Chakradharpur |
| **V** | **Madhya Pradesh** | |
| 48 | Balaghat | Kochewahi, Baihar, Lamta, Langi, Paraswada |
| 49 | Betul | Savani |
| 50 | Chhindawada | Amarwada, Damua, Patalkot |
| 51 | Hosangabad | Pipariya |
| 52 | Mandla | Pindarai, Bamhani, Mohgaon, Lingapondi, Bichhia, Chabbi |
| 53 | Narshinghpur | Chichli, Godarwara, Kalakhar, Kalyanpur, Nayakheda, Salechauka |
| 54 | Seoni | Bhimgarh, Chhopara, Dhuma |
| **VI** | **Maharashtra** | |
| 55 | Bhandara | Gobarwahi, Nakadongri |
| 56 | Chandrapur | Mul |
| 57 | Garhchiroli | Armon, Wadsa, Desai ganj |
| 58 | Gondia | Salekasha, Amgaon, Goregaon, Dhapewada, Tiroda, Navegaon Bandh |
| 59 | Wardha | Dhagabhawan, Navargaon, Hingni, Bordhara , Karanja range |
| **VII** | **Odisha** | |
| 60 | Balasore | Haldipada, Jaleshwar, Nilagiri |
| 61 | Keonjhar | Telkoi |
| 62 | Koraput | Ramgiri |
| 63 | Mayurbhanj | Jashipur, Kaptipada, Karanjia, Kusumi, Padampokhari, Sarat, Thakurmunda, Udala, |
| 64 | Nabarangpur | Chandahandi, Raighar |
| 65 | Sundergarh | Gurundia, Khandadhar, Kutra, Rajgangpur |
| **VIII** | **Rajasthan** | |
| 66 | Ajmer | Beawar, Kekeri, Kisangarh, Nasserabad, Puskar |
| 67 | Alwar | Bansur, Rajgarh, Ramgarh,Thanagazi |
| 68 | Churu | Sardarsahar, Sujangarh,Taranagar |
| 69 | Jaipur | Amer, Bassi, Chomu, Chaksu, Jamwa Ramgarh, Kotputli, Phagi, Phulera |
| 70 | Jhunjhunu | Chirawa, Khetri, Navalgarh |
| 71 | Jodhpur | Bilara, Luni, Phalodi, Shergarh |
| 78 | Nagour | Ladnu, Merta, Parbatsar |
| 79 | Pali | Bali, Sojat, Sumerpur |
| 80 | Sikar | Fatehpur, Neem-ka-Thana, Ringus, Sri Madhopur |
| 81 | Tonk | Niwai, Malpura, Uniara |
| 82 | Udaipur | Gogunda, Kotra |

## *Annexure IV*. Legal categorization of major gums produced in various states of India

| State | MFP Under State Monopoly |
|---|---|
| Andhra Pradesh including Telangana | Bamboo, tendu leaves and 25 other MFPs. |
| Chhattisgarh | Bamboo, tendu leaves, sal seed, myrobalans and gums of kullu, dhawda, khair and babool. |
| Gujarat | Bamboo, tendu leaves, mhowra flower, mhowra seed, gum *karaya* and other miscellaneous gums. |
| Maharashtra | Bamboo, tendu leaves, apta leaves and 33 other MFPs. |
| Madhya Pradesh | Bamboo, tendu leaves, sal seed, kullu gum and lac resin. |
| Rajasthan | Bamboo, tendu. State monopoly for other MFPs ceased to exist in 2003. |
| Odisha | Bamboo and tendu leaves. |
| Himachal Pradesh | No MFP has been put under State monopoly. |
| Jharkhand | Tendu leaves. |

***Source:*** *TRIFED, Ministry of Tribal Affairs, New Delhi*

# NOTES

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